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(Un)*grammatical*

HOW GRAMMATICALITY WORKS

BRETT REYNOLDS



[IMPRINT]

*(Un)grammatical
How grammaticality works

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COLOPHON

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For James

Preface

The * is used by linguists to mark a sentence as ungrammatical. Its origins trace back to philologists who used it to signify a word that, although unattested in ancient languages, was suggested by evidence to exist. It meant that no actual instance of such a word could be found in any source. Generative grammarians subsequently intensified the meaning of *, shifting it from ‘unattested’ to ‘impossible’. (I learned this from Chris Potts.)

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The asterisk

I'm often jealous, and slightly in awe, of folks with strong and detailed biographical memories. I can recall, for example, the feeling of watching an Indonesian harbour appear out of the fog: small rocky islands jutting out of a perfectly calm dark sea, not long after the sun had risen. I was working on an 18-metre yacht out of Darwin, Australia, but I have no idea which island it is that teases my memory, or when this was.

I don't remember where we were coming from or where we were headed. I could probably work out the year, but most other details may be irretrievable: the captain's name or that of the yacht, though I'm strangely sure of its size. I remember a married couple was also among the crew, and perhaps one of my crew mates may have been named Chris. Or perhaps Steve.

The plan, as I remember it, had sounded almost absurdly generous: a leisurely pinball from island to island, starting in Ambon and heading south-west, then up past Malaysia and Thailand, around India, through the Suez, lingering in the Mediterranean, and eventually on to North America. But the captain, an Australian who had never been outside Australia, wanted to abandon the island-by-island route, get to Bali quickly, and then hurry on to what he called civilization.

We argued, I know, about that change, and I was left somewhere on or near Komodo Island. I recall somehow explaining to a man in a small village that I wanted to leave the island. A wooden fishing boat took me out into the strait. Eventually a large ferry appeared and took me on board. And where did I go next? Bali? Flores? I certainly spent time on each island. But the sequence is lost to me.

To tell the story at all, I have to reconstruct it. I have a few fixed points, a foggy harbour, a quarrel, a boat, a ferry, and a set of gaps between them. Some parts were witnessed. Some are inferred. Some are just blank.

For one of those states, the inferred but unseen, historical linguists use the as-

terisk (*). If you have any association at all with that mark, it's likely connected to footnotes or perhaps the marking of birth dates. It's a symbol of additional information, a small detour from the main text. But within the world of linguistics, this little star became a mark for something missing.

August Schleicher became a linguist, or philologist, before dictionaries traced the history of our words with the precision they have today, before scholars confidently linked the English word *father* with German *Vater* and Sanskrit *pitar*, for instance. The background to Schleicher's use of the asterisk begins with Sir William Jones.

Jones was a British judge stationed in India in the late 18th century. While studying Sanskrit, the ancient sacred language of the subcontinent, he began to see a system in the parallels between Sanskrit, Greek, and Latin, in words for family members, body parts, and basic concepts that seemed to echo across millennia and continents. He theorized that these languages descended from a common ancestor, a prehistoric root from which the diverse branches of a language family had grown.

Schleicher inherited Jones's way of seeing. He wanted a system that could explain regularities in language change, and he wanted to show the missing links in the chain. That need gave the asterisk its first linguistic job: to mark a form that the scholar inferred, but hadn't directly seen.

Take the seemingly unremarkable English word *tooth*. The inherited word belongs to a well-established Indo-European cognate set: Old English *tōþ*, from Proto-Germanic **tanþ-* ~ **tunþ-*, beside Gothic *tunþus*, is cognate with Latin *dēns*, genitive *dentis*, Greek *odoús* / *odón*, genitive *odóntos*, Sanskrit *dánta-*, Lithuanian *dantis*, Welsh *dant*, and Old Irish *dét* (Oxford English Dictionary, 2026).

The details matter less here than the pattern. Several languages preserve related forms, and the missing ancestor has to be inferred from the family resemblance.

This is the kind of pattern Schleicher was after. While there's written evidence for words like *Zahn*, *tunþus*, and *dánta-*, he needed a way to mark the hypothetical: a reconstructed stem, **dant-*, from which they seemed to have descended (Schleicher, 1874, p. 53). He wasn't afraid to guess at its existence, but he wanted to be clear that it was only a guess.

The asterisk made the guess visible. It told the reader: this form is part of the argument, but the evidence for it is indirect. The symbol stood at the point where observation gave way to inference.

That's close to what memory does, at least when memory is honest. I have seen that Indonesian harbour. I remember the dark water and the small rocky

islands. The rest of the trip exists for me as a sequence of reconstructed positions, some firmer than others. In Schleicher's work, the asterisk held open that same distinction between what was witnessed and what was inferred.

Within linguistics, though, the asterisk now usually does another job. When I tell folks this, they imagine the linguist as a teacher marking up student papers with asterisks. But that's not it at all. Often the ungrammatical sentences linguists are "starring" are those they've made up themselves. Rarely is it to call attention to the kinds of mistakes that people make. In fact, it's often to illustrate the kinds of mistakes people *don't* make.

If this sounds bizarre, it is. The starred sentences are regularly so wonderfully off-kilter that I sometimes marvel at the ingenuity needed to come up with them. To give you a sense of what I mean, (1) presents six typical oddballs from *The Cambridge Grammar of the English Language*.

- (1) a. * *He offered her the steak fiendishly hungry.*
 b. * *An apparently bird hit the car.*
 c. * *The news his guilt was devastating.*
 d. * *It had been for you.*
 e. * *She lives here any longer.*
 f. * *She wrote more plays than he wrote five novels.*

These aren't ordinary schoolroom errors. They're probes, constructed to test where English allows a phrase to go, what kind of word can appear where, or which pieces of meaning can be combined.

This book follows that feeling you get when you read such examples: the little jolt that comes from reading a sentence English seems not to allow. But the asterisk does more than report that feeling. It turns it into a claim. So the small historical question opens into a larger one: how did a mark for reconstructed words become a mark for impossible sentences, and what kind of evidence can such a mark be taken to represent?

The answer begins with Henry Sweet (1845–1912). While Schleicher is the father of the asterisk in linguistics, he never used it to mean 'ungrammatical'. That use appears to be Sweet's innovation.

Sweet was a scholar of phonetics, Old English, grammar, and language teaching. Keenly observant of the mismatch between traditional grammar rules and the realities of everyday speech, he developed a descriptive approach to grammar. His *A new English grammar: Logical and historical* moved away from didactic notions of right and wrong usage. In the preface to that book, he wrote,

As my exposition claims to be scientific, I confine myself to the statement and explanation of facts, without attempting to settle the relative correctness of divergent usages. If an ‘ungrammatical’ expression such as *it is me* is in general use among educated people, I accept it as such, simply adding that it is avoided in the literary language. (p. xi)

With this focus on the language in use, Sweet repurposed Schleicher’s symbol. Where Schleicher had marked words not directly observed, Sweet used the asterisk to signal forms that could be constructed from English words but not accepted as ordinary English sentences. An early clear use appears in “thus from *these tall men* we can infer *these men are tall*, but we cannot make *some Englishmen* into **Englishmen are some*, or *half the island* into **the island was half*” (Sweet, 1892, p. 14).

That’s no small shift. A reconstructed word is missing because history failed to preserve it. A starred sentence is missing for another reason: an absence in the practice of speakers. It belongs to the grammar as a boundary case, a shape that helps us see where the system won’t go.

Today, the asterisk is as likely to mark an ungrammatical sentence as a reconstructed word. I used the symbol for years before I knew this history, proof of how deeply it’s woven into the field.

I didn’t arrive at the problem through Schleicher or Sweet. I arrived at it through teaching, and more particularly through the strange institutional confidence with which some rules are treated as grammatical boundaries. It feels like my own interest in grammar should have a single beginning, but the details are again frustratingly elusive. I have memories of hours, whole afternoons, spent on the sixth floor of the Kinokuniya book store in Shinjuku. This was the English-language floor, where you’d find the English novels along with translations of Japanese novels.

There was a section for language-teaching materials, for learning Japanese and for teaching English, and I made extensive use of both. I remember being surprised at the number of people standing and reading manga, magazines, and books in book shops and convenience stores. Somebody told me it was so normal they had a noun for it: *tachiyomi*, reading while standing in the shop.

It wasn’t clear to me what the social boundaries of *tachiyomi* were, but it seemed to license my going into Kinokuniya, picking up any book in the Cambridge Applied Linguistics series, and reading until my legs and back hurt from standing so long. I didn’t yet think of that as a grammatical question, but it was

already a question about rules, permission, tacit knowledge, and the difference between what people say is allowed and what people actually allow.

There was no way I was going to be able to buy any of those tomes, which typically sold for something like ¥8,000, a painful fraction of a month's rent. Standing time was the only rent I could afford, and a few chapters on language were what I got for it.

But none of this had been planned. I hadn't gone to Japan because I had a plan to teach English. I'd run out of money around the time I reached Singapore and was trying to get myself home. While I planned, an Australian backpacker sharing my room pointed out that Japan needed English teachers. I needed money, and I spoke English. We were compatible.

I arrived in Tokyo in late fall, 1993. I had no relevant training and little experience except for a few private English lessons I'd taught when I was in Chiang Mai, perhaps a year earlier. I had also arrived in Japan knowing literally nothing about its language, and I soon decided I was going to learn it faster and better than the other foreigners around me, a task at which I was only partially successful.

While I focused on memorizing vocabulary and sentence patterns, the sheer difference between Japanese grammar and the grammar I knew constantly nagged at me. Phrases were built differently, verb conjugations seemed baffling, and particles added layers of meaning I struggled to grasp.

Japanese didn't just give me new words for familiar sentences. It made different structural possibilities visible, and made some familiar English ways of packaging meaning feel less inevitable than they had before. In those days in Japan, I treated that as a problem of learning and teaching. It would take something else to turn my attention more to grammar itself.

I moved back to Canada with my wife and toddler in 2003, just before the first global SARS outbreak. I was looking for language teaching work in Toronto, but students were fleeing and universities were laying off English-language teachers. After four months of searching, I was finally interviewed for a full-time position at Humber College, a position I still hold today.

I remember, not long after arriving, perhaps a few months later, perhaps a few years, a rising sense of confusion and embarrassment over FANBOYS, a mnemonic for the "co-ordinating conjunctions" in English: *for*, *and*, *nor*, *but*, *or*, *yet*, and *so*. I was confused because I didn't understand why this idea seemed so oddly central in the Humber English-teaching ecosystem. I was embarrassed because I'd never encountered it before.

I recall turning to *The Cambridge Grammar of the English Language* to try to sort it out. I'd seen a copy years earlier at the Temple University library in Japan, but hadn't been able to make much of it. Humber also had a copy, so I

thought I'd give it another try. This time, I was able to make more sense of it and realized that I needn't have been embarrassed about FANBOYS.

The experience led me to start my blog *English, Jack*, where the first of roughly 800 posts, on Friday, July 28, 2006, was entitled "The myth of FANBOYS". It also led me into a deep dive of *The Cambridge Grammar*. It was around this time that I also encountered Geoff Pullum's work on the philosophy of grammar, which pushed me toward the foundational questions of what grammar is, how we know it, and what kind of evidence a judgment provides.

FANBOYS mattered because it felt like an asterisk without the star. Here was a rule treated as grammar, taught with institutional confidence, and used to sort good sentences from bad ones. But the more I looked at it, the less it seemed to name a real boundary in English. The question was no longer whether a particular sentence sounded right. The question was what kind of thing a grammatical boundary could be.

Over the next decade, I submerged myself in the study of English grammar, figuring out how it works and writing about what I learned. But a larger question started to press on me: what kind of thing is (un)grammaticality? Is an ungrammatical sentence like a social faux pas, a violation of an unwritten etiquette? Or is it more basic, a strain on the machinery by which we make language intelligible?

That question is what the asterisk hides. In 1973, Fred Householder (Householder, 1973) pointed out that the asterisk makes ungrammaticality look simpler than it is. Different speakers, or even the same speaker in different situations, might have different judgments on some sentences. The asterisk presents grammaticality as a simple yes or no, when the thing it marks is more complex.

One small star can cover several claims. It can mean that no competent speaker would produce the sentence. It can mean that some speakers reject it and others accept it. It can mean that the example needs a special context, or that the sentence is hard to process, or that it violates a classroom rule, or that it sounds wrong in this register but ordinary in another. The mark is slight. The work it does isn't.

Every asterisk in this book asks a version of that question. It may mark a sentence nobody says, a sentence some people say, a sentence someone has been taught to dislike, or a sentence that strains the machinery of understanding. The star looks like a single judgment. The book begins with the suspicion that it marks a family of them.

Part I

What the asterisk hides

A long long road

Consider an adjective like *long*. Repeat it, and its meaning intensifies. That's what The Hollies are doing when they sing:

*It's a long, long road
From which there is no return*

The reduplicative use of *long* emphasises the sheer immensity of the journey in a way that the more prosaic *it's a long road* wouldn't.

Another verse, though, goes like this:

*The road is long
With many a winding turn*

Now imagine reduplicating *long* here. Far from adding emotional depth, it would invite ridicule. As ordinary English, *the road is long long* is out. And what's fascinating is that we all intuitively grasp the difference between these two superficially similar uses of repetition. Why? Is there a universal rule at play, or is it something peculiar about the structure of English?

I teach English to international students studying in Canada, some of them beginners, and I've given examples like *a long long road* and *the road is long long*. There's never any reaction after the first, but laughter inevitably follows the second. The students respond as if they've already learned the boundary.

Before going further, let's pin down what's happening. The phrase *a long road* is a noun phrase, where *road* is the head noun and *long* is an adjective in attributive modifier function, a modifier sitting in front of the noun and ascribing a property to it. *The road is long* is a clause. *Long* is still an adjective, but here it functions as a predicative complement, the kind that comes after a verb

and says something about the subject. Other predicative complements are *you're beautiful*, *they all seem so happy*, and *I feel good*.

The doublet *long long* exemplifies a construction called REDUPLICATIVE INTENSIFICATION, “reduplicative” because the form is repeated and “intensification” because it pushes the property named by the adjective higher on its scale. The construction is also picky about which adjectives it accepts. *Long* is gradable: there are degrees of long. *Cambrian*, the geological period, isn't gradable, and **a Cambrian Cambrian fossil* is just as bad as **the road is long long*.

So a more careful statement of what we've seen is this:

- (1) In the intensificatory construction, a gradable adjective in attributive modifier function may be repeated to strengthen its meaning. The same pattern isn't generally available when the adjective functions as predicative complement.

I.1 SAME ASYMMETRY, OTHER LANGUAGES?

It's tempting to wonder if the pattern repeats in other languages. Japanese has the same asymmetry:

- (2) a. 長い 長い 道
nagai nagai michi
'a long long road'
b. *道 が 長い 長い
michi ga nagai nagai
'the road is long long'

So does Russian:

- (3) a. красивые, красивые глаза
krasivye, krasivye glaza
'beautiful, beautiful eyes'
b. *Её глаза красивые, красивые
Yeyo glaza krasivye, krasivye
'her eyes are beautiful beautiful'

Three examples don't establish a rule, but they suggest one. What would such a rule imply?

One answer is universalist: something about the human mind might make intensificatory adjective repetition impossible in predicative function. That answer predicts that no language should allow the construction there.

A second answer is semantic: maybe *long long* means something that works only when the adjective modifies a noun. That answer predicts that the restriction follows from the meaning of the repeated adjective itself.

A third answer is historical: English may simply have developed a construction for repeated adjectives in one position and not another. That answer predicts that other languages may do things differently.

Each answer makes a different prediction.

The universalist hypothesis falls quickly. Jamaican Creole has examples like (4).

- (4) *im want wan we blakblak*
 he wants one that black-black
 ‘he wants one that is very black’ (Gooden, 2003, p. 129).

North-East Ambae, an Oceanic language of Vanuatu, has the same pattern (5).

- (5) *Vai-gi ngire ra=u qi-qilodo*
 do-become things those short-short
 ‘Those things are very short.’ (Hyslop, 2004, p. 278)

So intensificatory reduplication in predicative position isn’t ruled out by anything in the human mind. Some languages permit it. English doesn’t.

If it’s not a universal of language, where does English’s restriction come from?

1.2 OLD ENGLISH

Intensificatory reduplication doesn’t seem to have been available in Old English. Two adjectives couldn’t really sit side by side modifying a noun directly. **Micel blac fugol* ‘big black bird’ would have been unusual; an Old English speaker might have rephrased it as *micel fugol & blac* ‘big bird and black’. But Old English did have a side door: compound nouns. You couldn’t put two modifiers in a row, but you could sneak one inside the noun.

Blackbird is a Modern English compound, a noun made up of two words: the modifier *black* and the head noun *bird*. *Blackbird* behaves like a single word: its meaning isn’t simply *black* plus *bird*, and its internal adjective is hidden from the surrounding grammar. *Really* can modify *black* in *a really black bird*, but it can’t reach inside to the *black* of **a really blackbird*. Female blackbirds are often brown, and a raven is a black bird, but most certainly not a blackbird.

Although we don’t write *citygate* today, *burggeat* was a compound noun in Old English, made up of the modifier *burg* ‘town/city’ and the head noun *geat*



Figure 1.1. A black bird and a blackbird. Image by DALL·E.

‘gate’. And just as *really* can’t “see” *black* in *blackbird*, an attributive modifier like *micel* ‘great’ would have been insensitive to *burg* in *burggeat*. With *burg* tucked safely inside the compound noun, *micel burggeat* wouldn’t have felt like two modifiers in a row.

At least, it wouldn’t to most speakers. Still, the odd burgher may have interpreted *burggeat* as two words instead of one, and so for them *micel burg geat* would have had two contiguous modifiers. It’s only a short step from *micel burg geat* ‘great city gate’ to something like *micel blæc geat* ‘great black gate’ or *fæst eald geat* ‘strong old gate’. Not only would that have put two modifiers in a row, but both of the modifiers would have been adjectives.

The path to two adjectives is clear when the first half of a compound noun can be mis-parsed as a stand-alone word. But when there’s no noun with a modifier to pry loose as a bridge, the path stays closed. That’s why English eventually got *old old tales*, and never *the tale is old old*.

1.3 MIDDLE ENGLISH

One of the bigger Middle English changes was the erosion of the case system. The Old English forms of *se bring* ‘the ring’ once carried distinct nominative, accusative, genitive, and dative endings. Over time those collapsed into the bare *the ring* and the genitive *the ring’s*. As the inflectional endings weakened, word order took on more grammatical work. Without those endings, (6) looks like the woman gave the queen a ring; with them, we can see that the queen was the subject and it was *to* the woman that she gave the ring.

- (6) *Dām wife sealde sēo cwēn bring*
 the woman-DAT gave the queen-NOM ring-ACC
 ‘It was to the woman that the queen gave a ring.’

For our purposes, what matters here is the growing stability of positions within the noun phrase that came with the loss of case. With the pre-nominal zone increasingly fixed as the place where modifier material lived, there was room for more of it to accumulate.

That laid the groundwork for a new kind of intensification: near-synonymous adjective stacking, as in (7), with minor spelling adjustments (González-Díaz, 2021).

- (7) *gosefoote an evill herbe: it is called pes anserinus, leaved like night shade, but more jagged in the said leaves, bearing [small little red flowers, with great plenty of seede, lyke unto arage].*
 (Bulleins bulwarke of defence against all sicknesse, 1579)

This isn’t yet *long long*. *Small little* is two near-synonymous adjectives stacked, with no single adjective repeated. But it prepares the same syntactic space: a place before the noun where more than one adjective-like unit can sit.

English also had other ways for repetition to intensify meaning. Repeated intensifiers such as *very very hot* belong to a different construction, but they show the same broad principle: reduplication can intensify meaning in more than one English construction (Méndez-Naya, 2017).

Repeated adjectives in the *old old* pattern begin to show up in Early Modern English, often in formulaic or affectionate settings. Falstaff says *old old master Shallow* in Shakespeare’s *Henry IV, Part 2* (1600). The degree use that sounds familiar in *an old old tale* was slower to settle; González-Díaz dates its establishment to Late Modern English (González-Díaz, 2018).

In short, the ungrammaticality of **the tale is old old* is more or less the historical default. The grammaticality of *an old old tale* is the result of a long, long chain of historical accidents on the left-hand side of the noun phrase.

1.4 WHY THE STUDENTS LAUGH

So what are my international students responding to when they laugh at **the road is long long*? Not a universal of mind: Jamaican Creole and North-East Ambae allow predicative repetitions of the same general kind. Not a meaning problem: *long long* would mean what *very long* means. They’ve extrapolated, from the evidence they’ve seen in English, that English doesn’t normally put stacked adjectives after *is* unless they’re joined by *and*.

After *is, she is kind, funny, and generous* and *he is tall, dark, and handsome* are fine because the adjectives are linked into a list. But stacked adjectives without *and* go in front of the noun, not after the verb. **The road is long steep* would be just as bad as **the road is long long*; the problem isn't reduplication specifically, it's the predicative position refusing more than one adjective unless the words form that kind of list.

1.5 THE DOOR REOPENS

In the 1970s, a new kind of adjective reduplication appeared in English, and this one can be predicative (González-Díaz, 2018). It looks more like compounding than like *long long*. You use it to pick out the meaning that's not just "right" but *right-right*, or to mark a coffee that's not just coffee but *coffee-coffee*, as opposed, say, to instant.

This isn't the same construction as intensificatory *long long*. It's contrastive, meaning something like 'the relevant or strict version', not 'more of this'. But its existence matters: it shows that the predicative position isn't permanently closed to reduplication. English can open that door when a different construction supplies a different meaning.

So the asterisk on **the road is long long* doesn't mark an impossible thought, or even an impossible form of English. It marks the limits of one English construction.

English learned to tolerate strings of adjectives on the left side of the noun phrase (*a long long road, an old old tale, small little red flowers*) through a chain of unrelated developments that opened up the pre-nominal zone. It didn't generalise the pattern to predicative complements. Other languages did. And newer contrastive reduplications show that even this boundary isn't fixed once and for all.

The asterisk, in other words, is doing less than it looks like it's doing. It isn't marking the edge of possible thought. It's marking where one English pattern happens, for now, to stop.

Stories from syntactic islands

The first islands in this book were literal. Those in this chapter are syntactic: locations isolated inside clauses that a word can occupy but can't easily leave.

2.1 THE SENTENCE THAT WON'T COME HOME

Consider what happens when you try to ask a perfectly ordinary question about a perfectly ordinary sentence. Start with *She doubted the idea that they had seen it*. The thing she doubted was a specific belief; *it* was the thing they had seen. Now try to ask about *it* by replacing it with *what* and moving it forward.

- (1) a. *She doubted the idea that they had seen it.*
- b. *She doubted the idea that they had seen what?*
- c. **What did she doubt the idea that they had seen __?*

Upon reaching *the idea* in (1c), we stumble. The thread of the sentence starts to unravel. We may read on, hoping that a solution will present itself. We may go back and check if we'd misread. At some point we conclude that this sentence is just ungrammatical. The syntax is snarled. It doesn't work.

Not only is it ungrammatical, but it's hard to fathom what it could possibly mean.

This isn't an everyday error like **How many people was there?* or **She went school*. Often, we understand such sentences intuitively and can even conjure a quick correction without much thought. But even a good deal of thought may not help us with (1c).

This kind of stuck sentence has a name. To see why it's stuck, and what kind of stuck it is, we should meet the person who named it.

2.2 THE MAN WHO NAMED THE WATER

SYNTACTIC ISLANDS is the name that John Robert Ross, widely known as Haj, gave to this phenomenon. Noam Chomsky had already noticed as early as 1964 that some natural-looking questions just can't be asked: you can ask *What did she see?* but not **What did she doubt the idea that he had seen?* Ross showed that these failures formed a pattern. His 1967 MIT dissertation, *Constraints on variables in syntax* (Ross, 1967), was the rare dissertation that became part of the terrain. It didn't merely add a few exceptions to Chomsky's machinery. It turned scattered failures of extraction into a theory of syntactic borders: domains that dependencies could reach into but not escape from.

Ross had a gift for seeing syntactic patterns as scenes of motion. Words got **PIED-PIPED**, like Browning's piper leading rats, when one word pulled other material along behind it. Clauses got **SLICED**, as if most of the sentence had been washed away and only the question word left behind, e.g., *Someone left. Do you know who left?* A verb leaving a coordinated structure left a **GAP**, e.g., *I chose one, and my sister chose the other.* A "heavy" noun phrase shifted to the end of its sentence under its own weight, e.g., *I gave ~~the book that I bought last week~~ to my mom the book that I bought last week.* The most consequential of his images involved a body of water that didn't exist, around a phrase that a word couldn't leave.

All this movement was to be done by the syntactic machinery that Chomsky and his followers, Ross among them, were conjuring. And so, when they pre-pended an asterisk to a sentence like **What did she doubt the idea that he had seen?*, they weren't merely reporting that it sounded wrong. They were saying the syntax couldn't have produced it. The mark was a structural verdict.

2.3 LEARNING THE LINE

We don't usually think of words moving around, but compare *What did they see?* with *They saw what?* *They saw what* might sound wrong, but you could ask it if I suddenly blurted out, *they saw it*, without providing any context.

The important thing to notice is that the position of *what* has changed. The first question, the normal one, starts with *what*, but in a sense *what*'s location in the second question, at the end, is its natural home. We say, *I see it*, not *I see*. First comes a verb (*see*), then the thing seen. So, we could think about *what* leaving its natural position after *see* and appearing instead at the beginning of the sentence. This becomes more apparent when the sentences are displayed one below the other, as in (2).

- (2) a. *They saw it.*

- b. *They saw what?*
- c. *What did they see __?*

In (2c), the ⟨__⟩ shows the position that *what* left.

Syntactic islands don't crop up all that often. It's not just a matter of adding in a few extra bits and you've got an island. Consider (3), where *what* has no problem leaving its home spot and appearing at the front.

- (3) a. *She doubted that I had seen it.*
- b. *She doubted that I had seen what?*
- c. *What did she doubt that I had seen __?*

And we can keep adding on without tripping up.

- (4) a. *She doubted that I had heard that you had seen it.*
- b. *She doubted that I had heard that you had seen what?*
- c. *What did she doubt that I had heard that you had seen __?*

This contrast is Ross's problem in miniature. The bad sentence in (1c) isn't just odd. It's bad beside a nearly identical sentence that works fine. Ross's move was to treat that difference as a clue. His dissertation didn't offer a list of impossible questions; it drew a map of failed escape routes and asked what sort of border had been crossed.

2.4 WHAT KIND OF BORDER IS THIS?

That pattern cried out for explanation. A good explanation has to do more than label the bad cases. It has to say why some long-distance questions pass so easily while others collapse. It should cover more than one island type, leave room for differences across languages, predict new cases, and stay simple enough to be worth believing.¹

The temptation is to take the asterisk at face value. The failure in (1c) feels hard and general. It's not about politeness, schooling, register, or situation. It feels structural. And so it looks like the asterisk is doing pure syntactic work: marking a sentence that the grammar simply couldn't have produced.

But a hard case can teach the wrong lesson. Once we've found a border, we still have to ask what kind of fact the border records. Is it architectural, interpretive, conventional, social, processing-based, or some mixture of these?

¹Discourse constraints may also limit long-distance dependencies in English and French (Winckel et al. 2025). Chapter 8 returns to this issue.

Ross himself wasn't sure. The man who gave the formalist program its most architectural-looking border later moved toward a view in which syntax and meaning couldn't be cleanly separated (Harris, 2021). He called his own approach SEMANTAX, on the conviction that syntax and semantics interpenetrate inseparably. The man who named the islands turned out to think the asterisks his discovery licensed didn't mean what most linguists took them to mean.

The chapter's question, then, is whether the asterisk in front of (1c) records one fact or several.

2.5 THE OLD WORD *UNGRAMMATICAL*

The word itself gives a hint. Grammar is a conventional system linking what's said to what's meant. The link runs both ways: form constrains meaning, and meaning constrains form (Futrell et al., 2020, p. 371).

When *ungrammatical* entered English in 1654, it didn't start as a complaint about word order. It started as a complaint about meaning.

It seems a great folly to declaim against us for denying the literal, natural sense, and yet that themselves should expound it in a sense which suffers a violence and a most unnatural, ungrammatical torture; for if they may change the words from the right sense and case to the oblique and indirect, why may not we? (J. Taylor, 1654)

Taylor's phrase doesn't accuse a sentence of being malformed. It accuses an interpretation of forcing a grammatical form to carry a meaning it doesn't normally bear. The torture is interpretive. The earliest known English use of *ungrammatical* already links form to meaning. It isn't a complaint about arrangement alone; it's a complaint that the arrangement has been made to say the wrong thing.

That gives us a way to read our island sentence. **What did she doubt the idea that they had seen?* isn't simply mis-arranged; the arrangement is fine piece by piece. What fails is the dependency between *what* and the place inside the bracketed clause where the question would have had to be answered. The asterisk marks a failed form-meaning pairing as much as it marks a failed structure.

2.6 WHEN WRONGNESS BECOMES HEARABLE

Form-meaning pairings aren't fixed forever. What sounded wrong yesterday may sound right tomorrow.

In 1865, Richard Wagner's "Tristan und Isolde" premiered with a chord that seemed to hang in the air without resolving. The Tristan chord, F, B, D $\sharp$, and G $\sharp$, made no ordinary sense to many listeners. Hector Berlioz, who had received

an early copy of the score from Wagner, found the Prelude confounding, noting, “There is no theme other than a kind of chromatic moan, filled with dissonant chords whose long appoggiaturas in place of the true harmonic note further intensify the painfulness.”

But even though it made no sense in the framework of musical harmony, some listeners were able to find a kind of narrative sense in it. For them, the unfulfilled tension mirrored the opera’s story of unrequited love. This innovative use of dissonance disrupted the conventions of Western music. Wagner’s composition didn’t just utilise the existing musical language; it expanded it, challenging the ear and setting the stage for the even more radical departures from tonality that the twentieth century would bring.

English speakers had done much the same with stacked adjectives. Phrases like *a long long road* hadn’t always been hearable; the bridge through Old English compound nouns let speakers reanalyse what had once been one structure as two. The community learned to hear the doubled adjective in front of a noun. It still hasn’t learned to hear the same doubling after a verb.

Some linguists read the asterisk on **What did she doubt the idea that they had seen?* as math: there’s a rule, the sentence breaks it, that’s the end. Others read it the way audiences came to read the Tristan chord: as a piece of dissonance that a community hadn’t yet learned to hear. What’s wrong now might not always be.

Ross’s own career runs both sides of this debate. The 1967 dissertation made the arithmetic reading look right; his later work, *SEMANTAX*, sits on the Tristan side. The man who gave the formal program its cleanest evidence didn’t end up believing the cleanness was the lesson. Through that later lens, the asterisk on **What did she doubt the idea that they had seen?* isn’t simply a structural verdict. It may also be a verdict on meaning.

2.7 ONE FORM, MORE THAN ONE MEANING

A string of words can have more than one structure, and so it can have more than one meaning. Consider the sentence *The farmer sold his prized bull with a tear in his eye* (Figure 2.1).

The phrase doing the work is *with a tear in his eye*. *With* is a preposition, and *with a tear in his eye* is a PREPOSITION PHRASE (a PP). The ambiguity comes from whether the PP attaches to the verb phrase headed by *sold*, so that it was with a tear in his eye that he sold the bull (the image on the left), or whether it attaches to the noun phrase headed by *bull*, with the bull doing the crying (the image on the right).

Linguists use a graph called a SYNTAX TREE to depict these kinds of struc-



Figure 2.1. The farmer sold his prized bull with a tear in his eye. Image by DALL·E.

tural relationships. The one for selling tearfully is Figure 2.2. Starting at the top, it shows that the whole sentence is a clause, and this clause has two main parts: the subject on the left (the noun phrase (NP) *the farmer*) and the head verb phrase (VP) on the right. The PP functions as a modifier in the VP in this interpretation.

The tree for the tearful bull is Figure 2.3. Again, the clause has two main parts, but this time, the *with a tear in his eye* PP is tucked inside the *bull* NP. The triangles show that some structural details have been omitted for simplicity.

The farmer sentence is grammatical because one form can pair with more than one meaning. Our island sentence fails the other way. It's not that the form is ambiguous; it's that the form isn't paired with any stable meaning at all. The dependency between *what* and the place inside the bracketed clause can't be recovered, so no interpretation comes home. Whether that failure is the kind a community could ever learn to hear is the next question.

2.8 SOME ISLANDS ARE ROCK

Some limits sit closer to the architecture of a language than to etiquette. The Left Branch Condition is a useful warning case. English is bad at pulling a modifier or possessor out of the left edge of a noun phrase: **How big did you buy a house?* is much worse than *How big a house did you buy?*

That doesn't mean every language draws the same boundary. Serbo-Croatian, for instance, permits left-branch extraction English forbids. But repeated exposure won't necessarily turn every structural mismatch into a new convention.

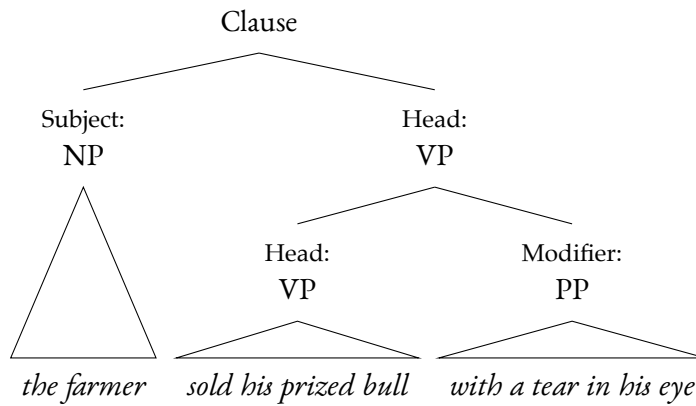


Figure 2.2. The structure showing the PP attached in the VP with the tearful farmer interpretation.

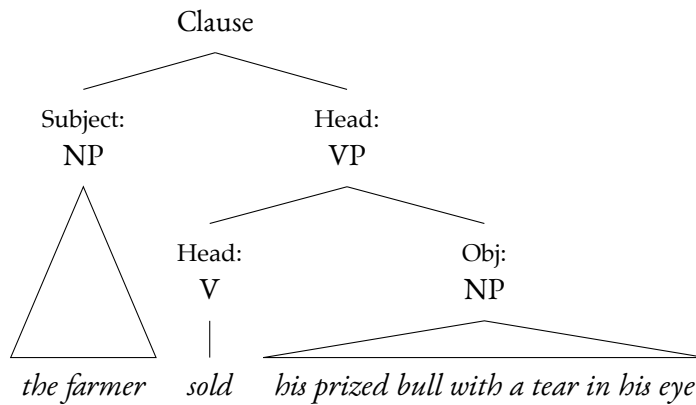


Figure 2.3. The structure showing the PP attached in the NP with the tearful bull interpretation.

Some violations are awkward because we aren't used to them. Others may be awkward because the grammar gives us no stable way to recover the intended dependency.

Whether the island in (1c) is rock or surface dissonance is a question the chapter doesn't have to settle. The book does, eventually. For now, what matters is that the asterisk doesn't tell us which.

2.9 THE ISLAND SEEN AGAIN

Grammaticality can't be found in the laws of physics. It's an emergent property. But if you believe in people and puppies, in the scent of rain on hot asphalt, in the first blooming lilac of spring, in orbits and orders and other such stuff, then there's every reason to believe that grammaticality is real.

That brings us back to Ross. His achievement was to make one kind of grammatical boundary visible. The danger was that the boundary looked so crisp on the page that it encouraged a larger conclusion: grammaticality itself might be syntax in its pure structural form, and the asterisk might apply, or fail to apply, in the abstract, for human language in general. The analysis worked so well that it opened a garden path.

This book starts from the suspicion that the conclusion was too quick. The asterisk in front of **What did she doubt the idea that they had seen?* marks a real failure, but the failure may be of several kinds at once: a failed dependency, a failed form-meaning pairing, a social expectation, a processing collapse, or a convention not yet learned. Grammaticality is real in the same sense Ross's islands are real: a pattern stable enough to guide inquiry, but never separable from the uses, speakers, histories, and situations that make the pattern matter.

The sentence is still bad. But its badness no longer looks simple.

Degrees of wrongness

The last chapter made the asterisk look crisp. A word could move out of one part of a sentence, or it couldn't. The dependency could escape, or it was trapped. That's part of what made syntactic islands so powerful: they made grammar look like a map with coastlines.

But asterisks don't always feel like coastlines. Sometimes they feel like slopes. One sentence is barely off. Another is bad, but easy to repair. A third falls apart so badly that the intended English has to be reconstructed from fragments.

3.1 A LITTLE WRONG, VERY WRONG

When children play, a surprising amount of the time is taken up establishing rules: what's allowed, what's forbidden, what counts as cheating, and what has to happen next. When my kids had friends over, I often overheard sentences like **you're not allowed double-bouncing somebody on the trampoline*.

That sounded wrong to me. For my English, *allow* licenses a *to*-infinitival complement, as in *you're allowed to bounce*. The present participle *bouncing* was off, though the neighbourhood kids seemed perfectly happy with it.

Still, it wasn't as wrong as it could have been. Consider the sequence in (1).

- (1) a. *You're allowed to have dessert.*
- b. *? You're allowed having dessert.*
- c. ** You're allowed that you have dessert.*
- d. ** You're allowed you have dessert.*
- e. ** You're allowed have dessert.*

For me, and perhaps for you too, (1b) is only a little off. Example (1c) gets my attention. Example (1d) shakes the frame more violently, and (1e) falls off the cliff.

The strange thing is that each underlined piece is a possible complement somewhere else in English.

- (2) a. *You want to have dessert.*
 b. *You like having dessert.*
 c. *We told them that you have dessert.*
 d. *We let you have dessert.*
 e. *You can have dessert.*

English has those forms. The verb *allow* simply doesn't welcome them equally. The wrongness comes in degrees.

3.2 THE GRAMMAR MACHINE

That's not how grammaticality looks in Noam Chomsky's early generative picture. In *Syntactic Structures*, Chomsky treated grammar as a system for generating sentences (Chomsky, 1957). If a string could be generated by the grammar, it was grammatical. If it couldn't, it wasn't.

A toy grammar makes the idea concrete. Give the grammar a small lexicon, say nouns such as *dog* and *cat*, determinatives such as *the* and *a*, and verbs such as *chases* and *sees*. Then give it rules for building noun phrases, verb phrases, and sentences.

1. $N \rightarrow \{dog, cat\}$
2. $D \rightarrow \{the, a\}$
3. $NP \rightarrow D N$
4. $V \rightarrow \{chases, sees\}$
5. $VP \rightarrow V NP$
6. $S \rightarrow NP VP$

With those rules, the grammar can build *the dog chases the cat*. Figure 3.1 shows the result.

The point of the toy grammar is the clean boundary. A string generated by the rules is in. A string not generated by the rules is out. There may be many questions about which rules to write, but once the rules are fixed, the answer is binary.

Computer languages give us another version of the same clean boundary. In high school, my friend Jamie Ramsden and I spent much of computer science

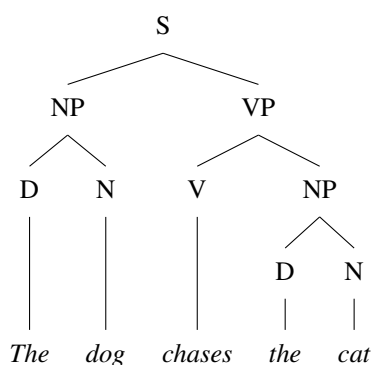


Figure 3.1. The generated sentence *The dog chases the cat*.

class writing simple games and helping classmates with their assignments. One assignment, perhaps a bubble sort, was due before we had left enough time for it. We wrote one program, printed two copies, drew the required flowcharts after the fact, and handed them in.

Our teacher, Mr. Boyko, noticed. The code was identical. That was suspicious because even a simple program can be written in many functionally equivalent ways. He asked whether he should give us each 50%. We agreed this was fair, then cheekily suggested that the same accounting should apply to the help we had given other students. I think we both got full marks.

Computer code resembles human language in one respect: the same broad task can be done in many ways. But a Basic interpreter isn't a human interlocutor. If a program runs, the code is syntactically acceptable to the interpreter. It may sort correctly, sort only some lists correctly, sort nothing correctly, or produce useless output. Those are failures of purpose, not failures of syntax.

Human languages don't work that cleanly. A sentence can be formed from English words in an English order and still feel more or less wrong for the meaning, setting, speaker, or construction at hand. The asterisk marks that failure, but it often hides how severe the failure is.

3.3 MORE AND LESS BROKEN

The easiest way to see the scale is to pile up damage. The extra stars in (3) are local theatre, not a standard notation. They stand in for a feeling: some sentences are worse than others.

- (3) a. *The big black dog fetched sticks at the park.*

- b. [?] *The black big dog fetched sticks at the park.*
- c. [?] *The black big dog fetched stick at the park.*
- d. * *The black big dog not fetch stick at the park.*
- e. ** *Black big dog no fetch stick at park.*
- f. *** *Black big dog no fetch park.*
- g. **** *Dog big no stick no no park fetch.*

For me, the line between ordinary English and broken English falls somewhere around (3c) and (3d). But that line isn't the only thing I feel. Example (3d) is much closer to English than (3g).

That may seem expected. The later examples have more things wrong with them. In (3c), the order of *big* and *black* is unusual and *sticks* has become *stick*. By (3e), determiners are gone, *not* has lost its auxiliary, and the sentence no longer marks past tense.

So we can hold the number of errors steady. Each example in (4) has just one main problem.

- (4) a. [?] *The black big dog fetched sticks at the park.*
- b. * *The big black dog didn't fetched sticks at the park.*
- c. * *Big black dog fetched sticks at the park.*
- d. * *The big black dog no fetched sticks at the park.*
- e. * *The dog big black fetched sticks at the park.*
- f. * *The big black dog fetched sticks at park the.*
- g. * *The big black dog fetched sticks the park.*
- h. * *The big black dog put sticks at the park.*
- i. * *The big black dog fetched sticks at.*
- j. * *The big black dog sticks at the park.*

I've sorted these from better to worse, as best I can. You may disagree with the order. You may even reject some of my stars. But I expect that the set doesn't feel flat. The sentences don't all fail in the same way, with the same force.

There's no simple hierarchy of error types here. Example (4c) and example (4j) both involve omissions. Example (4a) and example (4f) both involve order. The difference seems to lie in how much each disruption interferes with the construction we were expecting.



Figure 3.2. A model of an Archaeopteryx, an early bird with teeth. Photo by Dr. Nachtigaller, CC BY-SA 4.0, [via Wikimedia Commons](#).

3.4 BETTER AND WORSE MEMBERS

Gradience doesn't make categories unreal. It makes them more interesting. We already know this outside grammar. A robin is a very good bird. A penguin is a bird too, but less central to the category most children first learn. An ostrich is stranger still. None of that makes BIRD useless.

The old, classical view of categories treated membership as a matter of necessary and sufficient conditions. A bird has feathers, lays eggs, has a beak and wings, and lacks teeth. That works well enough until it doesn't. Birds occasionally grow atavistic teeth. These chicks don't survive, but they do exist, and earlier birds such as Archaeopteryx had teeth.

Prototype theory gives us a better way to think about this. Categories have central members and marginal members. A marginal member isn't outside the category simply because it's marginal. It's a worse example, not a non-example.

The same thing happens inside grammar. *Big* is a very good adjective. It names a property, it's gradable (*big, bigger, biggest*), and it combines with intensifiers (*very big*). It can appear before a noun, as in *a big deal*, or after a verb, as in *the world is big*.

Worth is a more awkward adjective. In *it's worth your time*, it takes something like an object, and objects normally go with verbs and prepositions, not adjectives. You can't say **this is a worth initiative*; you have to say *worthy* or *worthwhile*. One thing isn't **worthier* than another. *Worth* is still an adjective,

but it lives near the edge.

That matters for grammaticality. If categories have central and marginal members, then a sentence can be wrong because it violates a hard constraint, because it forces a marginal member into an uncomfortable position, or because it asks one construction to do the work of another. Those failures can feel different.

3.5 THE TWO FALSE EXITS

Two escape routes are tempting. The first is the clean generative one: a sentence is grammatical if the grammar generates it, and ungrammatical if it doesn't. That view captures something real. Speakers do know patterns, and some departures from those patterns are abrupt.

But the clean view misses the slope. It has trouble explaining why (1b) feels much closer to English than (1e), or why the failures in (4) don't all feel equally bad.

The second escape route goes in the opposite direction. Geoffrey Sampson and Anna Babarczy argue that the whole idea of grammaticality is misguided (Sampson & Babarczy, 2014). On this view, the question reduces to use: if people produce an expression, then it belongs to the language.

That view also captures something real. A construction may be rare, regional, specialized, old-fashioned, playful, or new without being ungrammatical. A sentence such as *I were there* may be outside Standard English while belonging perfectly well to another variety of English.

But use alone can't be the answer either. Speakers can produce errors. They can quote errors, joke with errors, teach from errors, and write impossible sentences in grammar books. We still need a way to say why *you're allowed having dessert* is easier to accept than *you're allowed have dessert*, even before we search a corpus.

3.6 THE SLOPE ISN'T THE CLIFF

Gradient grammaticality is easy to overstate. The sentences don't all sit on one smooth continuum. Several gradients are being compressed into the same small mark.

One gradient is felt severity: how hard the sentence hits the ear. Another is repairability: how easily we can see the intended sentence behind the one we got. Another is category centrality: whether the words involved are being used in ordinary or marginal ways. Another is social distribution: whether the pattern belongs to another dialect, register, age group, profession, or community.

Those gradients can point in different directions. A sentence may feel unusual because it's new to us, not because it's broken. Another may be common in casual speech and still avoided in formal prose. A third may be easy to understand and still fail as the English we were expecting.

The asterisk hides all of this. It makes one mark do the work of several judgments. A sentence can be more or less wrong without grammaticality becoming unreal. Gradience is one of the ways grammar becomes visible.

Real patterns

Amelie showed Basil Clara. English packs much more meaning into this terse sentence than is conveyed by the words alone. Sure, *show* tells us that there's a showing situation, and *Amelie* says that somebody named Amelie was involved, but we also understand that Amelie is the agent, the one doing the showing, that Basil is the recipient, and that Clara is the patient, the one being shown.

As English speakers, it's difficult for us to imagine that these could have been otherwise, but indeed they could. The same sentence in Azerbaijani, for example, would be *Amelie Basila Claranı göstərdi* (where *göstərdi* means 'showed'). The Cariban language Hixkaryana, spoken by roughly 500 people on the Nhamundá River in Brazil, would start the sentence with *Clara* (as the patient), followed by the verb 'showed', and only then would *Amelie* appear (as the agent), with *Basil* (as the recipient) in the final position.

But role shuffling isn't the only possible language trick. A role could be assigned without anyone being named. In the imperative *Show Clara to Basil*, the agent role is now yours: you're to do the showing. If you could have a role in the imperative without even being mentioned, why not in *Amelie showed Basil Clara* too?

English grammar is why not. It removes all role-related ambiguity, distilling the narrative of the showing situation, with all its roles and relationships, into a pattern: Agent + tensed verb + Recipient + Patient plus a mere quartet of words: *Amelie, Clara, Basil, and showed*.

It's the whole pattern that matters; the ordinal positions alone won't do. For instance, *Basil* being second doesn't automatically make him the recipient. In *Amelie showed Basil to Clara*, Basil retains his position but swaps roles with Clara. Even the first position, held by *Amelie*, isn't a reliable indicator of agenthood. In *Amelie was shown Basil*, Amelie becomes the recipient, and the agent

is left unspecified.

The pattern is grammar. And this grammar has a remarkable data-compression feature: it encodes all the role-related information, so that any English speaker knows without the need for words like *agent* and *patient* or *presenter* and *audience* who is showing who to whom. Somebody who doesn't speak English, even knowing what the verb *showed* means, wouldn't be able to extract the information reliably. A speaker of Azerbaijani may assume that Basil is the patient and Clara the recipient, while a Hixkaryana speaker might assume that Amelie is the patient.

In this way, the data compression feature of grammar works only for a particular social group: in this case, English speakers. And because of this, the pattern has another interesting feature: it allows us to predict how English speakers will behave. We can predict with near certainty that an attentive speaker of English will interpret the sentence with Amelie showing Clara to Basil. We can also predict with good certainty that they would express it this way.¹

4.1 SPRAYING AND LOADING

Linguists have noticed and puzzled over subtler patterns that rarely come up in popular discussions of grammar, perhaps because there's no obvious disagreement about how they're used.

There's a set of verbs, for instance, called the *spray/load* verbs. Like the verb *show*, which can have *A showed B C* or *A showed C to B*, *spray/load* verbs also have two patterns that mean more or less the same thing: *She sprayed the area with a fine mist* or *She sprayed a fine mist over/throughout/into the area*. Similarly, we have *They loaded the truck with hay* in alternation with *They loaded hay onto the truck*. Other verbs of this type are *dabbed*, *heaped*, *packed*, *smear*^{ed}, and *spread*.

Strangely enough, though, when we look at other semantically similar verbs – those that describe distributions or accumulations, such as *drench*, *soak*, or *flood* – we find that they don't offer the same kind of alternation. For example, while we can say *She soaked the area with water*, saying **She soaked water on the area* would be considered ungrammatical or awkward by most English speakers.

Linguists still don't have a simple account of why the split falls where it does. The hard part isn't just describing the two patterns. It's explaining why *smear*^{ed} and *spread* travel one path while *drenched* and *soaked* don't.

And yet, we can reliably predict that English speakers won't produce sentences like **He shoveled the garden with earth* while finding those like *She*

¹Based on a quick and dirty search of COCA, I'd say there's about a one in five chance that they would instead say *Amelie showed Clara to Basil*.

*smear*ed the bread with jam perfectly fine.

4.2 RAW ARBITRARY CONVENTIONALITY

Hurford (2012, p. 231) challenges the idea that grammaticality is essentially semantic with the examples in (1).

- (1) a. *It is likely/probable that John will leave.*
- b. *John is likely/*probable to leave.*

The first time I read this, a few years ago, it struck me as a strong challenge. My judgments match Hurford's. But recently, it occurred to me to check, and that's how I learned about *he's probable to play*.

The story starts with a bribery and gambling scandal in the 1946 National Football League championship game between the New York Giants and the Chicago Bears. After suspending the two Giants players indefinitely, commissioner Bert Bell realized that trust was ebbing and that something had to be done to protect the NFL from the negative influence of gambling.

There was no practical way to do away with gambling on NFL games, but one pernicious source of speculation revolved around inside information about player injuries. Bell realized that he could largely neutralize this aspect by mandating that all teams submit weekly injury reports.

In 1947, Bell introduced a requirement that the status of each injured player was to be listed as "probable", "questionable", or "doubtful" for the upcoming game. This information would be made available to the public, ensuring that everyone had access to the same information about player injuries.

Other major sports leagues followed suit, although the exact year of implementation varies. For example, the National Basketball Association has required teams to submit injury reports since the 1980s.

Perhaps coincidentally, I find no examples of *probable to play* before the 1980s. But on Nov. 18, 1986, *The Los Angeles Times* published a column by Mark Heisler that included the following example *This on a day that the Raiders had him listed as probable to play against the Browns*. Since then, in the sports media, the construction *probable to play/start/return/etc.* has gained currency. It's still not common, mostly limited to American sports media, but it can be found from *The New York Post* to the *LA Times*, *CBS Sports* to *ESPN*, and *The Toronto Sun* to the *Houston Chronicle*.

In this way, *he's probable to play* has come to mean something: that though an injury has been sustained, it's not currently severe enough to keep the player

from the game. Consider that, although almost every starting professional athlete is *likely* to play in any given game, only those who had been injured could be described as *probable to play*. In major-league baseball, pitchers typically have a number of non-playing games between each of their pitching appearances, but when it's a given pitcher's turn to play, he may be *likely* to play but he's probably not *probable*.

Interestingly, in the National Hockey League at least, it has become common to refer to injuries simply by naming the body part: *Matthews is out with an elbow*, for instance, doesn't mean that Matthews is having a night on the town with an arm joint but rather that he can't play because he may have a cracked radial head.²

As Hurford observes, one reason that *likely* and *probable* appear to have different structural possibilities may be that *likely* comes from Old Norse, *probable* from Old French. The *to* of the *to*-infinitival construction grammaticalized from the preposition *to*, and this process was already well on its way in Old Norse. At the same time, Old French had similar expressions with *à* 'to' but it was just one of a number of true prepositions doing this kind of work.

While it's not clear, *likely* may have already licensed the equivalent of a *to*-infinitive in Old Norse, while *probable* probably didn't in Old French. Thereafter, each word simply continued on its own linguistic rails until *probable* picked up this new sports-based meaning.

What this illustrates is that there's nothing inherent in the structure or meaning of the word *probable* that stops it from having a *to*-infinitival complement in the way that adjectives like *happy*, *ready*, and *likely* do. There's just a historical path that hadn't led that way. The injured-players list, though, provided just the right motivation because *probable* is the terminology that was adopted to signify a particular status. Most of the time sports announcers leave it at that: *the player is probable*. But if there's some lack of clarity about what's being discussed, it can be cleared up by appending *to play*. And we have a new construction.

This shows that the forces that cause grammaticalization aren't the same as those that give us the feeling of grammaticality.

Hurford notes that a word's "grammatical distribution does not follow completely from their meanings" (Hurford, 2012, p. 302). It seems to me that, in doing so, he may be confounding grammaticalization with grammaticality, what could become grammatical with what's already grammatical.

He cites the example of *the fifth of November/each month/the month in which your birthday falls* and contrasts it with *November the fifth*, which works,

²I thank Michael Dipetta for this detail.

and **each month the fifth* and **the month in which your birthday falls the fifth*, which don't.

In the Early English Books Online corpus, *November the* [ordinal number] occurs not once from the 1470s to the 1620s. Then, in the following decade, there are three examples. In the 1640s, there are 11, then 32 in the 1650s, and 67 in the 1660s. *January* follows roughly the same pattern. *February* has an interesting case in 1610, when we find *hee died with his brother the same moneth of february the twentieth day*, which seems to be an intermediate form, and *March* has a similar example in 1592 *dated from Groyne in march the thirtith day*.³

Perhaps the earliest example of the [month] *the* [date] format is from 1616:

therefore the best ground to sow hempe on, is the richest of all mixt earths, whose mould is driest, loosest, blackest, and quickliest ripe, with little earing, as namely with two ardors at the most, which vould be in october the first, and the last in march, which is the best and most conuenient time for sowing:

Notice the use of *in* with *october the first*, putting us in the month of October, where today we would likely say *on October the first*, putting us on a date.

At any rate, there appear to be no genuine examples before the seventeenth century, and the format seems to be growing through the middle of the century, and established before the following century rolls around.

Interestingly, Chaucer had *the eightetethe '18th' day of April* near the end of the fourteenth century, but *the* [date] of [month] format only really becomes established in the sixteenth century. Prior to that, it was also common to refer to particular landmark days such as feast days or the ides 'middle' of the month and then to calculate forward or back. Date formats seemed to be in flux.

Hurford claims that the *November the fifth* construction imposes a semantically unmotivated restriction to just the proper names for months. It's unlikely that a new semantic distinction was being made with the *November the fifth* format. It was more likely some kind of social or register distinction. The rise of bureaucracy and record-keeping could have contributed. Written documents benefited from brevity and standardization. The format could have initially been motivated by these bureaucratic reasons, subsequently becoming seen as more formal or "educated", particularly compared to the less standardized expressions of date, such as *Five days before the Kalends 'the first' of May*.

³There appear to be scattered instances from the 1550s, but these are spurious hits, such as *february the seconde yeare of her highnes most noble raygne*.

The counterexamples **each month the fifth* and **the month in which your birthday falls the fifth*, are notably the kind of thing that doesn't seem to suit bureaucratic standardization.

Part of Hurford's concern is that, for computational linguists, a large number of such rules that apply to only small groups of words such as the twelve month names is computationally inefficient. But the recent success with having Large Language Models, such as Chat GPT, at being grammatical, not only at the general grammatical facts, but also at particular micro-constructions such as *November the fifth*, suggests that perhaps we simply need to bite that computational bullet.

Chris Anderson's article titled "The long tail" appeared in *Wired* magazine in October 2004. Anderson developed the concept into a book, published in 2006. The term LONG TAIL refers to the strategy of businesses that sell a large number of unique items in relatively small quantities, in contrast to selling a small number of items in large quantities. This concept is particularly relevant in the context of digital goods and online markets, where the cost of inventory storage and distribution is significantly lower than in traditional retail settings.

The popularity of the "long tail" concept has led to its application in various fields, including statistics, business models, and information theory, reflecting its broad impact on understanding and leveraging the distribution of products, services, and information in the digital age. Hurford's concerns about the micro-grammar of words like *November* are similar to the problems that merchants faced before the internet. Computational linguistics didn't have the massive corpora or the sheer amount of compute needed to learn the micro-grammars, but

There seems to be an ability in these models to do few-shot learning, as long as some element of the situation is novel (Liu et al., 2019).

Part II

How grammar feels

How grammar feels

When you encounter an ungrammatical utterance, what does it feel like? Is it akin to stepping off the sidewalk unexpectedly – a sudden jolt to your cognitive equilibrium? Or sometimes more than just a simple misstep – a momentary disorientation, where the flow of language and thought breaks?

Is it sand in your shoe, a branch that scratches you as you walk by, an ice-cream headache, the sting of a mosquito, the glare of headlights? Does it feel heavy or light, salty or sour? Does it have a colour?

Emotionally, is this encounter more likely to evoke irritation and discomfort, fear and anger, or dismissal and superiority?

Or can it be all of those things at different times?

The idea that we have only five senses – sight, smell, hearing, taste, and touch – is nonsense. The traditional list of five ignores pain, pressure, itch, temperature, and chemical irritants like acids, the capsicum in hot peppers, and allyl isothiocyanate in radishes, wasabi and mustard. And that's just the exteroceptive senses – those that detect stimuli from outside the body.

We also have a rich array of interoceptive senses monitoring the internal states of our bodies, hunger for example, or sleepiness. When someone is dizzy, we literally say they've lost their sense of balance. Our sense of self relies on proprioception, the feeling we have of our bodies, their coherence, their locations, and their movements in space. We have senses of thirst, nausea, and fullness. We even have a sense of the levels of carbon dioxide and oxygen in our blood. We're sensitive creatures.

Sensations arise from the interaction between internal or external stimuli, our sense organs, and our nervous system. When our elbow is bent at 90°, intrafusal fibers in our muscles register the relative amount of stretch in our biceps and triceps and Ruffini endings in the joint capsule and ligaments detect changes in

joint angle. Nerves pick up the signals from these mechanoreceptors, and the sensations are sent as proprioceptive signals to our brain, informing it of the arm's position and movement. But positional sensations like this are typically evaluatively neutral. 90° should be neither more nor less pleasant than 120° .

When we do evaluate sensations as good or bad, we've moved on to talking about FEELINGS, the positive or negative experiences that come from sensing the world and ourselves. We sense hunger, pain, or high levels of CO_2 in the blood and we feel that they are unpleasant, problems to be righted. The sensation of warm touch on the arm, the sweetness of a peach, and the temporary relief that comes from scratching a mosquito bite, in contrast, foster positive feelings of relief, warmth, and connection, directing us to seek out such experiences.

No sooner had the warm liquid mixed with the crumbs touched my palate than a shudder ran through me and I stopped, intent upon the extraordinary thing that was happening to me. An exquisite pleasure had invaded my senses, something isolated, detached, with no suggestion of its origin. And at once the vicissitudes of life had become indifferent to me, its disasters innocuous, its brevity illusory this new sensation having had on me the effect which love has of filling me with a precious essence; or rather this essence was not in me it was me. ... Whence did it come? What did it mean? How could I seize and apprehend it? ... And suddenly the memory revealed itself.

– Marcel Proust, *In search of lost time*

Proust's vivid description highlights how a simple sensory encounter – the taste of a madeleine dipped in tea – can be intensely pleasurable. But the experience transcends his sensory delight. He perceives a feeling whose source he can't identify at first, one triggered by but distinct from his experience of the smells, flavours, and textures of the moistened cake. Proust recognizes an upwelling associated with this particular set of sensations even before the memory arrives. It announces itself the way a yawn does, with a gradual build-up of tension before the mouth opens wide and the breath begins to move. A pricking of the thumbs. Summer's ripening breath.

This kind of feeling is called a metacognitive feeling. Unlike the taste of a madeleine or the warmth of a cup of tea, metacognitive feelings are generated by our own thoughts and mental operations. Such impressions include the positive feeling of knowing you might experience when reviewing your responses on a test, the pleasant recognition of a friend's familiar gait, or the empowering

conviction of the rightness of your opinion in a debate. It's not the thoughts that we have but the feeling that we have thoughts, whether we can access them or not. When Faulkner writes, "memory believes before knowing remembers," he's evoking metacognitive feelings.

Metacognitive feelings aren't all positive. Tip-of-the-tongue feelings bedevil us. The thought that you've left something undone can be distressing. And then there's the feeling that this book is all about: ungrammaticality. This too I would place among our metacognitive feelings.

When we identify a sentence as ungrammatical, our response isn't primarily rooted in a physical sensation but in a cognitive evaluation. This process involves evaluating incoming linguistic constructions to ensure that they're familiar, conventional, and congruent, that they evoke the appropriate meaning for the situation, and that, if not, they deviate in understandable ways to good effect. It's difficult to identify necessary and sufficient criteria for feelings of ungrammaticality, but cognitive dissonance is often at their heart.

Our reactions to grammatical incongruences may vary widely, from irritation to curiosity, reflecting the diversity in our linguistic backgrounds and tolerances for variation. Ultimately, the sense of grammar underscores the cognitive foundations of language processing and our adaptive strategies for navigating linguistic complexity.

5.1 WHOSE GORILLA?

The article "Whose gorilla?", which appeared in *Linguistic Inquiry* in 1973, was a "squib" – a short publication that presented an interesting piece of data without proposing an explanation. Not even running a full page, its punch line was (1):

- (1) *My gorilla is over there drinking punch.*
**The guy whose you saw banging at the window is over there watering the rubber tree.*

They had fun in those days.

The authors of "Whose gorilla?" were Jorge Hankamer and Paul Postal. "Postal was a venerable force, ... second in reputation only to Chomsky in the prized domain of syntax" (Harris, 2021, pp. 127–128). He was also an outsider, spending most of his academic life at IBM instead of at a university, and something of a polemicist. His most famous paper was entitled "The best theory", and it supported a position called generative semantics that Chomsky would later deride as "the worst theory, in fact as the worst possible theory ... the worst imaginable theory" (Harris, 2021, p. 129). This despite the irony that "the

Chomsky of the 21st century reads like a reincarnation of the Postal of the 20th” (Newmeyer, 2003, p. 418).

As a graduate student, Postal was so sure of himself that he directed his own thesis. When he submitted it, his advisor wrote him a note saying “he was too busy with his work to read it for two years” (Hankamer, 2020). This mixture of brilliance and confidence made him one of the key combatants in the so-called Linguistics Wars of the late ‘60s and early ‘70s, a truly rancorous battle between the generative semanticists on one side and Chomsky and his supporters on the other.

In 1973, Hankamer had just completed his PhD at Yale and taken up a position at Harvard. His dissertation, on deletion and ellipsis in Turkish, had already established him as a rising star in the field. Like Postal, he was on the outs with Chomsky despite being far less bombastic and pugilistic.

In their article “Whose gorilla?”, Hankamer and Postal drew attention to an apparent gap in the pronoun system, specifically claiming that the word *whose* is defective.

In linguistic terminology, a DEFECTIVE word is one that lacks one or more of the expected forms. This is why Hankamer and Postal have starred that sentence, to draw attention to the *whose* that should – but seemingly couldn’t – be there.

What does it mean to lack a form? *Beware*, for example is missing every form but one; it’s highly defective. It has a bare form *beware of the dog*, but no present-tense **she bewares the dog*, no past tense **She bewore the dog*, and no participles **I’ve never beworn the dog*.

This missing form – the purported hole in the *whose* paradigm – should appear in the bottom row of Table 5.1, right where the question mark is.

Type	Nominative	Accusative	Dep Gen	Ind Gen	Reflexive
Personal	<i>I</i>	<i>me</i>	<i>my</i>	<i>mine</i>	<i>myself</i>
	<i>you</i>	<i>you</i>	<i>your</i>	<i>yours</i>	<i>yourself</i>
	<i>he</i>	<i>him</i>	<i>his</i>	<i>his</i>	<i>himself</i>
	<i>she</i>	<i>her</i>	<i>her</i>	<i>hers</i>	<i>herself</i>
	<i>it</i>	<i>it</i>	<i>its</i>	-	<i>itself</i>
	<i>we</i>	<i>us</i>	<i>our</i>	<i>ours</i>	<i>ourselves</i>
	<i>they</i>	<i>them</i>	<i>their</i>	<i>theirs</i>	<i>themselves</i>
	<i>one</i>	<i>one</i>	<i>one’s</i>	<i>one’s</i>	<i>oneself</i>
Interrog	<i>who</i>	<i>whom</i>	<i>whose</i>	<i>whose</i>	-
Relative	<i>who</i>	<i>whom</i>	<i>whose</i>	?	-

Table 5.1. Personal, Interrogative, and Relative Pronouns

Whose is often known as a GENITIVE PRONOUN, AKA “possessive pronoun”. The genitive pronouns come in two kinds: those that are DEPENDENT on a noun: *my* life, *your* name, *her* eyes, and those that are INDEPENDENT: *mine*, *yours*, *hers*, etc. Perhaps confusingly, dependent and independent forms can share the same spelling and pronunciation, as in *it’s his* job (dependent) or *the job is his* (independent). Such is the case with *whose*: You can ask *whose gorilla is this* or *whose is it*, at least when you use *whose* to ask a question.

The *whose* of questions is an INTERROGATIVE PRONOUN, but there’s another *whose*, as in *that’s the person whose name is on the door*. This is the relative pronoun. Relative *whose* isn’t interrogating anything; it’s “standing in for” *the person’s*. And the *whose* in (1) – the one that Hankamer and Postal marked as ungrammatical – is a relative pronoun. Putting it all together, then, the *whose* at issue is this independent genitive relative pronoun.

The point Hankamer and Postal were making was that, where most pronouns have paired dependent and independent genitives, *who* feels like it’s missing the independent one. All the other combinations in (2) are possible. Why not (d)?

- | | | | |
|-----|----|---------------------------------------|-----------------------------|
| (2) | a. | <i>Whose gorilla is this?</i> | [dependent interrogative] |
| | b. | <i>Whose is this?</i> | [independent interrogative] |
| | c. | <i>someone whose gorilla had left</i> | [dependent relative] |
| | d. | <i>* someone whose had left</i> | [independent relative] |

They didn’t know, and that’s why they published “Whose gorilla” as a squib. Generally, squibs don’t get cited much, and this one, true to form, has been mostly ignored. In 2002 the *Cambridge grammar of the English language* agreed with its judgment, marking (3) with an asterisk. And, nearly 20 years after that, almost 30 since the article was published, Cinque (2020) was still referring to the issue as a “residual puzzle”.

- (3) * *The police are trying to contact the person [whose it was].* (p. 472)

Linguistic discoveries often have unexpected origins. This particular journey began with a conversation about gorillas, meandered through book reviews and academic papers, and culminated in a cross-linguistic insight about the nature of possessive pronouns. Here’s how it played out.

Some years ago, my friend, co-author, and mentor Geoff Pullum told me about an intriguing paper from 1973. Two linguists, Jorge

Hankamer and Paul Postal, had made a bold claim in a short paper called a “squib”: English, they said, doesn’t include a particular kind of *whose*. This peculiar assertion stuck in my mind, resurfacing last summer as I reviewed Geoff’s most recent book *The truth about English grammar* (Pullum, 2024), in which he mentioned this missing *whose*.

As I embarked on writing this book, I found myself drawn to linguistic edge cases – those fascinating exceptions and rarities that push at the boundaries of our understanding of language. The case of the independent relative *whose* seemed like a perfect candidate for investigation.

The plot thickened earlier this year when I read a paper by Adele Goldberg and her graduate student Nicole Cuneo (Cuneo & Goldberg, 2023) about island extraction in relative clauses (See 2). Their work sparked my interest in the role of pragmatics in these linguistic puzzles. It occurred to me that similar forces might be at work on the missing *whose*, so I suggested to Adele that we team up on trying to figure this out. Although a collaboration didn’t materialize due to Adele’s existing commitments, I decided to pursue this line of inquiry on my own.

What followed was an intense two-week period of research and writing, during which several key insights – and a paper manuscript – emerged, sparked in a large part by a few key observations I found in Halliday and Hasan (1976) and Ariel (2014).

As the pieces fell into place, I realized that my findings might have implications beyond English. A preliminary investigation into French, German, Spanish, Persian, and Japanese indeed suggested that similar constraints might be at play across languages. But before we explore these broader implications, let’s go back to the beginning: Hankamer and Postal’s bold claim about a supposedly non-existent *whose* in English.

How had Hankamer and Postal determined that (1) was ungrammatical? Almost certainly by noticing their own metacognitive feelings. Did they search for instances or present relevant examples to a representative sample of English speakers to see how the independent *whose* was perceived? They don’t say, but it’s very unlikely. Such experimental methods were almost never seen as necessary at the time. A linguist’s personal grammaticality judgments were generally deemed valid and reliable. Rarely were they questioned.

It occurred to me, though, to do something they couldn't have done. In March of 2024, I asked OpenAI's GPT-4 and Google's Gemini Advanced to rate the grammaticality of this *whose* on a scale of 1–7, not because I questioned the feelings of Hankamer and Postal and the Cambridge grammar authors, but because I was curious to see how these large language models would respond. The sentence I gave them was (4).

- (4) *Mine was working, but I know someone whose wasn't.*

Surprisingly to me, both models rated it a 6, denoting high grammaticality, and cited the slightly unusual use of *whose* as the reason for not awarding a 7. I'd experimented with eliciting grammaticality judgments from large language models before with various constructions, and I'd mostly found their assessments to align surprisingly well with my own. I'd even asked them to produce examples at various points on the scale, and again, they'd fared reasonably well. This is why I was rather surprised at a 6 rating for (4).

Intrigued by this mismatch, it occurred to me to ask some non-linguist colleagues and family members what they thought of it. I sent out a few quick texts and within a day, ten of the twelve I'd asked had also given it a 6 or 7. Two gave it a 4. Many didn't mention *whose*, and even after I queried the pronoun specifically, two said it seemed fine to them. To my surprise, the large language models were a better fit for the naive humans than were the linguists. I was intrigued.

This issue arose while I was working on *Wikipedia's* [English pronouns](#) page. There, the other editors and I had reflected the position of Hankamer and Postal and the Cambridge grammar: that independent relative *whose* didn't exist. With these new grammaticality judgments, I started to wonder if I should revise that entry.

Wikipedia requires sources, though, and “Brett's friends and family along with some LLMs” doesn't cut it. It was then that it occurred to me to consult *The Oxford English Dictionary*. Had Hankamer and Postal done so, perhaps they wouldn't have published because they would have found that the lexicographers at the *OED* seem to take the same position as my informants and the large language models: the independent genitive relative *whose* is rare, but grammatical. The *OED* presents examples going back to the 1300s, including those in (5).

- (5) a. *After þe wille of him hos þe werkes bez.* ‘According to the will of him whose works are.’ [1325]
 b. *The man whose these are.* [1611]

- c. *The most generous patron ... continues to be our ... Pittsburgh and Allegheny auxiliary, whose also is the largest contribution to the rent of our Association's new office.* [1904]
- d. *Everything depends on the person whose this administration is.* [2018]

And (6) presents a few more I found online.

- (6) a. *In California you usually include a copy of the check with the offer. Earnest money despoit (EMD) can be anywhere from 1/2%-3% of the sales price. Mine was 3%, I just helped a person whose was 2%, I've never seen an EMD less than 1/2% in the past 3 years in California (from Shasta Co. to Orange Co.).*
- b. *I have met a person whose SSN was only 10 away from mine (yes only the second to last digit was one higher than mine) and he was born in a different state and was at least 7-10 years older than I was. And I met another person whose was kinda close to mine but again we had nothing in common pertaining to our birthday or location of birth/SSN registries.*
- c. *Mine is also submitted on June 30 and no updates yet, i know a person whose was filed on June 30 as well in Regular mode and he got approval on 8/16. I think we can expect a decision in the next few weeks.*
- d. *I'll be scouring eBay for an Alpha keypad. Debating whether to swap it out for the existing one or just use it for programming. The control panel isn't as hard to get to as the person whose was installed in the top of a closet.*
- e. *Head shaved? Mine wasn't but the guy whose was had to give a pubic sample.*
- f. *In my opinion, my 50%/50% bud/trim mix came in third place. One guy, whose was the best in my opinion, he treated his trim exactly like he did his buds.*
- g. *When I entered in 1977, I actually had the longest ash, but mine was bent. I came second to a woman whose was straight.*
- h. *I have vasovagal syncope (a specific thing triggers fainting) and it's not uncommon for a trigger to be urination or defecation. My trigger is medical stuff like needles, which mostly people are sympathetic to. But I did know another woman whose was set off by morning urination.*

i. *Meanwhile the original person whose this photograph is...*

Based on this evidence, I revised the *Wikipedia* article to include the two views. But, of the two positions, was it mine whose was right?

This is the problem with our feelings of (un)grammaticality. Hankamer and Postal are top-notch syntacticians, legendary even. And John Payne and Rodney Huddleston, the authors of the chapter of the *Cambridge grammar of the English language* that marks (3) as ungrammatical, are among the best English grammarians the world.

On the other hand, there's no theoretical reason to expect relative *whose* to go AWOL, there's the data from the OED and my web searches, there's the intuition of my friends and family, and there's whatever ineffable average over trillions of words of English is represented in the judgments of the large language models.

So, we have two seemingly contradictory forces at play. On the one hand, certain established linguistic authorities tell us that they feel the sentence *The person whose wasn't working* is ungrammatical. On the other hand, the feelings of various people, lexicographers and non-linguists alike, backed up by the vast echo chamber of everyday speech reflected in online examples and the judgments of generative AI models, suggest it might be fine.

Linguistics, like any field of study, bears witness to scholarly disputes. Yet, let truth be told, our squabbles over the fringe cases of English grammar – the quirky pronoun, the oddly placed phrase – are generally far removed from our everyday experience of language. Situations in which the grammaticality of the relative independent genitive pronoun might even be at issue, for instance, are vanishingly rare.

Unless I've messed up, every sentence I've composed in this book is uncontroversially grammatical. The same is true of almost everything you hear and read every day and all that you say, at least in a language you grew up using or are proficient in. Still, there are edge cases like this *whose* that aren't so clear cut, and others that aren't even so rare. What should we do with them? How should we adjudicate the situation when grammaticality is disputed, individually, as a community of language users, or as linguists and grammarians? What do we do when our feelings don't match up? Geoff Pullum has a suggestion.

Pullum was “the worst sort of lazy and undisciplined teenager”, eventually dropping out of high school. After a short stint as a shop assistant in a London bookstore, he joined a travelling band as the piano player. Somehow, this led to his appearing on two of the biggest selling UK albums of the 1960s. Despite his change in fortunes, he found touring tedious. The band broke up, and he

turned his focus on linguistics, where his intellect, broad interests, and workaholic tendencies could be fully engaged. He began writing for the public in the early 2000s on the *Language log* blog, which, at its peak, exceeded a million page views per month in part because Pullum is one of the best and funniest writers in the discipline. On top of that, he's also a philosopher of linguistics, and his philosophical contributions to the field are noteworthy.

Pullum (2019) suggests we employ REFLECTIVE EQUILIBRIUM. He says that we should carefully consider our initial feelings about the grammaticality of a sentence, and then weigh these against established grammatical principles and the judgments of other competent speakers, going back and forth between these sources of information until our position seems to have arrived at a kind of resting place. If our feelings conflict with these principles or with the consensus of other speakers, we should be willing to revise our judgments. At the same time, if a grammatical principle seems to contradict the strong feelings of many native speakers, it may be the principle that needs to be revised.

The idea of reflective equilibrium, along with the term itself, is often credited to John Rawls, who advocated it as a method for establishing principles of justice. But Rawls was building on the ideas of Nelson Goodman, who was more interested in science. Specifically, Goodman was trying to understand how we could justify deductive logic.

Pullum realized that the study of grammar by linguists involves a unique blend of these two perspectives. On one hand, like scientists and logicians, linguists must deduce grammatical patterns from actual data – in this case, the language produced by speakers, despite the presence of errors. On the other, similar to ethicists and political philosophers, they must also consider our feelings about how language ought to be used, while recognizing that our feelings of ungrammaticality may be shaped by idiosyncratic factors like dialect, education, or even momentary inattention. Navigating the tension between these two aspects requires a process of constantly refining both the observed patterns and the guiding principles until they align in a satisfactory manner.

Applying reflective equilibrium to the case of the relative independent genitive pronoun *whose*, we might proceed as follows: First, we note, as Hankamer and Postal did, our own feelings about a noun phrase like *the person whose wasn't working*. For some people, as we've seen, this will feel ungrammatical. We then consider this evaluation in light of the general principles and patterns of English grammar. We might wish to start by assuming perfect regularity within paradigms, but we would quickly be disabused of this idea.

Famously, plural and past-tenses forms can be highly irregular. Less commonly recognized is that *be* is the only verb with two past-tense forms: *was* and

were and that a few verbs are defective: *beware* for example. Among the modal auxiliary verbs, there are present- and past-tense forms for *will/would*, *can/could*, *shall/should*, and *may/might*, but no such counterpart for *must*. The pronoun *-self* forms are typically made up of the dependent genitive + *self*, as in *myself*, *yourself*, *herself*, and *ourselves*. But when it comes to *himself* and *themselves*, it's *self* together with the accusative form. The examples could be multiplied, but suffice it to say that rejecting an idea simply because it involves divergence from a pattern is a non-starter. Still, it's a piece of the puzzle.

We'd also need to consider the judgments of other speakers. If we find that a significant number of people find the sentence acceptable, we might need to question whether our initial intuition was correct, or whether the grammatical principle needs to be modified. We might also consider whether there are other constructions in English that seem to violate the principle in a similar way, and if so, how we judge their grammaticality.

Through this process of weighing intuitions, principles, and usage data, we can arrive at a more considered judgment. We may conclude that the sentence is indeed ungrammatical, and that our initial intuition was correct. Or we may decide that it represents a rare but acceptable variant, perhaps limited to certain dialects or registers. Or we may even conclude that the grammatical description we've relied on needs to be revised to accommodate this and similar constructions.

The key point is that reflective equilibrium provides a rational procedure for adjudicating disputes about grammaticality. It's not a mechanical algorithm that always yields a clear "yes" or "no" answer, but rather a method for navigating the complex interplay of intuition, systematic principles, and actual usage that characterizes our linguistic competence. By engaging in this process, both individually and collectively, we can arrive at grammaticality judgments that are not merely arbitrary opinions, but well-considered reflections of the deep patterns of our language.

Recall that Sampson's view that all is grammatical and Chomsky's early generative work, which assumes that something is grammatical if, and only if, it can be produced by the grammar. As distinct as they seem, these two ideas can be united. All we need to do is to create a new rule every time we observe a new pattern. If we don't like Sampson's view, then we need something to constrain us, but it can't be whether or not something is generated by the grammar because that's what we're trying to decide.

in the context of ethical reasoning, serves as a method for reconciling our intuitions with established principles. In ethics, it involves adjusting our moral

intuitions and theoretical principles until they cohere in a mutually supportive framework. Translated into the realm of linguistics, this means aligning our intuitive sense of what sounds right in language with the grammatical rules and theories developed by linguists. This process isn't static; it requires ongoing adjustment as new linguistic data emerges and our language intuitions evolve. By adopting this philosophical approach, we extend its application from moral philosophy to the dynamic and often subjective domain of language, offering a systematic way to navigate the complexities of grammatical judgment.

Picture being in a maze. Each choice you make is a bet between what feels right and what's traditionally accepted as right. The grammar guidebook tells you there's only one way through this maze. But what if this guide is outdated, or the maze itself shifts and changes, just like language does? Goodman's idea of reflective equilibrium is like having a compass in this scenario, guiding us not by the old map, but by a constant conversation between our gut feelings about language and the grammatical structures we've learned.

With the whose conundrum as our example, it's like starting off thinking, *The person whose wasn't working sounds alright to me*. But then, you bump into the expert opinions, the grammar guides, maybe even a linguistics paper or two that disagree. It's not about tossing your initial gut feeling aside; it's more about weighing it against these authoritative voices. This back-and-forth, this dialogue between what we instinctively feel and what we're told, is at the heart of reflective equilibrium. It might lead us to adjust our views a bit, maybe deciding the sentence is a tad off but not a grammatical catastrophe. Or, we might stick to our guns, now more aware of the broader conversation around it.

Yet, here's the thing: language is always on the move. It's alive, constantly evolving. New slang makes its debut, meanings drift, and yesterday's grammatical faux pas can become today's hot new phrase. Reflective equilibrium gives us a method to navigate this evolution, not with strict rules but with an open mind and a critical eye.

Consider the case of split infinitives, like the infamous *to boldly go*. Grammar purists might balk, but there's an argument to be made for such constructions if they enhance the sentence. Reflective equilibrium encourages us to balance traditional rules against the effectiveness and clarity of expression in specific contexts.

Reflective equilibrium, then, isn't about finding a one-size-fits-all answer. It's about understanding the dynamics behind our language choices and settling on what sits right with us, all while acknowledging that language is a constantly changing landscape.

As we move forward, we'll take a closer look at this balancing act, drawing on

real-life examples to illustrate how reflective equilibrium can help us navigate the complex terrain of grammatical judgment. It's about getting comfortable with linguistic ambiguity and appreciating language's rich, ever-changing nature.

5.2 FREE ADJUNCTS, DANGLERS, AND HOWLERS

Dangling participles haunt the dreams of grammarians. *Turning his attention back to his work, a dangler stared him in the face.* As you first read this, you develop the idea that somebody was turning his attention back to his work, and, right after the comma you see *a dangler*, and in your developing sense of the meaning, you entertain the idea that that dangler was that person. And then you encounter *him*, which forces a complete reassessment of the situation. This is the experience of reading a bad dangling modifier, or what linguists call FREE ADJUNCTS.

Consider (7), noticed by my friend, Jim Donaldson (2020).

- (7) * *Suddenly, he sits up and tugs my panties off and throws them on the floor. Pulling off his boxer briefs, his erection springs free. Holy cow!*
(E. L. James *50 Shades of Grey*)

This dangler is bad. It qualifies as a howler, and, in response, I've marked it with an asterisk. I think it's ungrammatical, but is it?

The first example cited by The Penguin Dictionary of American English Usage and Style (by Paul W. Lovinger; New York: Penguin Reference, 2000) is as clear a case as one could want, taken from a book about cannabis (if you could steel yourself, please, I want no politically incorrect giggling at this one):

- (8) a. *Although widely used by the men, Bashilange women were rarely allowed to smoke cannabis.*
b. *Rich and creamy, your guests will never guess that this pie is light.*

"Does this fall under the no dangling modifier prescription?," asks Rosanne, in a post on The X-bar. Yes, Rosanne, it does. Like participles, adjectives and also some idiomatic preposition phrases, when used as adjuncts, need an understood subject (or, it might be better to say, a target of predication) to be filled in if they are to be understood. The prescriptive tradition says that the subject filled in must be the one obtained from the subject of the matrix clause. Here that would be your guests, which makes a nonsense reading, so the sentence cited would be treated as an error.

But the prescriptivists have a problem. Sentences of this kind, which call for you to fill in the understood subject from somewhere

else (here, the subject of is light in the subordinate clause), are so common that when I and several friends have spent some time picking new ones up from print and radio sources, we get them at a rate of as many as one per day. That's in edited sources, where grammatical errors have almost entirely been screened out. This just can't be syntactic error. It's too frequent. (Pullum, 2003)

Donaldson's PhD dissertation, supervised by Geoff Pullum, examines dangling modifiers in detail. Pullum (2003, 2005a, 2005b, 2005c) and Donaldson argue that while dangles can be seen as "ill-written and discourteous", "inconsiderate", "stunningly inept", or "bad grammatical manners", they don't necessarily constitute ungrammaticality.

The construction means that his erection pulled off its boxer briefs, but that's obviously not the intended meaning. "Just about every rhetoric, grammar, and handbook written since the latter part of the 19th century warns the student against such constructions" (Gilman, 1989, p. 314).

Consider the sentence: *Having arrived late, the meeting was already over.* Traditional grammar would flag this as a dangling modifier since the subject of the participle phrase *Having arrived late* doesn't logically correspond to the subject of the main clause, *the meeting*. Pullum and Donaldson's perspective invites us to reconsider whether this incongruence automatically renders the sentence ungrammatical or merely less clear or elegant.

Their argument suggests that the judgment of grammaticality involves more than a binary distinction between right and wrong. It encompasses considerations of clarity, elegance, and the speaker or writer's intent. From this viewpoint, the use of a dangling modifier might reflect stylistic choices or conversational shortcuts rather than a lack of grammatical competence.

Reflective equilibrium offers a valuable framework for navigating this debate. It encourages us to balance our intuitive disapproval of dangles with the understanding that language flexibility often accommodates such constructions in everyday usage. This balancing act involves weighing the clarity and communicative effectiveness of language against the descriptive realities of how people speak and write.

Examining the acceptance of dangling modifiers in different contexts and registers can shed light on the fluid boundaries of grammaticality. For instance, creative writing and informal speech frequently employ dangles to convey a sense of immediacy or emotional resonance, prioritizing expressive impact over strict adherence to grammatical rules.

5.2.1 SINGULAR *THEY*

My daughter, Mia, had two roommates, one of whom, I'll call her Sasha, wanted others to refer to her using the pronoun *{they}*. This is easy and natural for Mia. It's fully grammatical for her. She can say *Jessica cleaned up her mess but Sasha didn't. I wish they would*, meaning not that she wishes both roommates would clean it up but that Sasha would.

That's not what that means for me though. When I hear this, I understand that Mia wants both of her roommates to clean.

Or consider this example that I heard on *This American Life*:

Things for Malinda and the kids are pretty good these days. Stella just turned 18. She says her climate nightmares only come every couple weeks instead of nightly. And she's off at college this fall, a milestone that, for a while, she never thought she'd reach. Emrys is 19. They work on a boat, teaching kids about marine life and maritime skills, a job they love, though they skip out on teaching the lessons about climate change. They said they're not ready to do that yet.
(DeKornfeld, 2021)

This understanding is “ballistic” for me, in the sense that it can't be stopped, at least not at the moment. It's like seeing *<cat>* and having the word *cat* pop into my mind. There's nothing I can do to stop it.

I can learn a new rule for *they*, but it's a clunky and artificial kludge slapped on top of what's an exquisitely tuned system, one that can do no better than to try to funnel the champagne into a different bottle as it foams forth. It's the effortful correction you make when you see *the colour of the following word is red*.

This isn't because *{they}* can't be singular. I have no trouble at all with singular *{they}*, as in *Someone forgot their bag*, but it doesn't compute for me when it refers to a named person.

My theoretical understanding of singular *{they}* tells me it should be grammatical. If this is confusing to you, recall that singular *{you}* is perfectly grammatical, even though *{you}*, like *{they}* started life as a plural pronoun.

The singular second-person pronoun that *{you}* displaced was *{thou}*. As in many Western European languages, *{thou}* didn't simply mean ‘the individual I'm addressing.’ It also conveyed a sense of familiarity even intimacy, or where that was absent, a sense of superiority of the speaker over the spoken to. Its complement, the second-person plural had the additional social meaning of respectful address.

Over time, the ‘familiar/intimate’ meaning was overpowered by the implication that the person being *thoued* was being spoken down to. And, so, in the name of politeness, {*thou*} was replaced with plural {*you*}.

{*You*} is still syntactically plural. The verb agreement is the plural agreement. But semantically, it can be singular or plural.

{*They*} is the same. There’s no reason that it couldn’t follow the precedent set by *you*. And yet, I can’t get there. I keep getting blindsided by cases like the messy roommate(s).

I’ve tried to reach a reflexive equilibrium, but updating my feeling of grammaticality for referring to a named individual, such as *Sasha*, as {*they*} isn’t as easy as learning the meaning of a new word. It requires unlearning half a century of practice. And that’s why we have gradient grammaticality in a population.

In trying to achieve a reflexive equilibrium, I did realize that singular {*you*} does differ from the new singular {*they*} in one somewhat related aspect. {*You*} isn’t interchangeable with a person’s name, not without infantilizing them. If I’m speaking to my daughter, I can ask *what do you think*, but I can’t ask *what does Mia think* without a great deal of implied condescension.

5.2.2 THE DOUBLE-IS CONSTRUCTION

There's a curious construction that you hear all over the place these days. In fact, I heard it this morning on the popular *EconTalk* podcast, hosted by economist Russ Roberts, and not for the first time. Roberts and his guests – people like Kevin Kelly, co-founder of *Wired* magazine, and Eric Hoel, a research scientist and author known for his work on consciousness and theoretical neuroscience – will say things like:

- (9) a. *But, the problem is, is that there's not just, say, the coasts versus the middle or the South versus the rest.*
 b. *But, my point is, is that, in America the bureaucratic infrastructure is not well-respected.*
 c. *But, the issue is, is that once the mask is on, it's very unclear.*

This construction, with its doubled *is*, has a fancy name: the double-*is* or ISIS construction. It's become surprisingly common in the last 40 years or so. Despite this, from a strictly structural viewpoint, it's a bit of an oddity.

At first glance, the ISIS construction might look like other common structures. For example:

- (10) a. *The question is is it dangerous?*
 b. *What it is is a tool for learning about language.*

But there's a key difference. In these examples, each *is* has its own subject and serves a clear purpose within its own clause. In (10a), the main verb of the sentence is the first *is* – the underlined one. In (10b), it's the second.

With the ISIS construction, though, the *ises* outnumber the clauses, and no-one can say with certainty which is the main verb.

So, why does this strange construction exist? One theory points to the focusing function of the constructions in (10a) and (10b) and suggests it has been misattributed to the coincidentally-shared *is is*.

What I mean by “focusing function” is that the examples seem a little strident, a little assertive. (10a) doesn't just ask “is it dangerous?” It focuses the listener on this question, asserting that this is *the* question. Similarly, (10b) isn't just saying “this is a tool for learning about language.” It's focusing on that aspect: That's what the thing is.

Each example focuses our attention in its own way, but what they share is *is is*. In each case, it's like a brief dramatic pause before the main event. The switching on of the spotlights before they converge on the key message.

Compare the simple statement *It's a fabulous thing* with the ISIS example *The point is, is that it's a fabulous thing*. If it were a question, it could be *The question is [is it a fabulous thing]*, but it's not a question – it's a point. And so *point* replaces *question* and out pops ISIS.

The ISIS construction is another example of conflicting signals of grammaticality. Roberts and his guests – polished writers and speakers all – use it relatively frequently, and it mostly goes unnoticed. Others, like me, though, notice it all the time. My grammatical intuitions may be contaminated, but even many years before I knew anything about the construction, I had noticed it in a friend's speech and found it ungrammatical. Formally speaking, a second *is* without a subject to connect to isn't found elsewhere in English. Also, ISIS doesn't turn up in edited writing.

I understand that it seems to be grammatical for many people, at least in spoken English. I also understand that there's no need for spoken English to conform to the grammatical norms of written English. I even know of constructions that I find grammatical despite their being unlike anything else in English, *I'd better try again*, *many's the time...*, *what with the way it went...*, *no matter the change...*, etc. And so, I don't think the extra *is* necessarily disqualifies it from being grammatical. In fact, it's my considered judgment that ISIS must be grammatical for people like Roberts and his guests.

And yet I never find myself saying it, and it always throws up a red flag when I hear it. I've tried over many years to overcome my feeling of ungrammaticality, but it seems I'm stuck with it.

5.3 WHAT DOES (UN)GRAMMATICALITY FEEL LIKE?

Sensations arise from the interaction between stimuli, our sense organs, and our nervous system. We've got the traditional five senses: sight, taste, smell, hearing, and touch. But it goes well beyond that. We have a sense of balance, regulated by nerves in our vestibular system in our inner ears. Our sense of hunger is triggered by nerves in the gastrointestinal tract and transmitted through the vagus nerve to the hypothalamus. We sense the location of our various body parts and the angles of our joints, pain, pressure, itch, hot and cold, and chemical irritants like capsaicin and allyl isothiocyanate in wasabi and mustard. We even have a sense of the levels of carbon dioxide and oxygen in our blood.

We're sensitive creatures, but we have no sense organs for grammar, so grammar isn't something we sense, and (un)grammaticality isn't a sensation. But it's a feeling, or at least part of it is. Feelings are sensations that have an evaluative aspect.

No sooner had the warm liquid mixed with the crumbs touched my palate than a shudder ran through me and I stopped, intent upon the extraordinary thing that was happening to me. An exquisite pleasure had invaded my senses, something isolated, detached, with no suggestion of its origin. And at once the vicissitudes of life had become indifferent to me, its disasters innocuous, its brevity illusory – this new sensation having had on me the effect which love has of filling me with a precious essence; or rather this essence was not in me it was me. ... Whence did it come? What did it mean? How could I seize and apprehend it? ... And suddenly the memory revealed itself.

– Marcel Proust, *In Search of Lost Time*

Proust describes a feeling triggered by but unconnected to any immediate sensation, a feeling about a memory but without, at least initially, access to the memory itself. It's a feeling about a feeling, a metacognitive feeling.

Language understanding is another such process we can have feelings about, and like Proust's reverie, those feelings come in different shapes and intensities. Consider the sentences in (11). All ungrammatical, but each elicits a different reaction. The first might cause a double-take – what does it mean for cheese to be made of green ideas? The second feels more jarringly wrong, with words simply in the wrong places. The third is word salad, scrambled past the point where you can even try to make sense of it.

- (11) a. * *The cheese green ideas sleep late furiously.*
 b. * *Late the ideas sleep furiously green cheese.*
 c. * *Sleep furiously late ideas the green cheese.*

Several things shape the felt severity: how much of the structure is broken, what level the breakdown is at (a semantic mismatch feels different from a morphological one), and how surprising the error was given what came before.

But across all the variability, one thing stays constant: ungrammatical sentences evoke a feeling, and grammatical ones rarely do. A sentence that hews to syntax – and almost all the ones we encounter do – doesn't elicit a feeling of rightness to match the wrongness of its ungrammatical counterpart. Grammaticality is the baseline. It just feels normal.

Exceptions exist. A mellifluous line of poetry, a sharp witticism, a sentence whose clarity makes a hard idea easy – these can provoke a feeling. So can a well-placed correction, for those who care about that sort of thing. But for the most part, grammaticality is like clean air or a healthy back: you don't notice it until it's

gone. It's homeostasis. Only when something perturbs it – a syntactic stumble, a semantic contradiction, a morphological mismatch – do we feel the friction.

So why this asymmetry? Because the feeling of ungrammaticality is a meta-cognitive feeling, and metacognitive feelings only register when the underlying process is in trouble.

Consider some other members of the family. The *feeling of knowing* is what tells you, before you can say it, that you know who played Ron Weasley in the Harry Potter films. The *tip-of-the-tongue* state is what tells you that you almost know it, but can't quite get to it. The *feeling of confidence* is what guides your hand on a multiple-choice test. Each is a feeling about a cognitive process that you can't see directly: memory retrieval, lexical access, judgement under uncertainty. Pablo Fernández-Velasco (2021) adds disorientation to the list. Disorientation is the feeling that rides on top of navigation. When navigation goes well, no feeling. When something goes wrong with it, you feel lost.

The same shape fits the feeling of ungrammaticality. The process beneath is parsing, and parsing works by prediction. As you read or hear a sentence, your brain runs ahead at every level – phonological, lexical, syntactic, semantic, pragmatic – predicting what comes next (Clark, 2013; Kuperberg & Jaeger, 2016). When the input matches the prediction, the process keeps moving and there's nothing for consciousness to do. When the prediction misses, the resulting error gets weighted, propagated, and registered (Rabovsky et al., 2018), and what surfaces in awareness is the feeling that something has gone wrong with the language.

The same asymmetry appears in music. Wagner's Tristan chord, F, B, D $\sharp$, and G $\sharp$, from "Tristan und Isolde", gave audiences a sonority they had no settled way to hear. It was unpredictable, and so it registered as wrongly dissonant, the musical equivalent of ungrammatical. But some listeners were able to find a sense in it, a narrative sense rather than a harmonic one.

Later, I'll get into how constructions and other linguistic forms can mean things other than denotations and predications, things like identity, status, deference, and interrogation. The point here is that meaning isn't a single thing. Because some people could see analogies between unfulfilled harmonic tension and the unrequited love central to the opera's story, the chord wasn't dismissed.

Almost certainly it was the case that people had to work to find such a sense, and they did so because the author of the chord was Wagner. This is crucial context. Wagner had come to be seen as a musical genius, and his previous operas – notably *Der fliegende Holländer* (1843), *Tannhäuser* (1845), and *Lohengrin* (1850) – had featured increased chromaticism and a more flexible approach

to tonality. He was also renowned for his concept of the *Gesamtkunstwerk*, or “total work of art,” which sought to unify all artistic elements within an opera. So, while the musical context of the Tristan chord evoked a sense of wrongness, this sense may have been attenuated by the context of having Wagner’s name attached to it.

A lesser musician – someone young and unknown, or one with other such prejudicial strikes against them – may have been seen as simply incompetent had they composed or performed such a thing. But, although the chord didn’t make sense, nor did it make sense to believe that the great Opernkomponist had made an error such as this. It had to be intentional, and so the task these listeners set themselves was to determine the intention. And they may have set off on that journey because the predictive brains had dampened the feeling of dissonance for these folks because, well, “it’s Wagner”?

Whether my stylized story is true or not, what’s indisputable is that the opera continued to be performed. While it wasn’t an immediate sensation, it gradually gained recognition and acclaim for its innovative music and dramatic storytelling.

The opening run itself wasn’t extended, but that was common for opera premieres at the time. Subsequent productions in other cities helped to build its reputation. By the 1870s and 1880s, “*Tristan und Isolde*” had become a staple of the operatic repertoire and was widely regarded as a masterpiece. And, as a result, more and more people were exposed to the chord. More composers started using it, exposing even more audiences, and folks got used to it. It came to reside in the predictable, and it came to sound no wronger than a minor chord in the right contexts.

So the feeling of ungrammaticality is the conscious side of a parser surprised by what came in. Like the tip-of-the-tongue or disorientation, it isn’t the cognitive work itself – it’s the signal that the work is in trouble. And because we don’t feel the work when it goes well, grammaticality has no felt counterpart. Smooth processing leaves no trace. We don’t feel the absence of trouble. We only feel trouble.

This account also explains why grammaticality judgments vary so much across speakers. If the feeling tracks parsing, and parsing draws on whatever statistical regularities a speaker has been exposed to, then speakers with different linguistic backgrounds will have different intuitions about what’s grammatical.

For example, double negatives such as *I don’t know nothin about that* or verb forms such as *I seen him yesterday* are familiar to most English speakers,

but they're not part of Standard English. As a result, in a context where Standard English prevails, they may provoke a feeling of ungrammaticality. But when the Kid Rock laments, "I ain't seen the sun shine in 3 damn days" or Lil West boasts, "I don't take no L's," we don't expect Standard English, and no feeling of ungrammaticality is evoked.

Young children over-regularize past tenses. So, instead of *I found you*, a child might say, *I finded you*. On the one hand, this is ungrammatical and we all know it. On the other, it doesn't feel as ungrammatical as if I said to you *I wrote two book*. Even if we hadn't expected the child to say *finded*, the reason for it is readily apparent in the context, and our sense of ungrammaticality is commensurately dampened. Similarly, should we hear someone with a noticeable foreign accent say, *I must to go*, it registers as ungrammatical, but at the same time, the level of surprise is attenuated by the inference that this is someone who has come to English later in their life.

Conversely, I speak Japanese with a reasonable level of conversational fluency, but if a Japanese speaker confidently said (12b) to me instead of (a) to explain why she had a cat, I probably wouldn't notice or experience any feelings of ungrammaticality. In (a), *neko* 'cat' is the topic – thus TOP – and the lack of a subject suggests that a first-person subject 'I'. COP means a copula, roughly *be*, so in a stilted overly direct translation it's 'as for cats, I'm fond (of them).'

- (12) a. 猫 は 好き です。
 Neko wa suki desu.
 cat TOP fond COP
 'I'm fond of cats.'
- b. 猫 が 好き です。
 Neko ga suki desu.
 cat SUBJ fond COP
 'Cats are fond of it.'

The second sentence, though, has *cat* as the subject (SUBJ), with no particular topic stated. That means the topic is probably whatever we've been talking about, and the sentence tells us that cats are fond of whatever it is. I know this *ga-wa* distinction, and I usually make the right choice when speaking. But English doesn't mark topics vs subjects this way, and so it might just slip by me if I wasn't primed for it.

The sense of ungrammaticality we feel isn't a binary, all-or-nothing affair. It's a graded experience, one that spans highs and lows, that reflect the uncertain

nature of prediction. A minor syntactic slip, something very close to expectations, might generate a small, localized prediction error, experienced as a fleeting sense of discomfort. A more serious grammatical deviation might generate a larger, more widespread error signal, experienced as a visceral sense of linguistic wrongness or confusion.

And yet what will register as serious isn't easy to pin down. Subject–verb agreement errors seem objectively minor, often rectifiable through the mere toggling of a single *–s*, but *this error feel* egregious. Other times, such disagreements might slide by with hardly a nudge. Perhaps you sensed the infelicity in the previous paragraph: *reflect* should have been *reflects*. If you did, I'd guess the intensity of ungrammaticality you felt was minor compared with *this error feel...*

5.3.1 GRAMMAR SENSITIVITY IN ANIMALS

The sensation of grammaticality we experience is often taken as evidence for the human-specific nature of language. But could this feeling have emerged from cognitive abilities that predate human language? Recent research with non-human primates suggests that the mental machinery underlying our sensitivity to grammatical dependencies may have evolved long before language itself.

In a groundbreaking study, Ravignani et al. (2013) demonstrated that squirrel monkeys (*Saimiri sciureus*) – New World primates whose lineage diverged from ours at least 36 million years ago – can detect abstract, non-adjacent dependencies in auditory sequences. The monkeys were habituated to tone sequences following an AB^nA pattern, where the first and last elements matched while being separated by a variable number of intervening elements. Not only did the monkeys distinguish between sequences containing dependencies and those lacking them, but they also generalized to previously unheard pitch classes and novel dependency distances.

This is remarkable precisely because squirrel monkeys lack vocal learning abilities and possess a relatively fixed vocal repertoire. These monkeys will never speak a human language, yet they possess the cognitive toolkit to recognize abstract patterns that form the foundation of many linguistic structures. As the authors note, “Although no squirrel monkey will probably ever speak a human language, these monkeys possess the cognitive potential to recognize the rule generating plurals of Turkish nouns, or many other linguistic phenomena” (Ravignani et al., 2013, p. 4).

What makes this finding so significant for our understanding of (un)grammaticality feelings is that it suggests a profound disconnect between the evolution of language and the evolution of dependency sensitivity. Rather than evolving specifically for linguistic purposes, our feeling of ungrammaticality

when encountering a malformed sentence may be an EXAPTATION – the repurposing of a cognitive ability that evolved for entirely different reasons.

Like the feeling of disgust that evolved to help us avoid pathogens but now extends to moral violations, our sense of grammaticality likely built upon pre-existing pattern-recognition systems that long predated human language. Our hominid ancestors may have used these skills for parsing complex environmental patterns, recognizing social structures, or even processing the temporal sequences in conspecific vocalizations, long before the emergence of syntactic language.

This perspective helps explain why our feelings of (un)grammaticality can be so context-dependent and variable. The neural circuitry underlying these feelings wasn't optimized specifically for language but was co-opted from more general-purpose cognitive mechanisms that had evolved for other adaptive functions. The mechanism – detecting violations of expected patterns – served our ancestors well in many domains of life, and language simply hijacked this pre-existing system.

The squirrel monkey study also emphasizes how these mechanisms are inherently predictive. The monkeys weren't simply memorizing specific sequences; they were extracting abstract patterns and using them to make predictions about novel stimuli. When these predictions were violated, the monkeys showed increased attention – the animal equivalent, perhaps, of our feeling of ungrammaticality. This aligns with our discussion of prediction error as a core component of grammaticality intuitions.

If a squirrel monkey can detect a violation in a tone sequence following the same abstract pattern as Hungarian vowel harmony, then our own feelings about well-formedness in language likely arose from deep cognitive foundations shared across primates. Our human language faculty didn't create sensitivity to dependencies; it inherited and specialized mechanisms that were already millions of years old.

This evolutionary perspective challenges the notion that grammar is inherently linguistic. Rather, grammar – and our feelings about it – may be better understood as the application of domain-general pattern-recognition abilities to the specific realm of language. What we experience as a feeling of ungrammaticality may be nothing more than the same surprise signal our ancestors felt when encountering any unexpected pattern in their environment, now directed at linguistic stimuli.

5.3.2 THE SOCIAL ROLE OF UNGRAMMATICALITY FEELINGS

If our sense of ungrammaticality arose before human language, what purpose might it have served? Evidence from non-human primates and other social mammals offers compelling clues. The studies on squirrel monkeys by Ravignani et al. (2013) along with others on rock hyraxes by Kershenbaum et al. (2012) reveal that syntactic processing capabilities may have evolved for social purposes rather than for language per se.

Rock hyraxes look like scruffy rodents but are actually more closely related to elephants and manatees, a biological surprise package about the size of a sturdy rabbit. These charming, vocal mammals live in colonies among rocky outcrops across Africa and the Middle East, where they spend their days sunbathing, foraging, and engaging in complex social interactions punctuated by elaborate songs. That such an unassuming creature produces vocalizations with sophisticated syntactic structure challenges our assumptions about which animals might possess complex communication systems.

In squirrel monkeys, the ability to detect abstract dependencies between non-adjacent elements suggests a cognitive capacity for recognizing patterns in vocal sequences. Similarly, hyraxes demonstrate complex syntactic structures with geographical dialects that correlate with social proximity. Both findings point to a potential social function of syntax detection: group recognition and cohesion.

What we experience as a feeling of ungrammaticality might have originated as a social signal – an alert to potential out-group communication. For our primate ancestors, detecting “incorrect” vocal patterns could help identify unfamiliar individuals or groups, triggering appropriate caution or interest. This mechanism would be particularly valuable in environments where visual identification was limited, such as dense forests or during low-light conditions.

The correlation between vocal similarity and geographical distance in rock hyraxes suggests that syntactic patterns may serve as markers of social familiarity. As Kershenbaum et al. (2012) observed, song syntax similarity degrades with distance, but only within a limited range corresponding to typical dispersal patterns. This pattern is consistent with a social signaling system that evolved to distinguish local group members from strangers.

Moreover, the mother-daughter vocal interactions in gibbons described by Koda et al. (2013) reveal how syntactic patterns might be socially transmitted. Their finding that “co-singing interactions between mothers and daughters may enhance vocal development” suggests that social learning plays a crucial role in acquiring group-specific syntactic patterns.

Our human sense of ungrammaticality, then, may be an evolutionary repurposing of a much older social recognition system. What was once a mechanism

for distinguishing in-group from out-group vocalizations has become a sophisticated tool for parsing linguistic structure. But the visceral nature of our reactions to ungrammatical sentences – that feeling of wrongness – may reflect this deep evolutionary history.

This evolutionary heritage may help explain the surprisingly emotional responses many people have to “grammatical errors” in modern contexts. Language policing – the practice of publicly correcting others’ speech or writing – rarely stems from genuine communication needs. Rather, it often functions as a social gatekeeping mechanism. The grammar pedant who bristles at a split infinitive or corrects someone’s *whom* usage may be unconsciously engaging an ancient social marking system, one that evolved to identify who belongs and who doesn’t.

The strong negative reactions to dialectal differences, casual speech patterns, or linguistic innovations reveal how deeply intertwined our sense of grammaticality is with social identity. When we feel that visceral discomfort at hearing someone “break the rules” of grammar, we may be experiencing the echo of an evolutionary adaptation that once helped our ancestors distinguish friend from stranger. In modern contexts, however, these instinctive reactions often serve to reinforce social hierarchies and exclude those who speak differently – suggesting that our feelings about grammaticality remain as much about social boundaries as they are about language itself.

5.4 GRAMMAR AND THE SACRED

The Tower of Babel story captured something true about language. God didn’t just divide people by making them speak differently – he divided them by making their grammars incompatible. Faced with patterns that felt profoundly wrong, they could no longer work together. Each group took their own grammatical patterns as natural and correct. The story tells us that grammatical differences aren’t mere technical distinctions but markers of fundamental social boundaries.

When sociologists talk about ‘the sacred’, they mean patterns that bind communities together and mark them off from others. Sacred patterns feel natural and right to insiders but strange or wrong to outsiders. They persist through enormous social investment but feel effortless to practitioners. They change, but slowly and from within.

Grammar works this way. German puts many verbs at the end of clauses. English doesn’t. Each pattern feels absolutely natural to its speakers. An English speaker encountering **I a walk take* recoils as spoken to by a creature from a swamp on the planet Dagobah. But German speakers do this naturally all day

long. The pattern that feels profane to one community is sacred to another.

Southern American English shows this clearly in its double modals. Speakers say *I might could help you* or *He should oughta be here by now*. These constructions are completely natural within their speech communities. Children acquire them without effort. Standard English speakers find them deeply ungrammatical – not just incorrect but somehow wrong. Both reactions reveal grammar's sacred character.

The profane appears when sacred patterns are violated. Consider verb complements in English. We say *decide to go* but *enjoy going*. Mix these patterns up (**decide going*, **enjoy to go*) and English speakers react viscerally. The violations feel not just wrong but almost unclean. This mirrors how sacred traditions treat violations of fundamental patterns.

The sacred status of grammatical patterns explains why some changes succeed while others fail. When English needed new ways to talk about possession, *his book* extended naturally to *Mary's book*. The innovation grew from existing patterns. But conscious attempts to change grammar usually fail. Invented pronouns like *ze* or *xe* never caught on. They violated the sacred character of grammatical patterns by imposing new forms from outside.

The sacred appears most clearly in how communities maintain their distinct grammars despite close contact. African American English keeps its aspectual system because these patterns carry deep community meaning. Take the remote past marker *been*, as in *She been married*. Insiders process this automatically. Outsiders systematically misunderstand. The patterns persist not through conscious effort but through their role in marking sacred social boundaries.

These boundaries work in subtle ways. Indian English speakers say *I am having two books* naturally. Many other English speakers find this wrong. Each group's patterns feel sacred through use. But 'sacred' here doesn't mean unchangeable. Consider how English developed the going-to future (*I'm going to leave*). The construction emerged gradually through speakers finding new ways to express meaning. Similar patterns appear across languages – French future tense from Latin *habere*, Mandarin perfective *-le* from 'finish'. Each change preserved sacred patterns while adapting to new needs.

Grammar differs from many sacred traditions in working below conscious awareness. French authorities campaign for linguistic purity. But basic French grammar patterns persist through implicit learning. Children master complex syntax without explicit teaching. The patterns become sacred through use rather than doctrine.

Grammar's sacred character explains why grammatical intuitions feel both universal and local. When English speakers judge **Book the him gave she* as

wrong, they're not consulting rules. Their reaction comes from deep patterns of community practice – patterns as automatic as any sacred tradition. But these patterns vary between communities because they emerge from shared histories of use.

The story of Babel got this right. Grammar divides us precisely because it unites us. The patterns that bind one community together mark boundaries with others. We experience violations of our grammatical patterns as profane because these patterns provide the sacred ground of shared meaning. Grammar isn't sacred in any mystical sense. But it takes on sacred characteristics because it enables the fundamental human task of making meaning together.

5.5 BARRETT'S THEORY OF CONSTRUCTED EMOTION: IMPLICATIONS FOR GRAMMATICALITY

5.5.1 OVERVIEW OF THE THEORY OF CONSTRUCTED EMOTION

Lisa Feldman Barrett's theory of constructed emotion offers a paradigm shift in how we understand emotions – and by extension, how we might understand other psychological phenomena, including our sense of grammaticality. Just as Barrett argues that emotions aren't innate, universal categories but rather constructed experiences, we might consider whether our intuitions about grammaticality are similarly constructed rather than innate or universal.

Barrett's theory posits that emotions emerge from more basic psychological ingredients: interoception (the brain's representation of sensations from the body), concepts (knowledge of the world based on past experience), and affect (feelings of pleasantness/unpleasantness and arousal/calmness). The brain, she argues, is constantly predicting and checking its predictions against sensory input, using concepts to categorize sensations and create meaningful experiences – including emotions.

Consider how this might apply to grammaticality. When we encounter a sentence, our brain isn't passively receiving input. Instead, it's actively predicting what's coming next, based on our past linguistic experiences (our 'concepts' of language). When we encounter something unexpected – say, a sentence like **The child seems sleeping* – our prediction is violated. This violation might generate an affective response (perhaps a feeling of unpleasantness or increased arousal), which we then categorize using our concept of 'ungrammaticality'.

This view suggests that our sense of what's grammatical might be intimately tied to our broader cognitive and affective processes, constructed in the moment based on our linguistic experiences and current context.

5.5.2 KEY CONCEPTS: INTEROCEPTION, CONCEPTS, AND AFFECT

To fully appreciate the implications of Barrett's theory for our understanding of grammaticality, we need to delve deeper into its key components: interoception, concepts, and affect.

Interoception refers to the brain's representation of the body's internal state. It's not just about conscious awareness of heartbeats or digestion, but a constant, usually unconscious process of monitoring and predicting the body's needs. In the context of language processing, we might consider how our physiological state influences our perception of language. Are we more likely to judge a sentence as ungrammatical when we're tired or stressed? Does the effort of processing a complex sentence structure create bodily sensations that we interpret as signs of ungrammaticality?

Concepts, in Barrett's framework, aren't static, dictionary-like definitions, but flexible, context-dependent predictions about what sensations mean. They're built from past experiences and constantly updated. Applied to grammar, this suggests our concepts of 'correct' language are shaped by every linguistic interaction we've had, from formal education to casual conversation. When we judge a sentence like '*The data is conclusive*' (some people prefer *are*), we're not consulting an internal rulebook, but rapidly constructing a prediction based on our accumulated experiences with similar phrases.

Affect is the basic sense of feeling good or bad, calm or agitated. It's the currency of allostasis – our brain's process of anticipating and meeting the body's needs. In grammaticality judgments, affect might manifest as the subtle feelings of rightness or wrongness we experience when reading or hearing language. The discomfort we feel at a 'garden path' sentence like *The horse raced past the barn fell* might be a form of affective response, signaling the need to reallocate cognitive resources.

Together, these components form a dynamic system. Our brain predicts what sensations (including linguistic input) mean for our body budget, categorizes these sensations using concepts built from past experience, and generates affect to guide action – all in service of maintaining allostasis. This process, Barrett argues, constructs our emotional experiences. And perhaps, it also constructs our experience of grammaticality.

5.5.3 PARALLELS BETWEEN EMOTION CONSTRUCTION AND GRAMMATICALITY JUDGMENTS

The parallels between Barrett's theory of constructed emotion and our experience of grammaticality are striking and potentially illuminating.

First, consider the variability of both emotions and grammaticality judg-

ments. Just as the experience of ‘fear’ can vary widely across individuals and contexts, so too can judgments of grammaticality. What feels ungrammatical to a copy editor might feel perfectly natural to a casual speaker. This variability is difficult to explain if we assume innate, universal categories of either emotions or grammatical rules, but it aligns well with a constructionist view.

Second, both emotions and grammaticality judgments involve a blending of ‘bottom-up’ sensory input and ‘top-down’ conceptual knowledge. In emotion, we interpret bodily sensations through the lens of our emotional concepts. Similarly, in language processing, we interpret strings of words through the lens of our grammatical knowledge. A sentence like *The old man the phones* feels ungrammatical at first because our initial prediction (based on frequent patterns) is violated, but can feel grammatical once we apply different conceptual knowledge.

Third, both processes appear to be deeply contextual. The same physiological state might be conceptualized as ‘fear’ or ‘excitement’ depending on the context; similarly, the same linguistic construction might be judged grammatical or ungrammatical depending on the context. Consider how our judgments of sentences like *Who do you wanna go to the movies with?* can shift depending on whether we’re in a formal or casual setting.

Fourth, both emotional and grammatical concepts seem to be learned and culturally influenced. Just as emotional concepts can vary across cultures, so too can concepts of grammaticality. What’s considered ungrammatical in Standard American English might be perfectly grammatical in another dialect, for instance.

Finally, both experiences of emotions and grammaticality judgments seem to involve an element of affect – a feeling of goodness or badness, rightness or wrongness. The ‘gut feeling’ we get about whether a sentence is grammatical might be analogous to the core affect that Barrett posits as a basic ingredient of emotion.

These parallels suggest that our experience of grammaticality might be constructed in much the same way as our experience of emotion: through a dynamic interplay of sensory input, conceptual knowledge, and affective feeling, all in service of making meaning and guiding action in our linguistic world.

5.5.4 REINTERPRETING GRAMMATICAL INTUITIONS THROUGH A CONSTRUCTIONIST LENS

If we take seriously the idea that grammaticality judgments are constructed in a manner similar to emotions, we’re led to some intriguing reinterpretations of familiar linguistic phenomena.

Consider garden path sentences like *The horse raced past the barn fell*. The traditional view might explain our initial difficulty with such sentences in terms of parsing mechanisms or structural ambiguity. A constructionist view, however, might see this as a case of prediction error. Our brain, drawing on past linguistic experience, makes a strong prediction about the structure of the sentence after encountering *The horse raced past the barn*. When *fell* appears, violating this prediction, we experience a momentary increase in arousal and perhaps a feeling of unpleasantness – core affect that we then conceptualize as ‘ungrammaticality’.

Or consider our varying intuitions about sentences like *The data is conclusive* versus *The data are conclusive*. Rather than appealing to competing grammatical rules, we might view this as a case of competing predictions, each associated with different conceptual knowledge (formal academic writing vs. everyday usage) and perhaps different affective states (the smugness of formal correctness vs. the comfort of colloquial familiarity).

The phenomenon of SYNTACTIC SATIATION – where repeatedly reading an ungrammatical sentence makes it begin to feel grammatical – takes on new significance in this light. Rather than a fatigue of grammatical judgment faculties, we might see this as a rapid updating of our predictive model. Each exposure adjusts our concept of what’s linguistically possible, gradually shifting our affective response from ‘wrong’ to ‘right’.

Even the intuitions of trained linguists, often treated as more reliable than those of naive speakers, might be reinterpreted. Perhaps what distinguishes the linguist’s judgments isn’t access to a more accurate internal grammar, but a richer set of linguistic concepts allowing for more nuanced prediction and categorization of language input.

This constructionist view also offers a new perspective on linguistic prescriptivism. The strong feelings that prescriptivists have about ‘correct’ language use might be understood not as defense of innate grammatical rules, but as affective responses constructed from deeply ingrained concepts about language, social status, and cognitive sophistication.

By reframing grammatical intuitions as constructed experiences rather than readouts of an internal rule system, we open up new ways of understanding the flexibility, context-sensitivity, and sometimes puzzling inconsistency of our linguistic judgments.

5.6 A NEUROSCIENTIFIC PERSPECTIVE

Ever catch yourself finishing someone else’s sentence? That’s your brain doing what it does best – making predictions. Our brains aren’t just passive receivers of language; they’re constantly trying to guess what’s coming next. When those

guesses are wrong, it's like a little alarm goes off in our heads. This predictive processing is key to understanding how our brains handle grammar.

Here's a fun example. Read this sentence:

- (13) *How many animals of each kind did Moses take on the Ark?*

Did you spot the error? It was Noah, not Moses. But don't feel bad if you missed it – most people do. That's because your brain, making predictions based on the words “animals”, “Ark”, and a biblical figure, fills in the blanks so efficiently that it often overrides your actual knowledge. It's like your brain is saying, “Close enough!”

This predictive power explains why we can breeze past some pretty glaring grammatical errors. Take this gem from *Fifty Shades of Grey*:

- (14) * *Pulling off his boxer briefs, his erection springs free. Holy cow!*

Grammatically, this is a howler. *Pulling off his boxer briefs* should describe what the subject of the sentence is doing, but unless this is a very different kind of novel, *his erection* isn't doing any undressing. Yet most readers don't even blink. The broader context creates such strong expectations that our brains just fill in the blanks and move on.

Our brains don't just passively receive language input – they're constantly making predictions about what's coming next. When those predictions are violated, it generates a neural surprise signal. This predictive processing framework offers a powerful lens for understanding how our brains handle grammar.

Recent research (doi.org/10.1038/s41586-024-07973-1) has given us an unprecedented look at how this process works at the level of individual neurons. In a groundbreaking study, scientists recorded the activity of single neurons in the brains of epilepsy patients who had electrodes implanted for clinical reasons. They found that neurons in the hippocampus and entorhinal cortex – areas crucial for memory and navigation – gradually changed their firing patterns to represent the structure of a sequence of images the patients were viewing.

This neuronal representation formed rapidly and without any explicit instructions. The patients weren't told to look for patterns; their brains just naturally extracted the underlying structure. This is remarkably similar to how we learn grammar – not through memorizing rules, but by implicitly picking up on patterns in the language we hear.

Even more intriguingly, when the researchers looked at the combined activity of many neurons, they could recover the entire structure of the image sequence. It's as if the brain had built a mental map of the sequence, much like the “cognitive maps” these same brain areas create of physical spaces. This suggests that

our brains might represent the structure of language – its grammar – in a similar way to how we represent the layout of a city or the order of events in a story.

This neuronal representation wasn't just a passive record of what had happened. It was predictive, encoding information about what was likely to come next in the sequence. This aligns perfectly with the predictive processing framework we discussed earlier. Your brain isn't just recognizing grammatical sentences; it's actively predicting how those sentences will unfold.

The researchers also found evidence of “replay” – the neurons would spontaneously reactivate in the same patterns they'd followed during the initial sequence, but speeded up. This replay happened during breaks, suggesting it might be a way for the brain to consolidate what it had learned. It's not hard to imagine similar processes happening as we internalize the patterns of our native language.

Let's bring this back to grammar. When you encounter a sentence like *The old man the phones*, your brain initially predicts that *man* is a noun, based on the familiar pattern of *The old man...* But as soon as you hit *the boats*, that prediction is violated. Your neurons are likely firing in patterns that reflect this violation, similar to the “surprise” signals seen in the image sequence study. Your brain then has to rapidly revise its interpretation, now understanding *man* as a verb.

Neuroscientists have actually measured these prediction processes in the brain. When we encounter unexpected words or structures, our brains generate distinct patterns of electrical activity. There are two main ones to know about:

- The N400: Think of this as your brain's “semantic speed bump”. It happens about 400 milliseconds after you encounter a word that doesn't fit semantically. If I say *I like my coffee with cream and socks*, your brain hits that speed bump at “socks”.
- The P600: This is more like a “syntactic pothole”. It occurs about 600 milliseconds after you hit a grammatical snag. You'd get a big P600 if you read *The man bit dog the*.

These brain blips get bigger the more surprising the word or structure is. A mild surprise is like a gentle speed bump, while a major violation is like hitting a pothole at 60 mph.

But here's where it gets really interesting: context can totally change how our brains respond. That Moses sentence earlier? In casual conversation, your brain might just sail right past it without triggering an N400 response. But in a psychology experiment, where you're primed to expect tricky questions and are paying

close attention, you're more likely to notice the error. In that case, your brain would probably generate an N400 response when it encounters *Moses*.

This flexibility ties into a broader idea in neuroscience: that our experiences, including our sense of what's grammatical, might be actively constructed by our brains. Lisa Feldman Barrett proposed this for emotions – that they're not fixed categories, but experiences our brains construct on the fly from more basic ingredients.

Kent Berridge extended this idea to our sense of good and bad – what neuroscientists call “valence”. He challenged the idea that specific brain regions are permanently dedicated to positive or negative feelings. Instead, he suggested that neural systems can have multiple “affective modes”, capable of producing both positive and negative feelings depending on the context.

We can apply this same thinking to grammaticality. When we encounter a sentence, we're not just mechanically applying rules. We're rapidly constructing an experience based on:

- How we're feeling physically (Are we tired? Stressed?)
- What we know about language
- Our gut feeling of “rightness” or “wrongness”

This explains why our sense of what's grammatical can be so flexible. Just like how the same physical sensation might feel like excitement or anxiety depending on the context, the same sentence structure might feel grammatical or ungrammatical depending on the situation.

Take a sentence like *That's the person who I saw*. In formal writing, you might get a twinge of wrongness – shouldn't it be “whom”? But in casual conversation, it'll likely sail right by. Your brain, in different contexts, is in different “grammatical modes”.

This view also sheds light on the phenomenon of syntactic satiation – where if you read an “ungrammatical” sentence enough times, it starts to feel okay. You're not wearing out your grammar module; you're updating your brain's predictive model. What was once surprising becomes expected.

It even explains why linguists often have different grammatical intuitions than the general public. It's not that linguists have access to a more accurate internal grammar. They've just developed a richer set of linguistic concepts, allowing for more nuanced predictions and categorizations.

This neuroscientific perspective invites us to see grammaticality not as a binary property of sentences, but as a dynamic, constructed experience. It arises from the interplay of prediction, sensation, and conceptualization. This view

embraces the messiness and flexibility of real-world language use, while still grounding our intuitions in the fundamental workings of our predictive brains.

Berridge's work provides some concrete examples of this flexibility. He found that stimulating neurons in the central amygdala – an area often associated with fear – could actually enhance positive motivation for rewards like cocaine or sugar. Similarly, in the nucleus accumbens, the same neural populations could drive both positive (eating) and negative (defensive) behaviors depending on the environment.

If we apply this to grammaticality, it suggests our neural circuits for language processing might be equally flexible. The same brain areas that spark a sense of wrongness when you say *Me and him went to the store* might, in a different context, give you a pleasant frisson when a poet breaks the rules for artistic effect.

This doesn't mean grammatical rules don't exist or don't matter. But it does suggest that our experience of those rules – our feeling of grammaticality – is far more dynamic and context-dependent than we might have thought. It's not about matching input to a stored set of rules, but about constructing a linguistic experience that best fits our current model of the world.

In this light, the feeling of ungrammaticality becomes less about detecting a rule violation and more about experiencing a prediction error. And just as our brains can rapidly update their predictions in other domains, they can also update their linguistic predictions. This is how new constructions gradually become acceptable, and how the language evolves over time.

This neuroscientific view offers a richer, more nuanced understanding of grammaticality. It helps explain why our intuitions can sometimes feel fuzzy or contradictory, why context matters so much, and why there's often not a clear line between "grammatical" and "ungrammatical". It suggests that our sense of grammar isn't a fixed, innate system, but a dynamic, constructed experience – one that's intimately tied to our broader cognitive and emotional processes.

5.7 FORM-MEANING MISMATCH

Open a dictionary to any random page to check the definition of a word, and you'll rarely find one – instead, you'll typically be presented with multiple meanings. Take, for example, the word *bank*. The *Oxford English Dictionary* lists three distinct entries for *bank* as a noun alone: 'the edge of a river', 'an institution that invests money', and 'a set or array of similar items'. Yet this only scratches the surface of its semantic complexity. Even within these broad categories, numerous sub-senses exist. The 'edge of a river' definition, for instance, is one of two major senses, alongside the earlier meaning of 'a mound or slope'. Together, these two broad senses encompass a total of twelve sub-senses.

The same multiplicity of meanings – or POLYSEMY – can be found in grammar. In my model, I claim that, at its most basic, a grammatical construction is one with an accepted form–meaning pairing. But I also note that a grammatical construction may evolve additional accepted form–meaning pairings. But this can be counter-intuitive.

Both in words and grammar, our intuition often suggests to us that everything should be monosemous – it should have only a single meaning. The other meanings, the scolds will tell us, are errors.

Take the case of *can*. Far too many children have been tormented by teachers who won't accept *can I go to the washroom* but insist on *may*. This stems from the teachers' wrongheaded belief that *can* has the meaning of ability, while *may* has the meaning of permission. Any dictionary will belie this fallacious position. As will some casual reading or observation of conversations. But the intuitive pull of the fallacy of monosemy is strong.

When it comes to grammar, belief in monosemy can be even more entrenched, likely because of the names we give to grammatical systems. Terms like THE PAST TENSE, SINGULAR, *definite*, etc. give us tunnel vision about the possible meanings of the forms in question. Clearly, the past tense should be for past time, singular nouns should denote individuals, and a definite NP should be specific and identifiable. But grammatical systems, like words, tend to be polysemous.

Consider the past tense. While its primary function is indeed to locate events in the past, it serves several other purposes in English:

1. Hypothetical situations: *If I won the lottery, I would buy a house.*
2. Politeness: *I wanted to ask you a favour.*
3. Reported speech: *She said she was busy.*
4. Narrative style in storytelling: *Once upon a time, there was a princess...*

Each of these uses demonstrates that the past tense form carries meanings beyond simple past time reference, illustrating the mismatch between its name and its full range of functions.

Similarly, definiteness, typically associated with the definite article 'the', exhibits a range of meanings:

1. Unique reference: *The sun is shining.*
2. Shared knowledge: *Did you feed the cat?* (assuming both speakers know which cat)

3. Generic reference: *The lion is a majestic animal.*
4. Superlatives: *This is the best restaurant in town.*
5. Unknown characters in anecdotes: *I saw this guy.*

Here, we see that definiteness encompasses not just a single concept of ‘being definite’, but a range of semantic and pragmatic functions related to specificity, shared knowledge, and even generality.

Innocent people can thus be lulled into persuading themselves that something is ungrammatical even when it doesn’t feel ungrammatical. At least, not initially.

5.8 ONE INTUITION OR MANY

Joshua Knobe (forthcoming from *Ergo*) argues that not only do different people have different intuitions but that individuals have conflicting intuitions.

Becoming (un)grammatical

In the late 13th century, *verray* (*very*) began its journey in the English language. It was an adjective with a clear mission: to denote what was ‘true,’ ‘real,’ and ‘genuine.’ Coming to English from the Anglo-French *verrai*, it was rooted in the Latin *verus*, meaning ‘true,’ the source of words like *verify*, *verdict*, and *warlock* (truly!).

When *very* was getting its start, *much* was already an English veteran, having survived several centuries and various incarnations. It likely started out in Proto-Germanic as something like *mekilaz* ‘great’, related to words like *majesty*, *major*, and *magnify*. But eventually, in Old English, it became *micele*, meaning ‘great in size’, as in (1) from Genesis 1:16.

- (1) *God geworhte twá micele leóht.*
 God created two great lights .
 ‘God made two great lights.’

This use of *much* is archaic, but it led to the Old- and Modern-English sense of ‘great in amount or extent’, as in (2).

- (2) *Oftor micle*
 oftener much
 ‘much more often’

Verray was going through its own development. The original meaning is still with us in expressions like *the very heart of the matter*, but as the years rolled on, *verray* explored new territories of meaning. By the late 14th century, it had expanded to signify ‘actual’ and ‘sheer’, gradually branching into a new role as an intensifier in the mid-15th century. This was a significant shift; *verray* was stepping into the realm of adverbs.

In this new role, *verray* began to encroach on the territory of *much*. In 1481, a published text could say,

On they took their way: on the right side, they left Jaffa, and through a great plain and an even way, they came into the city of Lde, where the body of the glorious martyr Saint George lies. In his honor, Justinian, who was Emperor of Rome, made there a very fair and rich church. But when the Turks heard news that our men were coming, they knocked it down and burned the timber work, which was very large. They feared that our pilgrims would take them to make engines for casting stones and castles to assault with. Then our barons heard say that nearby was a very noble city named Rames.

Except that in Middle English, it was like this:

on they toke their way: on the right side they lefte iaphe and by a grete playn and euen waye they cam in to the Cyte of lde where the bodye of the glorious martir seynt gorge lyeth: in thonoure of whom iustynyen that was Emperour of rome made there a moche fair chirche and ryche but whan the turkes herde tydynges that our men cam they bete it down and brente the tymbre werke whiche was moche grete for they doubted that oure pilgrims shold take them for to make engyns to caste stones: and castellis to assaile with thenne herde our barons saye that ther fas by was a moche noble Cyte named rames¹

Today, **much fair*, **much great*, and **much noble* are ungrammatical. *Much* can still modify adjectives, but it strongly prefers the comparative forms like *much fairer*, *much greater*, and *much more noble*, avoiding most plain-form adjectives; **much great* is no longer quite so great. *Very* has developed complementary tendencies, such that it's happy modifying a plain-form adjective, as in *very strange*, but modifying a comparative seems **very stranger*.

What might have happened, though, if *very* hadn't appeared? Perhaps we would still be using **much great*. But appear it did, and from the 1470s until the 1530s, all it did, it seems, was to usurp more and more of *much*'s turf. From the mid 1500s on, though, the use of *much* started to recover as the two words specialized. In Modern English, both *very* and *much* are ubiquitous. Occurring

¹Caxton, William, 1481. Here begynneth the boke intituled Eracles, and also of Godefrey of Boloyn the whiche speketh of the conquest of the holy londe of Iherusalem ... *Early English Books Online*.

roughly a thousand times per million words, *very* is the 105th most frequent word in American English, with *much* in 148th spot.² Still, like the poor child who makes good, or the team that comes from behind to win, *very* had a hard row to hoe.

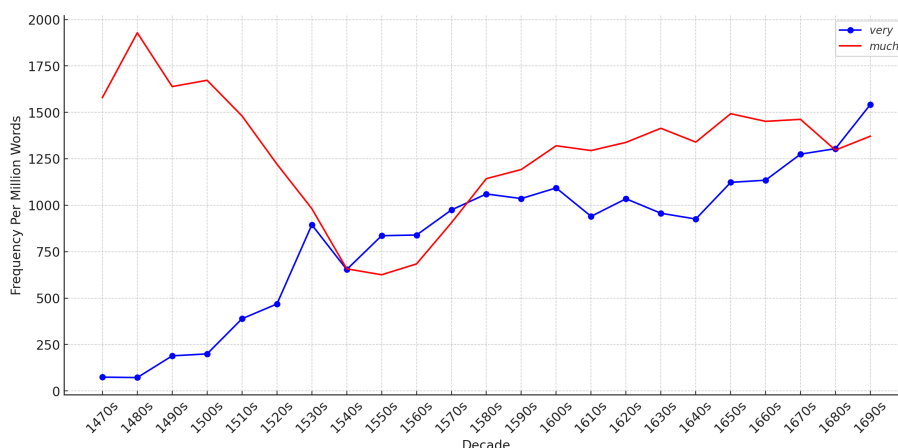


Figure 6.1. The frequency of *very* and *much* (in various spellings) in the *Early English Books Online* corpus, from the 1470s to the 1690s, shown per million words.

This is the story of *much fair*'s journey from being perfectly normal in Middle English to **much* ungrammatical today. But it's also a story about the very subtle and puzzling predilections that *much* has accumulated over the last thousand years or so. Why can we say *I don't know much about it*, while **I know much about it* is bad? And why is it saved when *much* is modified by *so* but not by *very*? Why can *much* almost never modify plain-form adverbs and yet *much too ...* is impeccable? How did all these *much* constructions become ungrammatical?

6.1 PAS DE ⟨ne ... pas⟩

In French, the negative has (or had) two parts: *ne* and *pas*, so that *it doesn't work that way* would be *Ça ne marche pas comme ça*. But it wasn't always so. In the early days of Old French, a simple placement of *ne* before the verb was all that was required to negate a sentence. Its use was straightforward and unadorned.

Today, though, **Ça ne marche comme ça* is totally ungrammatical, and it's also a great example of the life-course of a grammatical construction. The development of the clausal negation in French, particularly the emergence and transformation of *pas*, follow a pattern called Jespersen's Cycle. This cycle, conceptualized by the Danish linguist Otto Jespersen in 1917, outlines a three-stage pro-

²COCA

gression in the treatment of negation within languages.

The history of negative expressions in various languages makes us witness the following curious fluctuation: the original negative adverb is first weakened, then found insufficient and therefore strengthened, generally through some additional word, and this in its turn may be felt as the negative proper and may then in course of time be subject to the same development as the original word. (Jespersen, 1917, p. 4)

The early days of Old French, when a simple *ne* would do, represent the first stage of Jespersen's Cycle. As the centuries passed, though, the power of *ne* to convey negation began to diminish. It was as if the speakers of French sought greater emphasis, a stronger affirmation of nothingness. This need for reinforcement led to the second stage of Jespersen's Cycle: the birth of double negation.

Words like *rien* 'nothing', *point* 'at all', *personne* 'nobody', and of course *pas*, all of which had previously just intensified *ne*, began to demand to be heard. Interestingly, these words originally had concrete, quotidian meanings: *point* was 'point', *personne* meant *person*, and *pas* was literally 'step'. The phrase *Je ne marche pas* was akin to saying 'I won't walk a step'.

By Middle French (14th to 17th centuries), *ne* had weakened to the extent that its use as a "single negative" was becoming increasingly ungrammatical. As the language developed into Early Modern French (17th to 18th centuries), *ne* continued to weaken, and folks began to drop it: *Je ne marche pas* had started to become *Je marche pas*. The once-understudy *pas* has stepped up to eclipse *ne* in its role as the primary negator. This change marks the third and final stage of Jespersen's Cycle, where the original negator is overshadowed or even rendered obsolete.

A similar progression happens with slurs in what has been called the EUPHEMISM TREADMILL. A word like *stupid* can start out as a non-judgemental description of someone in a stupor, but eventually becomes insulting. As it becomes a meaner word, a kinder alternative is sought, a word like *retarded*, meaning simply slow. But as children get wind of this new use, they start using it to cruel and insulting purpose, and it too must be abandoned (S. H. Taylor, 1974). Although the euphemism treadmill pertains to the meaning of individual words and Jespersen's cycle to the grammatical system of negation, similar processes underlie both.

Jespersen's third stage isn't yet complete for *ne*, but it's well along in its trajectory. In the 20th and 21st centuries, the trend towards *ne* dropping in everyday spoken French was increasingly prevalent, especially in informal contexts. It's

observed across various dialects of French, including those spoken in France, Canada, and other Francophone regions. The formal rules of grammar laid down by the Académie Française still prescribe the use of *ne* for negation, and these tend to be followed in written and formal spoken French, but it's often missing from everyday spoken language.

The normality of double negatives

Despite their being stigmatized in Standard English, double negatives are actually quite common in the world's languages, to the extent that sometimes "single negatives" are out-and-out ungrammatical or impossible. For example, in Italian, the way to express "I never go" is *non vado mai*, which more or less translates as "I don't never go." **Vado mai* without *non* ain't no good.

The case of double negatives is a case where our sense of ungrammaticality may result from having been explicitly told that double negatives are both ungrammatical and illogical. (Illogical? Yeah, right!).

6.1.1 POLARITY ITEMS

If *pas* 'step' seems like a strange way to intensify negation, consider the English expressions, *I was so tired that I couldn't move* and then think about how that changes when you add *a muscle*. *I couldn't move a muscle* seems like a more complete inability to move, ruling out not just major limb motion but even single muscle contractions. This is the sense in which *pas* contributes to the negative meaning: it's not simply that I can't wander down to the shop – I can't even put one foot in front of the other. The more *pas* was used like this, the more it was bleached of its meaning, becoming something much closer to a pure negator.

A muscle, in contrast, still has much of its basic meaning, but consider saying, ³*I was so energized that I could move a muscle*. Grammatical, perhaps, but utterly bizarre. Other expressions of this sort are *drink a drop*, *sleep a wink*, *lift a finger*, *give a damn*, *spend a red cent*, *budge an inch*, *bat an eyelash*, *hold a candle to*, *miss a beat*, *show a spark of decency*, and *hurt a fly* (Israel, 2011, p. 24).

Israel (2011, p. 3) points out that

these are not the sorts of facts one is likely to notice about a language, but they are remarkable nonetheless. One would expect that anything one could affirm, one could also deny, and that anything one could deny, one could also affirm. But polarity items are subject to special constraints, the violation of which results in unexpectedly

unacceptable sentences.

This brings us back to *much*. In these constructions, *pas* and *a muscle* are what linguists call NEGATIVE POLARITY ITEMS or NPIs for short, or ... NEGATIVELY-ORIENTED POLARITY-SENSITIVE ITEMS for long (Huddleston & Pullum [CGEL], 2002). And so is *much*. Among the most common phrases with *much* are *much time*, *much money*, *much fun*, *much attention*, and *much trouble*. But while you hear *I don't really have much time* all the time, nobody says, ³*I really have much time*.

At first blush, it seems odd that *pas*, *a muscle*, and *much* would all be negative polarity items. A step or a muscle are the smallest possible ways to walk or move – the next closest thing to not walking or moving – which might explain their polarity preferences; being so energized that you can move a muscle isn't being very energized at all. *Much*, though, is all the way at the other end of the spectrum. It's great, majestic, magnitudinous, and magnificent. What might pull it down to negative levels?

Israel (2011, p. 82) points out that degree modifiers like *much* apply not just to positively oriented adjectives like *bigger*, but also to negatively oriented ones like *smaller*. If something is much smaller than something else, it

Consider, then, a weak intensifier like *pretty* as in *that's pretty good*. If you're pretty smart, you're smart but hardly brilliant. A pretty good try is one with some effort behind it, but probably not enough to succeed. And telling your romantic interest that you love them pretty much is never a winning strategy. *Pretty* is at the opposite end of the intensification scale to *much*, and it's also a positive polarity item – these things too exist – at the opposite end of the polarity scale.

In 2023, I was writing a paper on *much*, arguing that *The Cambridge grammar of the English language* had made an error in saying it was sometimes an adverb, and I sent the paper to John Payne for comment. John had worked with Rodney Huddleston and Geoff Pullum on the book, and he was one of the listed authors on the chapter that made the claim about *much*. As we went back and forth, looking at things from a variety of perspectives, John noticed something I hadn't, something perhaps nobody had before: *much* is an NPI in many constructions (like *not much time* or *I don't drive much*) but not in adjective phrases like *much bigger*. In other words, a positive assertion that *it's much better* is just as good as the negative *it's not much better*, while as we've seen *I don't drive much* is much better than **I drive much*.

And so I started looking at *much*'s behaviour at different points in the past, trying to see if it had always been like that. I went to the *Early English Books*

Online corpus, and there, in a random sample of 100 uses of *moche sorowe* ‘great sorrow’ from the late 1400s and early 1500s, found only a single example was in a non-affirmative environment. A typical positive example would be (3).³

- (3) *his peple for hym made moche sorowe.*
 his people for him made great sorrow.
 ‘His people caused him great sorrow.’

Now, you would certainly be correct to observe that such a situation doesn’t sound particularly positive, but it’s not negative in the sense that linguists have in mind when they speak about NPIs. Even in 1680, the last year of the corpus, only 16% of the occurrences of *much time* are in non-affirmative environments, certainly not what you would expect of an NPI. I switched then to the *Corpus of Historical American English*, which starts 140 years later, in 1820. In that decade, only 8% of my sample was in a non-affirmative environment. By the 1830s, 15% were, 28% in the 1840s, 27% in the 1850s, and 31% in the 1860s. Clearly, it was in the first half of the 19th century that *much* became an NPI.

Unfortunately, I have no idea why that happened then, but it suggests that people developed a sense that *much* was better in non-affirmative environments, without developing the feeling that it was ungrammatical there. Like a path in a forest that becomes more defined and specialized as more people walk it, *much* in non-affirmative contexts became more specialized simply through the natural progression of language use. These kinds of changes tend to happen the way one goes bankrupt, according to Hemingway, slowly and then suddenly. It’s like the “S-shaped curve” in Figure 6.2.

But we can say something else about how these develop because very similar NPIs develop in very different languages. This isn’t some kind of structural similarity that might point to a “universal grammar” but rather semantic similarities that point to a role for semantics in understanding grammaticality.

As the founder of modern linguistics, Saussure (1916) pointed out, in all languages we find both centripetal and centrifugal tendencies. The need to communicate creates the centripetal force of linguistic homogenisation, which is balanced by a centrifugal force of differentiation, driven by an impetus for local identity, by the need to exchange information without having the people from the next village or the other religious community overhear, by the desire of each new generation to mark itself off from the one before. Joseph,
 2015

³1515. The cronycles of Englund. *Early English books online*.

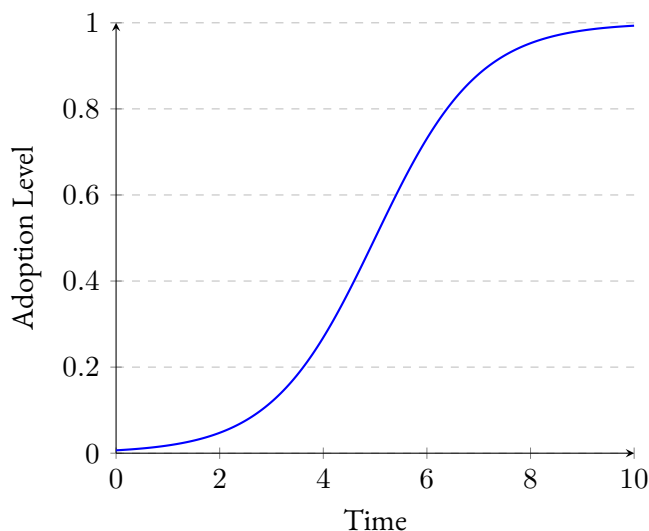


Figure 6.2. The typical adoption curve of language change.

6.2 WORDS

“There dwelled Abram in wealth and in frith.” This is translated from Middle English, keeping only the Middle English word *frith* meaning ‘peace’. How does it register with modern readers given that its use isn’t ungrammatical, but the word is entirely unfamiliar?

6.3 ASSERTION AND PRESUPPOSITION

A new sound, word, or construction comes into being by being used. One person simply **PRESUPPOSES** that it will be understood by others, and, generally, it is, after which it can be taken as common knowledge. Even if it isn’t fully clear, often we accommodate such innovations. In other words, it’s established just by being used.

In this way, it’s like a declaration of marriage, money, promises and apologies, or a pass code on a debit card.

In short, the continual attempt, by each of us, to determine our language’s nature is the force that shapes it over time. Everyone acts as if the rules are already common knowledge, and then – the strangest part! – they gradually try to work out what this common knowledge is. This is strikingly different to determining the position of a planet, or the weight of a large cake, where there is a fact of the matter totally separate from our beliefs about it, which can be re-

constructed by investigation through telescopes or scales. (Cohn-Gordon, 2024)

But we don't accommodate just anything.

Infants are born with an ability to distinguish between different accents. As you might expect, that ability isn't purely theoretical. By five or six months old, infants show a preference for people who speak with their native accent, and five-year-olds are more likely to want to be friends with children who have the same accent as theirs. This preference exists even if they can fully understand children with different accents (Kinzler et al., 2007). As they age, this kind of bias remains. For example, children are more likely to trust and learn from adults who speak with a similar accent to their own. This is demonstrated by their tendency to believe demonstrations of how to use new objects when those demonstrations are performed by adults with familiar accents (Kinzler et al., 2011).

And there's every reason to believe that's how it goes with grammar too. When a child hears an unfamiliar construction from an adult who shares their native language, they should be likely to look for a motivation for that construction. When they've heard it only from a non-native speaker, they're very likely to understand it, but they should also be very likely to see it as an error rather than as an innovation.

"This 'how-possibly' story for the evolution of vagueness is particularly interesting given previous game theoretic results due to Lipman [2009] indicating that, generally, vagueness is inefficient in common-interest signaling situations. The implication is that, while vagueness itself may not be evolutionarily beneficial, it may nonetheless persist in language because learning strategies that are themselves evolutionarily beneficial lead to vagueness" (O'Connor, 2014, p. 708).

What's ungrammatical

- There's a common way to say something, a construction that pairs meaning with form.
- When something fails to follow this, we assume some new function.
- If no such function can be found, it feels ungrammatical.

7.1 *I GO THERE YESTERDAY*

I go there yesterday sets up a clash between the present tense of the verb, which signals non-past time, and the past-time meaning of the modifier *yesterday*.

- (1) SON: *Hi, Mom.*
 MOM: *Hi, Honey! Where were you?*
 SON: *The new bookstore.*
 MOM: **Oh, I go there yesterday.*

Our default assumption is that Mom isn't just messing up here. We attribute intentionality to her choice of present-tense *go* + *yesterday*. Assuming a clash like this is intended, though, sends Son (and us) on a search for a reason, a quest that hits a dead end, as no justification presents itself. And so we register the underlined clause as ungrammatical.

But could *I go there yesterday* ever be grammatical?

- (2) SON: *Hi, Mom.*
 MOM: *Hi, Honey! Where were you?*
 SON: *The new bookstore.*
 MOM: *Oh, you won't believe this. So, I go there yesterday, and...*

In this iteration, Mom offers a vital interpretive key: she is initiating a story. The present-tense form *go*, clashing with the past-oriented *yesterday*, is now re-conceptualized as a story-telling technique. We're able to see an – likely *the* – intended purpose behind it: a sense of immediacy and involvement in the storytelling, making what was sharply ungrammatical in (1) indisputably grammatical in (2).

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As Old French became Middle French in the fourteenth century and gradually evolved into Modern French, the present perfect was becoming a past tense in a striking example of various pressures and motivations allowing the ungrammatical to become grammatical.

Until the fourteenth century, French and English had matching simple-past and present-perfect tenses. The simple past looked like this: *Je changai mon avis* 'I changed my mind', and the present perfect like this: *J'ai changé mon avis*. 'I have changed my mind'. Though these mean basically the same thing, *changai* 'changed' is a past tense while the *ai* of *J'ai changé* is the present 'have'. That means *J'ai changé* is a present tense, albeit a retrospective one. Like the English present perfect, it's anchored in the now but encompasses things past. This makes it very compatible with a word like *recently*, which covers the present and not-too-distant past.

A word like *hier* 'yesterday', though, whose meaning is completely separated from the here and now sets up a clash. In Old French, **J'ai changé mon avis hier* meant both present (*ai*) and not present (*hier*), making it as ungrammatical as **I have changed my mind yesterday* in English.

To say that the Old-French present perfect turned into a past tense, then, is to say that it lost that 'present' meaning, becoming fully compatible with a word like *hier* 'yesterday'. In Modern French it's perfectly grammatical to say *J'ai changé mon avis hier* because it now means 'I changed my mind yesterday,' with no taint of the present. In short, what was the present perfect became the *PASSÉ COMPOSÉ* 'the compound past tense'.

Although the meaning of *j'ai changé* has changed, the form really hasn't. It still includes an auxiliary verb, usually *avoir* 'have' in its present-tense form, along with the past participle of the verb to be put into the past tense, here *changer* 'change'. The full set of forms is shown in Table 7.1.

There's no simple semantic or syntactic explanation for how this set of verb forms changed its meaning. But when social and phonological factors are accounted for, a potential story begins to emerge.

Table 7.1. Formation of the Passé Composé in Modern French

Subject	<i>Avoir</i>	<i>Changer</i>	Translation
<i>j'</i>	<i>ai</i>	} <i>changé</i>	'I have changed'
<i>tu</i>	<i>as</i>		'You have changed'
<i>il/elle</i>	<i>a</i>		'He/She has changed'
<i>nous</i>	<i>avons</i>		'We have changed'
<i>vous</i>	<i>avez</i>		'You have changed'
<i>ils/elles</i>	<i>ont</i>		'They have changed'

In Old French, as in its ancestor Latin, stress was typically placed on the penultimate syllable. Over centuries, this resulted in an erosion of the final syllable. For example, Latin *fundamENTum* became Old French *fondeMENT*, preserving only the ⟨t⟩ of the last syllable. Even that has been lost in Modern French pronunciation /fɔ̃d.mɑ̃/, though the fossilized spelling remains. This kind of “Matthew effect”¹ where unstressed elements fade away is pervasive and a common driver of change across languages.

Erosion at the end of words is a problem for Latinate languages because that’s where the verb conjugations are, and those conjugations are numerous. The simple past tense’s troubles start here. Table 7.2 shows the robust Old French simple past verb endings and their diminished state in Modern French.

Person	Old Fr. (IPA)	Modern Fr. (IPA)
1st Singular	/navi'zai/	/na.vi.ʒe/
2nd Singular	/navi'zas/	/na.vi.ʒa/
3rd Singular	/navi'za/	/na.vi.ʒa/
1st Plural	/navi'zames/	/na.vi.ʒɑ̃/
2nd Plural	/navi'zastes/	/na.vi.ʒat/
3rd Plural	/navi'zarənt/	/na.vi.ʒɛr/

Table 7.2. Simple Past Tense Conjugation in Latin, Old French, and Modern French

These sound shifts were influenced by various factors. One might have been the social life of teen girls, a demographic that often plays a significant role in accelerating and popularizing linguistic change (Labov, 2001). Their desire to express their unique identities, their experimentation with language, and their influence within tight-knit social circles often make them drivers of linguistic

¹From the Gospel of Matthew 25:29, “For to everyone who has, more will be given...but from the one who has not, even what he has will be taken away.”

innovation. Imagine a group of girls subtly tweaking the established stress patterns, perhaps for emphasis within their conversations or simply as a playful new way of speaking. These variations may have caught on within their circle, eventually spreading more broadly.

Whatever the vector, stress shifted, and the waning syllables lost their final consonants like so many autumn leaves. Not in writing, but in speech. Because Old French relied on verb endings to distinguish between, for example, first-person singular and third-person plural within a given tense, not to mention distinguishing among tenses, things got complicated. Pronouns, previously optional as in Modern Italian or Spanish, began to gain importance.

For Old French, words were generally pronounced as they were spelled. The Old French pronunciations in the table are somewhat speculative, but they're reasonable approximations for the verb *aimer* 'love'. This clearly shows that the Old French conjugations are much more distinct than the Modern ones.

Person	Old French	Modern French
1st Singular	<i>aim</i> /aim/	<i>j'aime</i> /ʒɛm/
2nd Singular	<i>aimes</i> /aiməs/	<i>tu aimes</i> /ty ɛm/
3rd Singular	<i>aime</i> /aimə/	<i>elle aime</i> /ɛl ɛm/
1st Plural	<i>aimons</i> /aimõs/	<i>nous aimons</i> /nu zɛ.mõ/
2nd Plural	<i> aimez</i> /aiməz/	<i>vous aimez</i> /vu zɛ.me/
3rd Plural	<i>aiment</i> /aimənt/	<i>ils aiment</i> /il zɛm/

Table 7.3. Comparative Verb Conjugations in Old French and Modern French with IPA Transcriptions

As a result, many elements of the simple past began to sound very much alike. Luckily, the irregular nature of the auxiliary verbs for the present perfect (before it became reconceptualized as the *passé composé*), meant that they were far less subject to this phonemic erosion, and so far less subject to conjugational confusion.

Person	Passé Composé	IPA
<i>j'</i>	<i>ai aimé</i>	/ɛ ɛ.me/
<i>tu</i>	<i>as aimé</i>	/a ɛ.me/
<i>il/elle</i>	<i>a aimé</i>	/a ɛ.me/
<i>nous</i>	<i>avons aimé</i>	/a.võ ɛ.me/
<i>vous</i>	<i>avez aimé</i>	/a.ve ɛ.me/
<i>ils/elles</i>	<i>ont aimé</i>	/õ ɛ.me/

Table 7.4. *Passé Composé* Conjugation of *aimer* with IPA Transcriptions

Since the *passé simple* and the present perfect could generally be used to refer to the same events, as long as a specific time word like *hier* ‘yesterday’ wasn’t used, people started to avoid the more phonologically ambiguous *passé simple*. If that meant being vague about when something had happened, so be it.

Though this series of changes took roughly 300 years in Paris, they spread much more slowly in the surrounding areas, the result being that the use of the *passé simple* began to take on a whiff of provincialism, at least in certain uses. Once this had happened, the pressure for Parisians to avoid it was enough to overcome the sense of incompatibility between present nature of the form and the past-nature of the meaning. For them, it had become the *passé composé*.

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We can see just the type of problems that verb-form ambiguity can cause in the famous sentence – at least relatively famous among linguists – *The horse raced past the barn fell*. If you’ve never encountered it before, you likely judged it to be ungrammatical. But I haven’t starred it because it only gives the illusion of ungrammaticality, an illusion produced by the overlap between the past tense and the past participle, both of which are *raced*.

Past-tense forms are generally more common than past participles, and this is particularly true directly after a noun; the past participle usually follows an auxiliary such as *be* or *have*, as in *the horse was raced*. And so we are tricked into believing that *raced* is the past-tense form. But there’s a structure in which the participle doesn’t need an auxiliary: a post-head modifier, as in, *the horse seen on the hill*. Because *see* distinguishes *saw* from *seen* the illusion collapses. In case it’s still not clear, the original sentence was *the horse (which was) raced past the barn fell*.

Despite this occasional overlap between the past tense and past participle in English, our present perfect hasn’t become a past tense. A true past tense, like the *passé composé* or the simple past in English (again the “preterite”), means ‘completed before now’. The English perfect, though, means ‘looking back from a point’, and the *present* perfect establishes that point as ‘now’. We can often freely choose between the two for any given past event: *I’ve seen her* or *I saw her*, but **I’ve seen her yesterday* is still impossible. The ‘past-time’ meaning of that explicit *yesterday* clashes with the ‘present-time’ meaning of present-tense *have*, and this kind of meaning clash gives rise to feelings of ungrammaticality.

This leads to the question of why we don’t get the same ungrammatical feeling when the paradox is introduced by word choice rather than word forms. Statements such as *the water was so dry* or *it was too light to pick up* provoke curiosity rather than discomfort.

Perhaps, lexical choices like this open up many possible interpretations, such as metaphor, exaggeration, and humour, and so we assume that the paradox was intended. The use of the perfect with a past-time reference like *yesterday*, though doesn't suggest such a purpose, and so immediately registers as a grammatical error.

Consider **I have two cat*. The intended sentence could be *I have one cat* (wrong word) or *I have two cats* (wrong word form), but my guess is that no editor would change *two* to *one*; they would assume the error is in the grammar. Similarly, *I've seen it yesterday* could be *I've seen it recently* (wrong word) or *I saw it yesterday* (wrong tense). Again, I assume that most editors would blame the tense.

In agreement errors like *a group of students are coming*, we assume that the verb form is incorrect rather than that *group* should be *number* or *majority*. In an example like *give the form from Sarah*, lacking context, we set aside the possibility that the lexical word *give* should have been *get* and assume that the "grammatical word" *from* should be *to*.

In part, this may be because the degrees of freedom are almost always much smaller with grammar than with lexis. To make this concrete, consider that, across the world's languages, the system of number may mark plurality as it does in English – singularity being unmarked – or duality and plurality, as in Arabic. A very few, such as Wamesa, mark triality on pronouns. A few other systems exist, such as marking paucality (the state of being few in number), but this mostly exhausts the possibilities for marking grammatical number, apart from not marking number at all. Now consider tallying the possible nouns to which any system of number applies. The number will be thousands of times greater.

Think of language as a set of Lego blocks. You have a limited number of shapes, and they can only connect in specific ways – the grammar. If something doesn't fit, you would default to shifting, turning, or trying to connect it in one of those ways rather than assuming you've got the wrong piece. You'd assume you made a placement or "grammar" mistake rather than a selection or "lexical" mistake. This reflex is born of an understanding that the grammar rules, much like the physical constraints of Lego, are relatively inflexible. A block misplaced by a few degrees or millimetres simply won't connect. But as long as you've got the angle and position right, you've got a lot of latitude about which blocks you use.

This balance isn't inevitable. In constructing an Ikea desk, the "vocabulary" of pieces is roughly as constrained as the "syntax" of combinations. So, if something's wrong, you might as well guess you've got the wrong piece as that you've put it together the wrong way. Languages, though, are much more like Lego

sets than Ikea kits, at least in this regard.

Another way to look at the question of why we default to assuming ungrammaticality is caused by grammar, not words, is simply to ask how likely any given sentence is. Back in 1957, Chomsky attempted to illustrate that answering such a question is simply an impossibility with the famous pair of “sentences” in (3).

- (3) a. *Colorless green ideas sleep furiously.*
- b. *Furiously sleep ideas green colorless.*

At the time, he asserted without proof that “in any statistical model for grammaticality, these sentences will be ruled out on identical grounds as equally ‘remote’ from English” (Chomsky, 1957, p. 15). Not so!

Most people find (3a) to be nonsensical but perfectly grammatical. They realize that the words don’t add up to anything they could really make sense of, but that other words might, *colorful green leaves rustle gently*, for instance. Consider, though, that for most people at most times, many sentences seem nonsensical: Werner Heisenberg’s observation that “the more precisely the position [of an electron] is determined, the more imprecisely will the impulse be known, and vice-versa” (Heisenberg, 1927, p. 5), or Martin Heidegger’s claim that “the most thought-provoking thing in our thought-provoking time is that we are still not thinking” (Heidegger, 1993), or Friedrich Nietzsche’s aphorism, “when you stare into the abyss, the abyss stares back at you,” (Nietzsche, 1886), or Simone de Beauvoir’s thought that “one is not born, but rather becomes, a woman” (de Beauvoir, 1949). Indeed much of what has been said throughout human history might seem meaningless to most of us. These examples illustrate that we’re quite familiar with words being put together to mean things we haven’t thought or find difficult to think, but that could, with effort, be rendered sensible. And so, though any one of these confounding strings is highly unlikely in itself, the experience of encountering such unexpected collections of words arranged expectedly (or so it would seem) is just a commonplace, not an aberration as Chomsky claimed.

What’s truly remote from our experience is seeing strings like (3b). Pereira (2000) demonstrates this with a simple bigram model. Imagine language as a network of pathways between pairs of words (bigrams). Some are well-worn superhighways between words that often occur together (*and the*), others overgrown trails between words that almost never co-occur (*my the*). Roughly speaking, Pereira looks at likelihood as the total traffic between any two points. Sentences with the most traffic over the shortest distances are more likely in this model. *Colorless green ideas sleep furiously* doesn’t take the main streets, but it uses established routes – people actually do talk about *green ideas*, though *green* means

‘environmental’ here. Meanwhile, *Furiously sleep ideas green colorless* strikes out on its own, forging new paths at each turn. Under this bigram model, the apparently grammatical but meaningless (a) isn’t just more likely but roughly 200,000 times more likely than the word salad in (b) (Pereira, 2000, p. 7).

Returning, though, to the difference between lexical meaning and grammatical meaning, it’s clear that the words in Chomsky’s sentence don’t make semantic sense, but the grammar is fully coherent. Its meanings all work.

There’s a subject, meaning that the clause is about something – it has a topic. It also says there’s an agent available to perform the action denoted in the predicate. And the subject is an NP, which is conventional.

The noun phrase (NP) has two adjective phrase (AdjP) modifiers, attributing properties from the lexical adjective heads to the head noun. The absence of a determiner means that the NP is indefinite, and nothing in the sentence contradicts this interpretation. The NP is plural, as marked by the morphology of *ideas*. This plurality aligns with the systems of determiners, quantification, and definiteness, all conveying coherent meanings. Even the order of adjectives conforms to standard expectations: a general property adjective (*colorless*) followed by a color adjective (*green*) attributed properties to the entity *ideas*.

The verb phrase (VP) exhibits similar coherence in its grammatical meaning. Placed after the subject, the VP indicates that the subject performs the action denoted by the verb. Specifically, *sleep* requires an Agent role, and the subject NP fulfills this by actively engaging in the action. The present-tense verb form suggests that this situation is a general one, not a particular past or future action, which aligns well with the indefinite nature of the subject.

The verb *sleep* requires a subject, and the grammatical relationship – that the referent of the subject NP performs the action – is clear and unviolated. The adverb *furiously*, positioned at the end of the VP, attributes a manner to the action, irrespective of whether sleeping can logically be done furiously. The verb agrees with its plural subject, maintaining grammatical agreement. Crucially, although we might question the plausibility of ideas sleeping (especially furiously), this doesn’t violate the grammatical meaning of predication: the action denoted by the verb is attributed to or performed by the subject’s referent. The tense marking is appropriate, the word order adheres to English syntax, and all grammatical subsystems operate without contradiction. The sentence’s oddness arises entirely from conflicting lexical meanings, not from any failure in grammatical meaning conveyance.

In short, we have sound justifications for our intuition that some sentences are ungrammatical while others are just puzzling, and that, to fix an ungrammatical string such as *two cat*, it makes sense to fix the grammar rather than the

choice of words.

Another linguistic realm where we encounter the unusual without it registering as ungrammatical is in pronunciation and accent. If one hears talk of a “bi-O-pic” about Robert Oppenheimer, one assumes the fault is in the stress pattern (BI-o-pic) and not the choice of words (*myopic*). Again, hearing “bi-O-pic” could lead to a moment of confusion, but never a feeling of ungrammaticality.

In this example, the interpretation is constrained by meaning (it’s about Oppenheimer) and syntax (*biopic* is a noun, and *myopic* isn’t), and we can even reason our way towards an explanation about how the speaker had perhaps only seen *biopic* in print and guessed – incorrectly – about an analog with the pronunciation of the similarly spelled *myopic*. But even so, we know that pronunciation is highly variable, or at least we are accustomed to the fact.

It’s plain that the vowel in a British *can’t* is quite unlike a Canadian *can’t*. But what we may not be so aware of is that our own individual vowels have a certain amount of variation. The /æ/ in a Canadian *can’t* is realized slightly differently than the one in *bat* which differs again from that in *bag*. Although we mentally classify each as /æ/, there are measurable differences. In fact, your own /æ/ in *bat* differs slightly each time you say the word, perhaps because you’re a little more tired, or a little more thirsty, or speaking a little more loudly, or for any of hundreds of other reasons.

An overly simplistic view of this is to liken our perceptual categories to watersheds. The rain that falls in the broad Columbia-River watershed in British Columbia flows into that river, and that which falls on the neighbouring Fraser-River watershed into the Fraser. Similarly, vowels falling anywhere within one perceptual range “flow down” into /æ/, while those with sufficiently different frequencies will flow down into the neighbouring /e/. Those perceptual ranges are quite broad, varying by hundreds of hertz or more. And just as minor variations in where the rain falls don’t change the broader classification into Columbia or Fraser-River watersheds, minute variations in how we pronounce a phoneme don’t alter our classification of that particular sound.

Here’s where the analogy breaks down though: rain that falls into one watershed never flows out of it into the neighbouring river, but phonemes are far more fickle. There’s actually a good deal of overlap between some vowel sounds, as illustrated in Figure 7.1. The way a sound is perceived depends a good deal on the surrounding context and what we’re expecting. And just as we can coerce certain constructions into being grammatical we can coerce sounds into being perceived as this or that phoneme.

Another reason that a form-meaning clash is unlikely at the phonemic level

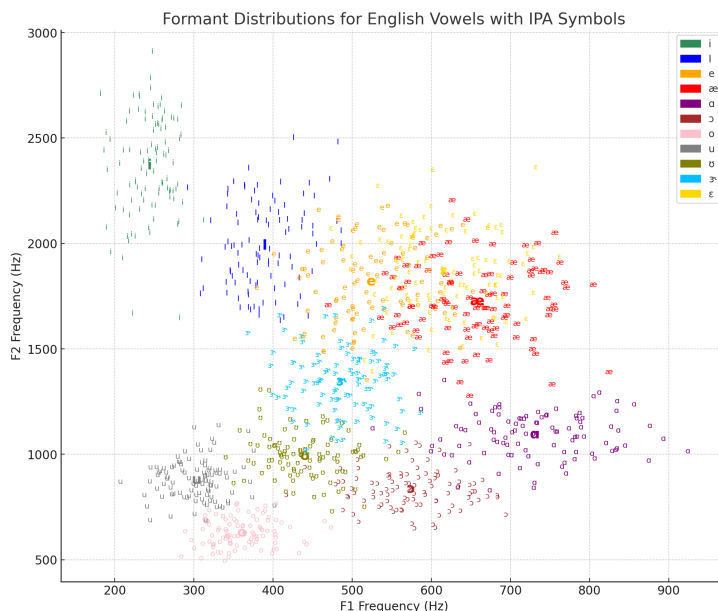


Figure 7.1. Simulated overlapping distributions of English vowels. Distributions are illustrative and not based on empirical data.

is because, for the most part, phonemes don't mean. Despite claims throughout history for an iconic connection, such as those of Cratylus in Plato's *Cratylus*, words don't have a natural correctness; they're arbitrarily, not inherently, linked to the concepts they denote.

Cratylus argues that every word isn't just a random label but is crafted with an understanding of what it denotes. This implies that a kind of philosophical insight into the nature of things is embedded in the very language we use. For example, according to his perspective, the phoneme /s/ in *sun* could be seen as naturally indicative of the qualities or essence of the sun itself. He might argue that /s/ captures an aspect of the sun's essence, perhaps its hissing, radiant heat, or the sharpness of its light. This phoneme, under Cratylus's view, is iconic. It's linked to the concept of the sun by some inherent resemblance between the two.

But this can't work. If someone like me who knows no Persian encounters a word like *siḅ*, I have no way of working out its meaning from the pronunciation /si:b/. The presence of the /s/ is entirely unhelpful, let alone reliable in signalling some kind of hissing, radiant heat, or sharpness of light. It's simply impossible to deduce from phonology alone that *siḅ* means 'apple' in Persian but 'goat' in Zoogocho Zapotec. And if I thought that its meaning might be similar to the meaning of /zi:b/ or /sib/ or or /zi:p/ because they sound very similar, well, I'd

just be wrong. That's not how it works. The association between a word's meaning and its pronunciation is mostly arbitrary.

I say "mostly" arbitrary because some phonemes really do appear to be iconic. In 2022, Winter et al. (2022) published a massive, cross-cultural, multi-language set of four studies of the textural iconicity of a trilled /r/. If asked about the texture that phoneme evokes, what would you guess?

Their studies involved participants from diverse linguistic backgrounds who, despite their varied languages, consistently linked the trilled /r/ with a sense of roughness or vibration, akin to the texture one feels when touching a coarse surface. This convergence across cultures indicates that our sensory experiences can shape and be mirrored in our linguistic practices. But don't forget that trilled /r/ was the first such pattern shown to apply across a wide range of the world's languages, as well as to various words within those languages. The lack of other documented cases suggest that it's special. Even so, its iconicity doesn't apply consistently (consider *creamy*, *murmur*, and *serene*), and its applicability is limited to sensory words.

What it all comes down to is that phonetic variability is just part and parcel of speaking a language. We know it's there, and we try to cope. We understand that phonemes, unlike words, have no clear meaning but that, like words, have a lot of degrees of freedom, and we expect that freedom to be evident in what we hear. It's quite unlike the form-meaning mismatches that lead to a sense of ungrammaticality.

It does seem odd, though, that we'd look for meaning in changes in grammar, attributing ungrammaticality only when we come up empty and that we would seek meaning in changes in lexis, even when it's not apparent to us, and yet that we would attribute nothing at all to deviations from expected pronunciation. Aren't we "meaning-seeking animals" (Clifford Geertz, 2016, p. 636)?

But of course, we do assign meaning to differences in pronunciation, as discussed in this very chapter, just not semantic meaning. If a speaker says *I can't do it* as /kɑ:nt/, that *means* they haven't grown up in Canada, and if they say /kænt/ then it *means* they could have. If a French speaker dropped word endings in 1400, it likely *meant* they were Parisian and stood higher on the social ladder than provincial speakers who didn't. If a man in California in the early 2000s said *pan* more like *pen* and *boat* more like *bought*, it might *mean* that he's gay (Podesva, 2011). And if he also pronounces an /ɛ/ before words like (*e*)*school*, (*e*)*spoon*, and (*e*)*still*, it probably *means* he grew up speaking Spanish. What pronunciation means, what it signifies, more than anything else, is identity.

But it goes beyond that. If a speaker's voice rises at the end of each sentence,

it might mean they want to hold the floor and you shouldn't interrupt. A sustained high pitch, exaggerated vowels, and slow tempo may signal condescension. In contrast, a lowering of overall pitch, reduced pitch variation, and a slightly harsher or rougher voice quality often marks a kind of negation of the meaning through sarcasm. Yeah, I bet.

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Recall *So, I go there yesterday ...*, from the conversation between the mother and son? I claimed that this use of the present tense for past time didn't feel ungrammatical because the motivation is clear. Conversely, past tense for present time is also possible, as long as it's motivated.

Before going on, we should address the fallacy of monosemy, the mistaken idea that any given word, tense, or structure must have a single "true" meaning – *mono-* for single and *semy* from *sēma* 'sign or portent'. Quite the contrary! In language, polysemy is the rule.

A familiar example of this fallacy can be found in the self-appointed grammar maven invidiously pretending to misunderstand the meaning of *can* in requests. *Can I leave now* is never a question about ability, and the choice of *may I leave now* isn't about the correct meaning. Rather, *may*, in this question, is a sign that the request is a formal one and an acknowledgement of the greater power of the person being asked. As with the Parisian discomfort with the provincialism of the *passé simple*, the insistence on *may* for requests is a way to assert the social status of the person being asked, a way to insist on deference.

So, it's a mistake, then, to assume that words are monosemous, and the same is true of tenses. Just as *may* can signify social standing along with permission, possibility, or purpose (*You may go; It may rain; ... that they may prosper*), the past tense can convey deference, formality, likelihood, and impossibility (*Could you help me?; Did you wish to reconsider?; If I did go, ...; If she was a client, ...*). Perhaps the common element motivating these seemingly diverse meanings is the idea of distance, temporal, social, possible, or factual.

Compare (4a) with (4b).

- | | | | |
|-----|----|---|-----------------|
| (4) | a. | <i>I'm wondering if this is a good idea.</i> | [PRESENT TENSE] |
| | b. | <i>I was wondering if this was a good idea.</i> | [PAST TENSE] |

It's difficult to set aside our learned associations between the past tense and tentativeness, but looking as strictly as we can at the basic time meaning, (a) implies a current discomfort with the idea, while (b) suggests that discomfort was in the past and may have been overcome. As a result, (b) is less "pushy" than

(a), more open to allowing the listener their own opinion. This kind of example then provides a motivation for interpreting the past tense more generally as less direct, as more polite.

Another motivation for using the past tense to indicate politeness is that it provides a kind of point-of-view shift. By using the past tense, the speaker “moves *as if* into the future” (Brown & Levinson, 1978, p. 204), thus making the statement or question less immediate, less of an imposition on the listener.

These uses are so appealing that, in some cases, the past-tense forms of modal auxiliary verbs have left behind their past-time meanings. *Could*, as the past-tense of *can* ‘is able to’, rarely means ‘was able to’ in today’s English, though that was its typical meaning for hundreds of years. The use of *could* for requests, which seems so natural to us, appears to be an innovation of Modern English. According to the *Oxford English Dictionary*, the first recorded example of *could* used in a request is in a letter from Lord Chesterfield in 1748, *Could you send me... some seed of the right Cantelupe melons?* Once the request use got going, though, it took off and took over. Presumably, its politeness meaning was so obvious that it wasn’t felt to be ungrammatical, at least, not by many.

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To reiterate, grammar conveys meaning, and when the meaning clashes with the meaning of some other element of the sentence, we experience that clash as ungrammaticality. We don’t get that same feeling when the meanings of words appear to clash mostly because we’re used to words telling us stuff we’re not used to. At the same time, differences in phonology don’t provoke feelings of ungrammaticality because phonology is extremely variable and is generally not a carrier of semantic meaning.

Here’s another example of a form-meaning mismatch that results in ungrammaticality, one that, as far as I know, has never been described, and one that seems difficult to attribute solely to structural factors.

A few days ago, as I was working on this chapter, my friend Geoff Pullum emailed me to comment on the strange sentence in (5) that had been published online by the BBC. The star is Geoff’s, but I completely agree that this is ungrammatical.

(5) * *The Rubymar is the ship to have been sunk by the Houthis.*

What, he wondered, could be wrong with this sentence. The hypothesis he suggested was that the *to have been ...* part needs something to “license” it. What he had in mind was a process like the one that allows the *than* phrase in cases

like *It is more radical a plan than that*. Here, *more* licenses *than that*. Without *more*, you get the completely awful **It's radical a plan than that*. And Geoff wondered if *to have been ...* might need a word like *first*, *only*, or *largest* to render the sentence grammatical as in (6), just as *than ...* needs *more*.

- (6) a. *The Rubymar is the first ship to have been sunk by the Houthis.*
 b. *The Rubymar is the only ship to have been sunk by the Houthis.*
 c. *The Rubymar is the largest ship to have been sunk by the Houthis.*

One problem this poses is that these three words aren't all the same grammatically. Sure, they're all adjectives, but consider that neither *large* nor *larger* work, and they're just as adjective as *largest*. This suggests that it might be the superlative suffix *-est* that's doing the licensing. With this hypothesis in mind, we can experiment with other superlatives to see how they fare. And, indeed, they fare very well: *the best ship*, *the fastest*, *the strongest*, *the most expensive*, *the most recent*, etc.

But that doesn't explain why *first* and *only* work. While only *largest* is a true superlative, the other two words only hint at superlative qualities. *First* can imply the 'most important' or the 'most desirable' position, and *only* hints at being 'the best' or 'the most exclusive'. And when it comes to the forms, *first* is already the base form, like *large*, not like the superlative form *largest*. *First* isn't the superlative of a base **firs* and comparative **firser*. Nor is *only* the superlative of *on* or anything else. There's no word from which it could be inflected, no word indicating a state of *limited* exclusivity or a *preliminary* form of uniqueness, ideas that seem entirely paradoxical. So, it can't be that the superlative is licensing *to have been ...* in the way that the comparative licenses the *than* phrase.

Another problem for the licensing hypothesis is that we can construct similar sentences that work without any apparent licensing words. Before moving on, let's look at the structure to see what Geoff and I were considering "similar sentences".

The key part seemed to be the noun phrase *the ship to have been* So we have the definite article *the* plus a head noun, and then a modifier in the form of a *to* infinitival. This includes *to* plus *have* to mark the perfect, along with a past participle. In this case, the past participle was *been*, but it could just as well have been *gone*, *seen*, *found*, *made*, *called*, *smoked*, *minced*, or *barbecued*. So, the part we're interested in is *the NOUN to have PAST PARTICIPLE*.

And so, I went looking for sentences of this kind and found a small number.² This was surprising, because we'd felt that there was something wrong with the

²You can see the results at <https://www.english-corpora.org/coca/?c=coca&q=119073728>.

structure. And so we decided to zoom out a bit to see what context the examples appeared in. But when I prefaced the search with *is*, the tool I was using wouldn't allow a search that long with so many common words.

To get around this, I went back to the previous list of results and started browsing the full concordance of sentences. As is common when you do this kind of research, you find that lots of matching strings aren't also matching phrases. *Is it a mistake [for the president] to have joined the lawsuit* has the *to* infinitive as the complement of *mistake*, not the modifier of *president*. *City records show the officer alleged [in the lawsuit] to have fired the fatal shots joined the CPD in 2014* has it as a complement of *alleged*, etc.

Eventually, though, I turned up *She seems to be the one to have changed the most*, which seemed fine to both of us. At this point, the whole premise that there was something structurally wrong with *the NOUN to have PAST PARTICIPLE* was in tatters. After staring at it for some time, I realized something new: *is* is present tense, while *be* has no tense at all. It's infinitival. But, we asked ourselves, why should (7a) be better than (7b).

- (7) a. *She seems to be the one to have changed the most.*
- b. **She is the one to have changed the most.*

I also found examples (8–13).

- (8) *Again, he was the one to have handled it correctly.*
- (9) *...you were probably also the type to have memorized them*
- (10) *He should have been the one to have run for president*
- (11) *Petraeus is said to have been the one to have broken off the extramarital affair*
- (12) *Mrs. Stanton was not the type to have “stayed up all night dancing”*
- (13) *[he] should have been the one to have given the land to the boy*

In no case was the (underlined) verb before this phrase in the present tense. This directed our attention back to the *to* infinitival, and we realized that *have PAST PARTICIPLE* was an infinitival perfect.

The present perfect has already come up a lot in this chapter. As a present tense, it's anchored in the now, but as a perfect, it's retrospective. The infinitive perfect, on the other hand, has no anchor in the present. It's just retrospective,

the semantic equivalent of a past tense. And I think the feeling of ungrammaticality is coming from a meaning clash between the present-tense of *is* in (5) and this effective past tense.

This suggests that the problem is a clash between the present tense in the matrix clause and the infinitival perfect in the subordinate. But that doesn't explain why adding in a modifier like *first*, *only*, or *largest* should fix things. (And, yes, I know that "like" is still undefined here.)

Other things need considering. Interrogatives seem better than declaratives.

- (14) a. ? *Who is the one to have betrayed the people of India?*
 b. ? *Is he the one to have betrayed the people of India?*
 c. * *He is the one to have betrayed the people of India.*
 d. *He is the one who betrayed the people of India.*

Also if you change the token to a type, you can coerce it into grammaticality:

- (15) a. * *She's the one to have read a handful of books from a century ago.*
 b. *She's the type to have read a handful of books from a century ago.*

7.2 THE EX-LAX CONUNDRUM

Sometimes, what appears ungrammatical at first glance can become perfectly acceptable given the right context. To illustrate this point, let's consider a striking example from the Nixon years in the US. This example, which appears in a dialogue between James McCawley and Herman Parret (1974, p. 252), builds on insights about the grammaticality of responses to questions, an idea developed by Jerry Morgan (1973). Together, these contributions illuminate the complex relationship between grammar and context, foreshadowing the kinds of issues we'll encounter in the next section.

McCawley presented the following sentence:

- (16) * *Spiro conjectures Ex-Lax.*

At first glance, most syntacticians would agree with the asterisk, marking this as ungrammatical. The verb *conjecture* typically doesn't take a direct object denoting a physical substance. However, Morgan then posed a question that dramatically shifts our perception:

- (17) *Does anyone know what Mrs. Nixon frosts her cakes with?*

In the context of this question, the previously ungrammatical sentence suddenly becomes acceptable. We interpret it as an elliptical response, implying a structure more like:

- (18) *Spiro conjectures [that Pat Nixon frosts her cakes with] Ex-Lax.*

(In case you're unfamiliar with the historical context, Ex-Lax is a chocolate-flavored laxative, and Spiro Agnew was the Vice President under Richard Nixon. I'll say no more.)

Geoff Pullum reminded me of this example and how it challenged a fundamental assumption in syntax at the time: that it was possible to categorize sentences as grammatical or ungrammatical without considering their communicative context. Morgan's insight was that grammaticality judgments can't be made in isolation from the sentence's intended use and situational context.

This example demonstrates how context can dramatically alter our perception of grammaticality. As we'll see in the next section, this principle extends beyond simple question-answer pairs to more complex grammatical structures. The case of the supposedly missing independent relative *whose* provides another striking illustration of how our judgments of grammaticality can be influenced by contextual factors.

7.3 THE CURIOUS CASE OF THE MISSING ⟨whose⟩

Earlier in our exploration of linguistic intuitions (Chapter 5), we encountered a puzzling case: the supposedly missing independent relative genitive *whose*. You'll recall how this apparent gap in English grammar challenged long-held assumptions and demonstrated the importance of empirical investigation in linguistics. Now, as we dig deeper into the nature of grammaticality itself, I'd like to return to this curious case, which provides a rich example of how various linguistic factors intertwine to shape our judgments of what's grammatical and what isn't.

To briefly recap, in 1973, the same year Morgan published his Ex-lax example, linguists Jorge Hankamer and Paul Postal claimed that English lacks this particular form of *whose*, ruling out constructions like:

- (19) * *My gorilla is over there drinking punch. The guy [whose you saw banging at the window] is over there watering the rubber tree.*

This judgment, as we saw, was influential and long-lasting. Even the authoritative *Cambridge grammar of the English language* (Huddleston & Pullum 2002) largely agreed, as did Pullum in his recent book *The truth about English grammar* (Pullum, 2024).

Our previous discussion revealed that this apparent grammatical impossibility isn't as clear-cut as it first seemed. In this chapter, we'll build on that foundation, using the case of *whose* to illustrate how syntax, semantics, pragmatics, and information structure can all come together in determining (un)grammaticality.

To recap, in 1973, Hankamer and Postal boldly claimed that English doesn't include a particular kind of *whose*. It may not be obvious that kinds of *whoses* exist, but they do. The kind they claimed was missing is the kind that appears in these strange ungrammatical sentences they gave, the 70s being something of a strange time.

- (20) * *My gorilla is over there drinking punch. The guy [whose you saw banging at the window] is over there watering the rubber tree.*
- (21) * *Melvin, [whose is banging at the window], is over there watering the rubber tree.*

They reasoned that the sentences were ungrammatical because this particular *whose* didn't exist. And how did they know it didn't exist? Well, they'd been looking for it without success, and every time they tried to create a sentence using it, the result was ungrammatical.

But as we'll see, the story of this missing *whose* is far more complex – and far more interesting – than Hankamer and Postal could have imagined. We begin with the question “what kind of *whose* is this anyhow?”

To understand that, we need to look at the *whoses* we do have. First, there's the interrogative *whose*, as in:

- (22) *Whose gorilla is drinking all the punch?*

Then there's the relative *whose* that introduces a phrase about something, like:

- (23) *The guy [whose gorilla is drinking all the punch] should probably take it home.*

In both of these cases, *whose* is followed by a noun: *gorilla*. But in Hankamer and Postal's examples, *whose* stands alone, without a noun to keep it company. That's the kind they claimed was missing from English.

Now, you might be thinking, “Wait a minute, I've heard *whose* used without a noun before!” And you'd be right. We do it all the time in with interrogative *whose*:

- (24) *Whose is this half-empty punch glass?*

But Hankamer and Postal weren't talking about questions. They were talking about *whose* in relative clauses – those constructions that give us more information about something, like *the gorilla that's drinking punch* or *the guy whose gorilla is making a scene*.

So, they argued, while we can say:

- (25) *The guy [whose gorilla is making a scene] should leave.*

We can't say:

- (26) * *The guy [whose is making a scene] should leave.*

And for years, most linguists agreed. It seemed like a clear case: this particular *whose* – let's call it the “independent relative *whose*”, independent because it doesn't need the noun and relative because it's in a relative clause – just didn't exist in English.

But language has a way of surprising us. As it turns out, this missing *whose* isn't quite as missing as Hankamer and Postal thought. It's more like a linguistic ivory-billed woodpecker: long thought extinct, occasionally glimpsed, hotly debated, but stubbornly refusing to be definitively ruled out.

Consider this sentence from the *Cambridge Grammar of the English Language*, a book that's about as authoritative as it gets in the world of English grammar:

- (27) *I was going to visit Lucy, [a friend of whose had told us of the accident].*

Here we see our elusive *whose*, standing proudly without a noun in tow. It's right there in the relative clause *a friend of whose had told us of the accident*. No gorillas in sight, but it's the same type of *whose* that Hankamer and Postal claimed didn't exist.

So what's going on here? How can this *whose* both not exist and exist at the same time? Well, that's where things get really interesting...

The key to understanding this linguistic puzzle lies in the interactions of syntax, semantics, and pragmatics – the rules of sentence structure, meaning, and contextual language use. While Hankamer and Postal were correct that this independent relative *whose* is extremely rare, they missed some crucial factors that allow it to exist in very specific circumstances.

First, let's consider the structure. In the sentence from the *Cambridge Grammar*, *whose* is part of a larger phrase: *a friend of whose*. This structure, called the “oblique genitive”, seems to provide a comfortable habitat for our rare *whose*.

It's like finding an ivory-billed woodpecker not just anywhere, but in a particular type of old-growth forest.

But structure alone doesn't explain everything. After all, we can't say **The guy of whose is making a scene*. There's more at play here.

The second factor is information structure – how we package information in a sentence. This is a bit like how we organize items in a suitcase; some things are right on top (easy to access), while others are buried deep inside.

In language, the items that are “right on top” are said to be topical. They're the main focus of what we're talking about.

In the Cambridge Grammar example:

(28) *I was going to visit Lucy, [a friend of whose had told us of the accident].*

Both Lucy (the possessor) and her friend (the possessed) are highly topical:

- Lucy is important because she's the person being visited.
- The friend is important because they're the one who told about the accident.

This high topicality of both elements – the possessor (Lucy) and the possessed (her friend) – seems to be a necessary condition for our independent *whose* to appear. It's like our rare linguistic bird needs both large tracts of old-growth bottomland hardwood forests and an abundance of large, recently dead trees.

To understand this better, we need to get up to speed on some fundamental concepts about how language works, particularly pronouns, ellipsis, and their antecedents.

7.3.1 PRONOUNS, ELLIPSIS, AND THEIR ANTECEDENTS

In language, we often refer back to something we've already mentioned. This backward reference is called anaphora, and the thing we're referring back to is called the antecedent. In the following example, *that guy* refers back to Aden.

(29) Aden just bought a new car. That guy sure loves his rides.



Noun phrases like *that guy* can perform this referential function, but pronouns – small words like *they*, *it*, *he*, and *she* – are the true specialists in anaphoric reference. For this linguistic magic to work, though, the antecedent must be readily available in the discourse. The surest way to achieve this availability is for the antecedent to be topical or recently mentioned. English provides some clever mechanisms to extend our referential reach: we can distinguish between

singular and plural (*she* vs *they*), personal and non-personal (*she* vs *it*), and even masculine and feminine (*he* vs *she*). These distinctions act like linguistic signposts. If we encounter a feminine pronoun, but the most recently mentioned potential antecedent is masculine, it's a cue to search further back in our mental discourse for a more suitable referent. Interestingly, each time we use a pronoun, we're not just referring back – we're also boosting the prominence of its antecedent in the ongoing discourse. It's as if we're a linguistic juggler, periodically tossing each referent high into the air to keep it in play.

Another way to do this is a bit counter intuitive. It involves referring back to an antecedent without making any explicit mention of it at all. As its name suggests, ELLIPSIS involves leaving something out. But doing so, we're indicating that the elided phrase is so obvious and available that it goes without saying. And by not saying it, we paradoxically make it even more salient, as if we were tossing it up above the other semantic elements we're juggling.

Let's look at an example:

- (30) *Aden can speak French, and Bella can _____ too.*



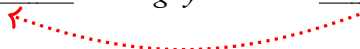
Here, *speak French* is missing after *Bella can*, but the meaning is unmissable. Ellipsis, then, is like an invisible pronoun. It's pointing back to something, but instead of using a small word to do it, it's no word. And just like with pronouns, the antecedent needs to be available and ideally topical or nearby. Ellipsis can't happen just anywhere:

- (31) * *This is Bella. Bella can ____.*
(Without prior context)

Now, you might be wondering what all this has to do with our elusive independent *whose*. Well, it turns out that independent *whose* is doing both these tricks at once. It's acting like a pronoun, pointing back to a possessor, but it's also involved in a kind of ellipsis, leaving out the thing that's possessed. And this dual nature is part of what makes it so rare and tricky to use. But even that's not the whole story.

There's another kind of anaphoric relationship at play here. Let's look back at our earlier example:

- (32) *Aden just bought a new car. That guy sure loves his rides.*



In the previous example, Aden and that guy are the identical person. This is how anaphora usually works, with two phrases referring to the same individual or item, but here *his rides* doesn't refer just to the identical car. Instead, it references a type: vehicles owned by Aden, and *a new car* is just one example of this type.

Here's another example:

(33) *Aden took his car and I took mine ____.*



We'll say that pronouns like *my* are DEPENDENT, because they can't appear without a following noun, but pronouns like *mine* are INDEPENDENT; they won't appear with one. (Remember that, above, we saw that there's an independent *whose*. Unfortunately, while *my* and *mine* are easy to tell apart, *whose* – like *his* – is a bit confusing because the dependent and independent forms are spelled the same and sound the same.)

When I use the independent *mine* in this example, I mean “my car”, and that meaning comes through an anaphoric relationship between the missing word and *car* in the previous clause. But, again, it's not the identity-anaphora that exists between *Aden* and *that guy*; It's a type-anaphora. The meaning is that I took the same type of thing that Aden took – a car – but not the identical car that he took – not his car.

We're almost at the end of this journey, but we need to pull this all together. Before we do so, let's quickly recap what we've uncovered in our linguistic investigation. We've seen that anaphora allows us to refer back to previously mentioned entities, whether through pronouns or ellipsis. We've distinguished between identity-anaphora, where we refer to the exact same entity, and type-anaphora, where we refer to the same kind of thing. We've also explored the difference between dependent pronouns like *my*, which require a following noun, and independent pronouns like *mine*, which stand alone. Each of these concepts plays a crucial role in understanding the behavior of our elusive independent relative *whose*. Now, armed with these insights, we're ready to solve the mystery of why this particular *whose* is so rare and why it appears only under very specific conditions.

Because *mine* is a pronoun, it has an antecedent too. There's the elided “car-type” antecedent, and there's me myself. It's my car. We'll say I'm the possessor and the car is the possessed (shades of Stephen King's *Christine*). So, the *mine* ____ phrase has the double anaphoric relationship shown here. The possessed car-type antecedent is related to the gap and the possessor antecedent *I* is related to *mine*.

- (34) *Aden took his car and [I] took [mine] ____.*
-

Bringing this back to independent *whose*, we can now see that it also has this double anaphora. So why do we see *mine* all the time, but not *whose*? The thing about *mine* is that, as the first person, *I* is/am always topical (and so is/are *you*, dear reader, as the second person). So as long as the possessed thing (or at least that type of thing) is topical, then we've met the conditions for comfortable use of *mine*.

But with independent relative *whose*, it turns out that the possessor and possessed are rarely both topical enough at the same time. People are more likely to be topics than things, and if we're using *whose*, then it's because the person is topical. But if that's the case, then the possessed usually isn't. On top of that, possessors are more likely to be people, the possessed more likely to be things, which are inherently less topical than people. Also, topicality can be spread around a little, but mostly it likes to pick out one thing. And that's fine; when the possessor is the topic, we can always use dependent *whose* and avoid eliding the non-topical possessed. But it does make it a lot less surprising that independent relative *whose* is so rare.

What about independent interrogative *whose* you may be wondering. *Whose is the brown car?* But the thing with interrogatives is that they don't refer to antecedents – they ask about things. In the case of interrogative *whose*, it asks about the possessor, leaving the possessed to bask in the full glow of topicality. This interrogative context presents a much less demanding topicality condition for *whose* to meet.

So, let's bring this all together. The independent relative *whose* that Hankamer and Postal claimed didn't exist in English does exist, but it's exceedingly rare. Its rarity is due to a burdensome confluence of linguistic conditions that need to be met:

1. It needs to function as both a pronoun (referring back to a possessor) and involve ellipsis (leaving out the possessed noun).
2. Both the possessor and the possessed need to be highly topical in the discourse.
3. It needs to occur in a relative clause, not an interrogative context.

When all these conditions align, we get sentences like the one from the Cambridge Grammar:

- (35) *I was going to visit Lucy, [a friend of whose had told us of the accident].*

Here, Lucy (the possessor) is topical because she's the person being visited, and her friend (the possessed) is topical because they're the one who told about the accident. Notice that the possessed is, unusually, a person. The oblique genitive construction *a friend of whose* provides the right structural environment. And it's all happening in a relative clause.

But I've found and constructed other examples. Consider

- (36) *His gorilla isn't much fun at parties, but I know someone whose is.*

This use of *whose* has a number of factors in its favour. It's contrasting, which elevates the topicality of whatever is being contrasted. Contrasts are generally made between types – here two members of the gorilla type – rather than individuals, which is favourable for ellipsis. And, finally, the possessor of the gorilla isn't topical. It's just someone I know, it doesn't matter who. This allows the topical spotlight to shine on the possessed.

7.4 A WORLD TOUR OF ⟨whose⟩

Our investigation of independent relative *whose* in English raises an interesting question: How do other languages deal with this linguistic puzzle? Let's take a look at a few languages to see how they handle similar constructions.

We'll start with German. German has words that function similarly to our *whose*: *dessen* for masculine and neuter nouns, and *deren* for feminine and plural. Like English *whose*, these can appear without a following noun in certain contexts. Consider this example:

- (37) *Meins funktionierte, aber ich kenne jemanden, dessen _ nicht funktionierte.*
 mine worked but I know someone whose _ not worked
 'Mine was working, but I know someone whose _ wasn't.'

The underscore shows where we might expect a noun, but it's been left out. This mirrors our English construction:

- (38) *His gorilla isn't much fun at parties, but I know someone whose _ is.*

In both languages, we see that this construction works well in contrastive contexts, but becomes problematic in other situations. For instance, the German equivalent of our ungrammatical English sentence is equally troublesome:

- (39) * *Die Person, dessen _ du vergessen hast, ist mein Cousin.*
 the person whose _ you forgotten have is my cousin
 *'The person whose _ you forgot is my cousin.'

Spanish shows a similar pattern with its word *cuyo*. It's comfortable in contrastive contexts but starts to falter in constructions like this:

- (40) * *La persona cuyo _ olvidaste es mi primo.*
 the person whose _ forgot.2SG is my cousin
 *‘The person whose _ you forgot is my cousin.’

Interestingly, even languages that structure their sentences quite differently show analogous patterns. Persian, for example, doesn't have a direct equivalent to *whose*, but it uses a possessive construction that behaves in a similar way. It's fine in contrastive contexts but becomes problematic in constructions parallel to our troublesome English examples.

Japanese takes us even further afield. It doesn't have relative pronouns at all, instead using a different strategy to build relative clauses. Yet even without a *whose*-like word, Japanese still expresses these ideas in ways that echo the patterns we've seen in English and the other languages.

What can we learn from this linguistic world tour? A few key points stand out:

1. The quirks we've observed with English *whose* aren't unique to our language. Other languages show similar patterns with their possessive words in relative clauses.
2. The preference for contrastive contexts seems to be a common thread, appearing across languages with different structures.
3. The need for both the possessor and the possessed item to be highly salient in the conversation appears to be a shared feature.
4. Even when languages lack a direct equivalent to *whose*, they often follow similar patterns in how they express these ideas.

These cross-linguistic patterns suggest that what we've discovered about English *whose* isn't just a peculiarity of our language. It seems to tap into something more fundamental about how languages work – how we package information, how we track what's important in a conversation, and how we balance the need for clear structure with the complexities of real-world communication.

It's hardly surprising that Hankamer and Postal, and many other linguists after them, thought this *whose* didn't exist. Ferreting one out is like trying to spot a rare bird that only emerges under very specific conditions. If you're not looking at exactly the right time and place, you might conclude it doesn't exist at all.

But language is full of these rare constructions that push at the boundaries of what we consider grammatical.

Part III

Whose grammar

Whose grammar?

8.1 MORALITY AND GRAMMATICALITY

8.1.1 IDEAS

1. The Pirahã
2. Alice Evans and [link](#)
3. A difference between moral intuitions and grammatical intuitions is that it's typically less self-serving to adjust your grammatical intuitions. (true?)

Could speech have real-world negative consequences by virtue of its being ungrammatical? When Eve eats of the fruit of the tree of knowledge of good and evil, she already speaks.

8.1.2 UTILITARIANISM

Utilitarianism is ...

Bentham's musings on language, writing, and grammar offer another dimension to his character. Living in an era where legal and philosophical texts were often mired in complexity, Bentham championed clarity and precision. He was particularly critical of the legal jargon of his time, viewing it as a barrier to justice and understanding. "Prolivity is the soul of tedium," he once remarked, emphasizing the need for simplicity in expression. For Bentham, language was a tool, its value derived from its utility in conveying ideas. This pragmatic view extended to his thoughts on grammaticality. While he acknowledged the importance of grammatical norms, he believed that the ultimate aim of language was clear communication. Any rule that hindered this objective was, in his view, superfluous.

Language frameworks

Many language assessment schemes, such as the Canadian Language Benchmarks, or the Common European Framework of Reference for Language take a pragmatic, utilitarian view of grammaticality. Language proficiency, under these schemes, is very much about what you can achieve, what you can do, what your instrumental outcomes are when communicating in the language. For example, at CLB 7 (out of 12), “Adequate Intermediate Ability”,

The speaker can: Communicate with some confidence in many daily routine social, educational, and work situations, and present concrete and some abstract information on an expanding range of familiar topics.

And in so doing, the speaker demonstrates an

increasing variety of grammatical structures, with developing control of complex structures, [although] grammar, vocabulary and pronunciation difficulties may occasionally impede communication.

Cooperation and grammar

Jahangir Khan is anything but cooperative. A titan in the world of squash, he dominated the sport in the 1980s. His unparalleled winning streak of 555 matches remains unmatched in professional sports. Khan’s style of play was relentless, often wearing down his opponents with his exceptional fitness and tactical acumen. Yet, his success wasn’t built on cooperation. In the court’s confines, he was a fierce competitor, always seeking an edge, always looking for a way to outplay his adversary. His non-cooperative stance in the game wasn’t a sign of moral failing but a strategic choice, essential for triumph in such a competitive arena.

Jahangir Khan’s life and career are replete with instances that showcase his resilience, determination, and humility. Born into the squash-playing Khan family of Pakistan, Jahangir’s introduction to the sport was almost preordained. But his journey was anything but smooth. At a young age, he faced health challenges, including a couple of hernia operations. Few believed he would ever play professionally, let alone dominate the sport.

Tragedy struck when Jahangir’s older brother, Torsam Khan, died suddenly during a game due to a heart attack. Torsam had been a guiding force in young Jahangir’s life, mentoring him in the nuances of the game. Devastated by the

loss, Jahangir considered quitting the sport, but he chose to persevere, turning his grief into motivation. He once remarked that he felt his brother's presence with him during matches, guiding and watching over him.

Off the court, Jahangir was known for his humility. Despite his unmatched success, he was always approachable and took time to engage with fans and budding players. Numerous accounts describe him offering tips to young players and sharing his knowledge of the game. His sportsmanship was also legendary.

Khan was clearly a deeply moral and ethical person, but that didn't prevent him from intense competition on the court. Many spheres of human interaction are built on the bedrock of cooperation. Consider the intricate dance of a symphony orchestra, where each musician must not only master their instrument but also listen intently to their fellow players, synchronizing their efforts to produce harmonious sound. Here, cooperation isn't just beneficial; it's essential.

Yet, the principle of cooperation isn't universally applicable. Just as Khan found success in non-cooperation on the squash court, some contexts in life make competition crucial: sports, business, arguing legal cases, science. This isn't to say that all interactions in those fields are competitive – a lawyer has to cooperate with their client – but particular interactions make cooperation inimical to the proper functioning of the institution.

going against the grain, challenging the status quo, or simply choosing a different path can lead to innovation and progress. Think of the mavericks in the world of art or science who defied conventions and, in doing so, reshaped entire fields.

Language, too, offers a rich tapestry of cooperative and non-cooperative dynamics. At its core, communication is a cooperative endeavor. Speakers and listeners engage in a mutual dance, with each party making assumptions, drawing inferences, and adjusting their behavior based on feedback. The Gricean maxims, which emphasize the importance of being clear, relevant, truthful, and concise, underscore this cooperative nature of communication.

However, just as in squash or music, breaking the rules or flouting conventions in language can be powerful. Think of the poets, playwrights, and novelists who play with words, bending and twisting them to create new meanings or evoke specific emotions. Their deviations from linguistic norms aren't errors but deliberate choices, much like Khan's tactical decisions on the court.

This brings us to the concept of grammaticality. In certain circles, especially academic or formal settings, adhering to the norms of Standard English is expected. But these norms aren't set in stone. They evolve over time, shaped by societal changes, technological advancements, and cultural shifts. What's deemed

“ungrammatical” today might be the standard of tomorrow.

Moreover, the idea of a “standard” itself is fraught with complexities. Who decides what’s standard? Why are certain dialects or forms privileged over others? These questions hint at the power dynamics at play in language, where issues of prestige, class, and identity intersect with grammar.

In everyday interactions, people often switch between different linguistic registers, choosing the one that’s most appropriate for the situation. A casual chat with a friend might see the use of slang, colloquialisms, and non-standard forms, while a job interview would demand more formal language. This linguistic flexibility is a testament to our ability to navigate different social contexts, recognizing when to cooperate with linguistic norms and when to deviate.

Returning to Khan, his non-cooperative strategy in squash was a response to the specific demands of his sport. Similarly, our linguistic choices are shaped by the demands of our communicative context. Recognizing when to adhere to linguistic norms and when to challenge them is a skill, honed over time and through experience.

In conclusion, while cooperation is a guiding principle in many areas of life, it’s not an absolute. Just as Khan’s dominance in squash was built on a foundation of strategic non-cooperation, there are times in language, and life more broadly, when going against the norm can be not just beneficial, but transformative.

The key lies in discernment, in understanding the nuances of each situation, and in making informed choices. Whether on the squash court, in a concert hall, or in a conversation, recognizing the interplay of cooperation and non-cooperation is crucial.

8.1.3 LAWS AND MORALITY

Moral philosophers will sometimes point out a distinction between laws and moral

Double modals in English

Linguists, like bird watchers eagerly seeking rare bird species, are fascinated by unusual language patterns. One such rare pattern in English is the double-modal construction, exemplified by phrases like (1).

- (1) *The flavouring within soda can come from a wide variety of places, but one you would might not expect would be from tree bark.*¹

¹YouTube

I can't ever recall hearing a single instance of a double modal in my daily life, but Steven Coats, of University of Oulu in Finland, got tired of waiting and went looking. He started by developing the 1.29-billion-word *Corpus of North American Spoken English* (Coats, 2023) from YouTube videos. Then, he searched this massive corpus for double modals like *would might*. Next, he had to check each one and determine whether it was a real double modal or just some kind of transcription error, self correction, change of speaker, subordinate clause (*those of us [who can] [must make ...]*) or any number of other false positives. After all that, just 54 true examples of *would might* remained (Coats, 2022). That's about one for each 24 million words. For comparison, *interstitial* occurs about once per 2 million words, while you'd have to go to a word like *exigence* to find one roughly as rare. Coats tells me he finds just three in my home province of Ontario.

For most English speakers, double modals are ungrammatical. Not only are they ungrammatical, but they are so rare in most communities that most of us are entirely unaware that they even exist. We've probably never heard them deprecated, unlike constructions such as *him and me tried it*, *my and Geoff's book*, *I ain't gonna try that*, *I done it already*, and a host of others.

There's a good grammatical reason for this. In Standard English, the modal auxiliary verbs (*may*, *might*, *can*, *could*, *shall*, *should*, *will*, *would*, and *must*, along with their negative forms, *can't*, *won't*, etc.) are highly defective verbs. To call a verb defective isn't to render some qualitative or moral judgment, but rather to indicate that it lacks one or more of the regular verb forms. A normal verb like *jump* has six forms: a plain form (*jump*), a plain present-tense form (also *jump*), a third-person singular present-tense form (*jumps*), a past-tense form (*jumped*), a past participial form (also *jumped*), and a present participial form (*jumping*).

Contrast that with a modal auxiliary verb like *will*, with only four: present-tense *will*, past-tense *would*, negative present-tense *won't*, and negative past-tense *wouldn't*. The modal auxiliaries lack a plain form, the form used in infinitival clauses (*It makes you jump*.) You can't say **It makes you will jump* because *will* is the present-tense, and *make* requires the plain form. If a present-tense form worked, *jumps* would be fine in **it makes him jumps*, and it's not. Other places that require the plain form are after *to* (*I want to (*can) jump*), after *let's* (*Let's (*can) jump*), and – here's the crucial part – after a modal auxiliary verb (*I should (*can) jump*). In Standard English, you don't find *you would might not expect it* because *would* expects a non-tensed verb, and *might* is past-tense.

8.1.4 ETHICAL PERSPECTIVES ON GRAMMATICALITY

Grammaticality judgments, often presented as objective facts about language, frequently reflect underlying ethical stances. These judgments intersect with social values, power structures, and cultural identities in complex ways.

Derrick Parfit's late-career effort to unify disparate moral theories offers a useful lens through which to view these varied approaches to grammaticality. Parfit's "climbing the mountain" metaphor suggests that seemingly conflicting ethical frameworks might converge at a higher level of understanding. Similarly, our varied perspectives on grammaticality might be seen as different paths up the same mountain.

Consider how grammatical "errors" are often treated as moral failings, or how grammatical norms typically align with the speech of socially dominant groups. These phenomena suggest that our notions of grammaticality extend beyond mere linguistic description into the realm of ethics and social justice.

The following sections explore seven ethical frameworks and their implications for understanding grammaticality. Each offers a distinct perspective on what makes language use "right" or "wrong." Yet, following Parfit's insight, we might find that these perspectives aren't as irreconcilable as they first appear.

This exploration has real-world implications. Our conceptualization of grammaticality shapes language policy, educational practices, and social attitudes. By seeking a unified understanding of grammaticality's ethical dimensions, we might develop more nuanced and effective approaches to language variation and change in our diverse linguistic landscape.

Pragmatism: Effectiveness Over Prescription

Pragmatism in grammaticality prioritizes effective communication over strict adherence to prescriptive rules. This approach aligns closely with the utilitarian view of language proficiency exemplified by frameworks such as the Canadian Language Benchmarks (CLB) or the Common European Framework of Reference for Language (CEFR).

Consider the CLB description for level 7, "Adequate Intermediate Ability":

The speaker can: Communicate with some confidence in many daily routine social, educational, and work situations, and present concrete and some abstract information on an expanding range of familiar topics.

Notably, this description focuses on communicative outcomes rather than perfect grammatical form. It acknowledges that at this level, a speaker demon-

strates an “increasing variety of grammatical structures, with developing control of complex structures,” while recognizing that “grammar, vocabulary and pronunciation difficulties may occasionally impede communication.”

This pragmatic stance echoes Jeremy Bentham’s views on language. Bentham, living in an era of often impenetrable legal and philosophical texts, championed clarity and precision. For Bentham, language was a tool, its value derived from its utility in conveying ideas. While he acknowledged the importance of grammatical norms, he believed that the ultimate aim of language was clear communication. Any rule that hindered this objective was, in his view, superfluous.

The pragmatic approach to grammaticality raises interesting questions about the nature of linguistic cooperation. Paul Grice’s cooperative principle and its associated maxims (quality, quantity, relevance, and manner) suggest that effective communication relies on a kind of tacit agreement between speakers and listeners. However, as we’ve seen with examples like Jahangir Khan in squash, non-cooperation can sometimes be strategically advantageous.

In language, this non-cooperative stance might manifest as deliberate deviations from grammatical norms for rhetorical effect. Consider the following examples:

1. Poetic license: “Anyone lived in a pretty how town” (E.E. Cummings)
2. Advertising slogans: “Think different” (Apple)
3. Political rhetoric: “We shall fight on the beaches, we shall fight on the landing grounds, we shall fight in the fields and in the streets, we shall fight in the hills” (Winston Churchill)

Each of these examples breaks conventional grammatical rules, yet each is highly effective in its context. From a pragmatic perspective, they could be considered grammatical because they successfully achieve their communicative goals.

The pragmatic approach, though, isn’t without challenges. One significant issue is the potential conflict between immediate communicative effectiveness and broader societal needs for standardization. While “u ok?” might be a perfectly effective text message, accepting such forms in all contexts could lead to communication difficulties across different groups or in more formal situations.

Moreover, judgments of communicative effectiveness can be subjective and culturally bound. What constitutes clear communication in one context might be opaque or inappropriate in another. This subjectivity can make it difficult

to establish consistent standards, particularly in educational or professional settings.

Consider the case of double modals in English, such as "might could" or "used to could." These constructions, while rare in most varieties of English, are grammatical in some Southern United States dialects. A strictly pragmatic approach might accept these forms as grammatical wherever they're understood, but this could lead to difficulties in contexts where speakers are unfamiliar with the construction.

The pragmatic approach to grammaticality also intersects with questions of linguistic identity and power. Non-standard forms often serve as markers of group identity or solidarity. Judging these forms solely on their communicative effectiveness to outsiders might miss their important social functions within the community.

In conclusion, while the pragmatic approach offers valuable insights into the nature of grammaticality, it can't be applied in isolation. A comprehensive understanding of grammaticality must balance communicative effectiveness with considerations of standardization, cultural context, and the social functions of language variation. As with many aspects of language, the key lies in recognizing the complex interplay between these factors rather than seeking a single, universal criterion for grammaticality.

Virtue Ethics: Cultivating Linguistic Character

Virtue ethics shifts our focus from rules or outcomes to the character of the language user. It's less about whether a particular utterance is right or wrong and more about developing linguistic virtues like clarity, precision, and eloquence.

This approach aligns with how we often judge writing. We don't just look at whether the grammar is correct, but at how well the author uses language to convey their ideas. A virtuosic writer isn't just following rules; they're wielding language with skill and insight.

But what does this mean for everyday language use? Let's consider an example:

A: "The data don't support your conclusion."

B: "The data doesn't support your conclusion."

Prescriptively, A is correct ("data" is plural). But B is increasingly common, even among educated speakers. A virtue ethics approach might ask: Which speaker is exhibiting better linguistic character? Is A showing precision and respect for traditional forms? Or is B demonstrating adaptability to changing usage?

There's no clear-cut answer. It depends on context, audience, and the speaker's overall command of language. A scientist writing a paper might lean towards A, showing mastery of field-specific conventions. A politician addressing a general audience might prefer B, displaying an ability to communicate clearly without sounding pedantic.

This perspective encourages us to view grammaticality not as a binary state but as a skill to be honed. It's about developing judgment about when to adhere strictly to rules and when flexibility serves communication better.

Consider how we judge a chef. We don't just check if they followed a recipe correctly. We look at how they combine ingredients, adapt to available materials, and create dishes that suit their diners. Similarly, a virtue ethics approach to grammaticality looks at how speakers combine linguistic elements, adapt to communicative contexts, and craft utterances that serve their audience.

This view of grammaticality as a cultivated skill rather than a set of inflexible rules echoes how linguists understand language acquisition. Children don't learn their native language by memorizing rules. They develop an intuitive grasp of the language's patterns through exposure and use.

The virtue ethics approach also offers an interesting perspective on language change. If we see grammaticality as a matter of cultivated skill rather than rigid rules, it becomes easier to understand how and why languages evolve. Changes aren't "errors" that have become widespread, but shifts in what the language community considers skillful usage.

Take the ongoing change in how we use "literally." Prescriptivists bemoan its use as an intensifier ("I literally died laughing"), but this usage is becoming increasingly accepted. A virtue ethics approach might say that skillful use of language now includes knowing when this intensifier use of "literally" is appropriate and when it isn't.

This perspective on grammaticality isn't without challenges. It's subjective and context-dependent, which can make it difficult to teach or assess. It also risks reinforcing linguistic prejudices if we're not careful about whose "virtues" we're promoting.

But despite these challenges, a virtue ethics approach to grammaticality offers a nuanced way of thinking about language use. It encourages us to move beyond simple right/wrong judgments and consider the broader skills involved in effective communication.

Consequentialism: Judging Language by Its Outcomes

Consequentialism in grammaticality focuses on the results of language use rather than adherence to rules. It's about whether an utterance achieves its intended effect, not whether it follows prescribed norms.

This view aligns with how we often judge communication in real-world contexts. In business, for example, we care more about whether a message persuades a client than if it splits an infinitive. In casual conversation, we're more concerned with understanding and connection than perfect subject-verb agreement.

Let's look at an example:

"Me and him went to the store."

Prescriptively, this is incorrect. It should be "He and I went to the store." But a consequentialist approach asks: Did the message get across? Did any misunderstanding occur? If not, then from this perspective, the sentence worked. It was grammatical enough.

This doesn't mean "anything goes." Consequentialism in grammar isn't about abandoning all standards. It's about judging those standards by their effects. Some prescriptive rules genuinely aid clarity and prevent misunderstanding. Others don't. Consequentialism gives us a framework for distinguishing between them.

Consider the "rule" against ending sentences with prepositions. Does following this rule actually improve communication? Often, it doesn't. It can lead to awkward constructions like "This is the sort of arrant pedantry up with which I will not put." A consequentialist view might say this rule isn't worth following most of the time because its outcomes aren't positive.

On the other hand, the rule about using "fewer" for countable items and "less" for uncountable ones can prevent ambiguity. "Fewer calories" and "less calorie intake" convey different meanings. Here, the rule serves a purpose. It leads to clearer communication.

The consequentialist approach to grammaticality also offers insight into language change. New forms that effectively serve communicative needs tend to stick around and spread. Those that don't, don't. This is why "u" for "you" in text messages has caught on, but most of us aren't dropping our definite articles despite their frequent omission in headlines.

This view of grammaticality isn't without problems. It can be hard to measure the consequences of language use objectively. It might also undervalue the social and cultural roles of language that go beyond mere information transfer. But it offers a practical framework for thinking about grammaticality, one that aligns well with how we often judge language in day-to-day life.

Deontology: Grammar as a Moral Duty

Deontology in grammaticality posits that following grammatical rules is a duty, regardless of outcomes. This view aligns closely with prescriptivism in linguistics.

For a deontologist, the sentence "Me and him went to the store" is wrong not because it fails to communicate or because it leads to bad consequences, but simply because it breaks the rules of Standard English grammar. The correct form, "He and I went to the store," is right because it follows the rules.

This approach has some appeal. It offers clear, consistent standards. It aligns with how grammar is often taught in schools. And it reflects a common intuition that there's a "right" and "wrong" way to use language.

But deontology in grammar faces significant challenges:

1. Which rules? English has many varieties, each with its own norms. Even within Standard English, experts often disagree on specific rules.
2. Rule conflicts: What happens when rules contradict each other? The injunction against splitting infinitives often clashes with the desire for clear, concise expression.
3. Language change: Deontology struggles to account for how language evolves. Today's error often becomes tomorrow's standard usage.
4. Linguistic diversity: A strict rule-based approach can delegitimize non-standard dialects, even when they're systematic and effective for their speakers.

Consider the double modal construction we discussed earlier:

- (2) *The flavouring within soda can come from a wide variety of places, but one you would might not expect would be from tree bark.*

For most English speakers, this is ungrammatical. The deontological view would say it's wrong, full stop. But this ignores the fact that double modals are grammatical in some dialects. It also misses the potential communicative value of the construction, which can convey nuances of meaning not easily expressed with single modals.

Deontology in grammar often manifests as "grammar purism," the belief that there's one correct way to use language and all deviations are errors. This view can lead to linguistic prescriptivism and even language discrimination.

Yet, the deontological approach isn't without value. Standardized rules can facilitate clear communication, especially in formal or professional contexts. They

can provide a shared framework for language learners. And for many people, following grammatical rules is part of showing respect for the language and its traditions.

The key is recognizing the limits of this approach. Rules can guide us, but they're not absolute. They're tools for communication, not moral imperatives. A nuanced view of grammaticality needs to balance respect for linguistic norms with an understanding of language's inherent flexibility and diversity.

Elitist Power: Grammar as Social Currency

The elitist power view of grammaticality treats “correct” language use as a marker of social status and authority. Under this framework, adherence to prescriptive grammar rules isn't just about clear communication – it's a way to signal membership in educated or elite social groups.

But it's not just elites who use language to discriminate. People across the social spectrum create and use non-standard forms, particularly slang, to set themselves apart. This linguistic differentiation serves to strengthen in-group bonds and exclude outsiders.

Consider the reaction to a sentence like:

- (3) *Me and John was talking about that yesterday.*

An elitist approach might not just label this as ungrammatical, but use it to make judgments about the speaker's education, class, or fitness for certain social roles. The “correct” version – *John and I were talking about that yesterday* – becomes a shibboleth, a way to identify who belongs and who doesn't.

This perspective often manifests in the enforcement of grammatical “rules” that have little to do with clarity or effective communication. The prohibition against split infinitives, for instance, originated as a way for educated elites to signal their knowledge of Latin grammar. It persists not because it aids understanding, but because it serves as a social marker.

The elitist view of grammaticality poses several problems:

1. It often privileges the dialect of dominant social groups, treating linguistic differences as deficiencies.
2. It can be used to justify discrimination in education, employment, and other areas.
3. It ignores the linguistic validity and communicative effectiveness of non-standard forms.

4. It tends to resist language change, treating evolving usage as “decay” rather than natural evolution.

Yet this approach persists, often implicitly, in many institutions. School systems that penalize students for using non-standard dialects, workplaces that judge employees’ competence based on their mastery of prescriptive grammar, and social circles that use language as a gatekeeping mechanism all reflect aspects of this elitist view.

It’s worth noting that linguistic identity and social identity are deeply interconnected. Speakers of prestige dialects are unlikely to broadly adopt elements of non-prestige dialects, though certain “cool” forms may be “appropriated.” This interconnection makes changing linguistic hierarchies particularly challenging.

Social Justice by Any Means Necessary: Grammar as Liberation

This radical approach to grammaticality views traditional grammatical norms as tools of oppression and seeks to dismantle them. It’s less about defining what’s grammatical and more about questioning why certain forms are privileged over others.

Consider these lines from “Fight the Power”:

- (4) *Gotta give us what we need*
Our freedom of speech is freedom or death
We got to fight the powers that be

Standard grammar labels this ungrammatical. But a social justice approach might argue that this form is valid and that labelling it “incorrect” perpetuates linguistic discrimination. It recognizes that these non-standard forms are deliberately chosen for their expressive power and as markers of group identity.

This perspective often manifests in:

- Advocating for acceptance of non-standard dialects in a broader range of contexts, including formal contexts
- Criticizing the notion of a single “standard” language
- Promoting “code-meshing” (blending dialects) over “code-switching”
- Questioning the cultural biases in grammar textbooks and standardized tests

The social justice view of grammaticality faces several challenges:

1. It can sometimes overlook the practical value of a shared linguistic standard
2. It might inadvertently reinforce linguistic isolation of marginalized groups
3. It struggles to define boundaries – if all forms are equally valid, how do we maintain mutual intelligibility?
4. It can be seen as threatening by those who've invested heavily in mastering standard forms

Interestingly, the elitist power view and the social justice view can be seen as mirror images of each other, with the groups reversed. While the elitist view seeks to maintain linguistic hierarchies that benefit dominant groups, the social justice view aims to dismantle these hierarchies and elevate non-standard forms. Both approaches use language as a tool for group identity and social positioning, albeit with very different goals and beneficiaries.

This mirroring extends to the challenges each view faces. Just as the elitist view can reinforce social inequalities, the social justice approach risks isolating speakers of non-standard dialects if taken to extremes. Both views also grapple with the tension between group identity and broader communication needs, albeit from opposite directions.

Moreover, both approaches often align with identity politics, which can take a zero-sum view of power and prestige. While economic resources can be expanded, it's less clear that linguistic prestige can be similarly enlarged. This raises questions about the feasibility of truly equalizing linguistic status across all varieties.

There's also a tension between the desire for linguistic liberation and the practical reality that certain forms of linguistic capital are more valued in many social and professional contexts. While advocating for broader acceptance of diverse language forms, this approach must grapple with the fact that mastery of standard forms often remains a pathway to social mobility.

Yet the social justice approach offers valuable insights. It highlights how grammatical “rules” often reflect and reinforce social power structures. It encourages us to examine our linguistic biases and consider the rich diversity of language use across communities.

Consider the “singular *they*.” Long criticized as ungrammatical, it's gaining acceptance partly due to social justice arguments about gender inclusivity. This shows how this approach can drive language change.

Ultimately, both the elitist power view and the social justice view of grammaticality push us to think critically about language norms, albeit in very different ways. They serve as counterbalances to each other, encouraging a more nuanced understanding of what counts as “grammatical” and how language intersects with social power structures. However, both must also reckon with the complex realities of linguistic identity and the practical implications of linguistic diversity in a world that often demands standardization.

Conclusion: A Unified Theory of Grammaticality

Our exploration of ethical perspectives on grammaticality has revealed approaches that often seem at odds. Yet, Parfit’s work on unifying ethical theories offers a model for reconciling these varied views on language norms. Just as Parfit argued that consequentialism, Kantian deontology, and contractualism are “climbing the same mountain,” we can view our grammatical frameworks as different routes to a unified understanding of language norms.

This unified theory of grammaticality would blend elements from each perspective:

Rule Convergence. Different grammatical approaches often yield similar judgments in practice, much as Parfit proposed that seemingly different moral rules often recommend the same actions. For instance, both prescriptivists and pragmatists might prefer standard forms in formal writing, albeit for different reasons. Our unified theory would recognize these convergences, focusing on the common ground rather than surface disagreements.

Underlying Principles. Across grammatical perspectives, we can identify shared concerns: clarity, social cohesion, identity expression, and adaptability to change. Our unified theory would articulate these common principles, showing how they manifest in different approaches to grammaticality.

Triple Theory. Parfit’s “Triple Theory” combined Kantian ethics, consequentialism, and contractualism. A unified theory of grammaticality would similarly blend rule-based, outcome-focused, and social contract elements of language norms by:

- Acknowledging the role of conventional rules while recognizing their historical contingency
- Evaluating grammatical choices based on their communicative outcomes in specific contexts
- Treating language norms as implicit social agreements that can be renegotiated as society changes

For example, consider the sentence "Me and him went to the store." Our unified approach would:

- Recognize the prescriptive rule preferring "He and I went to the store"
- Assess whether the non-standard form impedes understanding in its context
- Consider how acceptable this form is within the relevant speech community

Wide Value-Based Objective Theory. Parfit argued for moral realism based on irreducibly normative truths. Similarly, our unified theory would posit that objective truths about language use exist, but that they're complex, context-dependent, and not reducible to simple prescriptive rules. It would recognize that grammaticality judgments involve both universal linguistic principles and culturally specific norms.

This unified approach views grammaticality not as a binary state, but as a multidimensional space where linguistic, social, and ethical factors intersect. It acknowledges the validity of different perspectives while seeking common ground.

In practice, this theory would lead to more nuanced language policies and pedagogies. Instead of teaching absolute rules or embracing complete relativism, it would foster critical language awareness. Language users would learn to navigate the complex landscape of grammaticality, making informed choices based on context, audience, and communicative intent.

For instance, in language education, this approach might:

- Teach standard forms while explaining their social and historical context
- Explore how grammatical choices affect communication in various settings
- Discuss how language norms reflect and shape social relationships
- Encourage students to analyze and question grammatical judgments

In language policy, it could lead to:

- More flexible standards that account for linguistic diversity
- Policies that balance the need for a common standard with respect for non-standard varieties
- Recognition of the legitimacy of multiple language varieties in different contexts

Challenges remain, of course. Balancing standardization with linguistic diversity, reconciling descriptive and prescriptive impulses, and addressing ingrained language attitudes are ongoing tasks. But by recognizing the common ground among different approaches to grammaticality, we can work towards a more comprehensive, ethically informed understanding of language norms.

This unified theory doesn't eliminate the tensions between different perspectives on grammaticality. Rather, it provides a framework for understanding these tensions and navigating them productively. Like Parfit's approach to ethics, it seeks not to declare one view correct and others wrong, but to show how seemingly conflicting views can be seen as different aspects of a more complex, multi-faceted truth about language and its role in human society.

8.1.5 SYNTHESIS: ETHICAL FRAMEWORKS AND LINGUISTIC REALITIES

Having examined grammaticality through the lenses of consequentialism, deontology, and contractualism, we now turn to synthesizing these perspectives and grounding them in linguistic realities. This synthesis reveals both the strengths and limitations of applying ethical frameworks to language, while also highlighting crucial factors in grammaticality that these frameworks might overlook.

8.2 PRESCRIPTIVISM ON THE RIGHT AND ON THE LEFT

It will be news to nobody that, in English, there has been a longstanding project to remove gender-marked nouns such as *waitress* or *policeman* from the language. The project dates back to the 1960s and 1970s, when the feminist movement began to gain momentum and raise awareness about gender equality and the impact of language on societal attitudes (Cameron, 1998). This period saw the emergence of various efforts to challenge and revise the use of gender-specific language in English (Lakoff, 1973).

The feminist movement sought to address sexism in language by advocating for more inclusive and gender-neutral terms (Miller & Swift, 1976). These efforts were aimed at dismantling traditional gender stereotypes and promoting a more equal representation of both men and women in various aspects of life, including the workplace, education, and government (Spender, 1980).

Over the years, the project has made significant progress: Style guides, dictionaries, and educational institutions have increasingly adopted gender-neutral language (Baron, 1986), and many organizations and businesses have implemented policies to promote the use of gender-inclusive terms (Maggio, 1998). The continued efforts to make the English language more gender-neutral reflect a broader commitment to fostering a more inclusive and equitable society.

What may come as a surprise to many, as observed by Susan Neiman in her book *Left is not woke*, is that in Germany, one way to avoid being sexist is precisely the opposite (Neiman, 2023). There, at least in the context of professions and occupations, the use of gender-specific forms is considered important for promoting gender equality and inclusivity (Diewald, 2018). The rationale behind this approach is that, by using gender-coded nouns, the German language can more accurately represent and acknowledge the presence of women in various roles, which were traditionally dominated by men. Additionally, research has shown that the use of gender-specific language in German can have a positive impact on the visibility of women and challenge traditional gender stereotypes (Vervecken & Hannover, 2012).

The irony in this context is that, while the left is typically associated with more tolerant grammar norms and against grammatical prescriptivism, it's also much more intent on policing the use of individual words, especially slurs and words used to refer to traditionally marginalized groups. This emphasis on using inclusive language and avoiding offensive terms is seen as a way to promote equality and respect for all individuals, regardless of their background or identity. But just as there's no objectively correct way to form a subordinate interrogative in English, there's no objectively correct way to refer to individual racial groups.

As usual, that's not to say that anything goes. Some words are clearly used with malicious intent, and such use is, rightfully, socially sanctioned. We need to consider the complexities of language use, including in-group versus out-group dynamics and the distinction between use and mention of potentially offensive terms.

Take the example of the word *fucker*, which is generally considered offensive and vulgar. In some cases, friends or in-group members might use this term in a lighthearted or even affectionate manner, without intending any harm. This highlights the significance of context and intent when assessing the appropriateness of language.

Moreover, we need to differentiate between using a term and merely mentioning it. Discussing the word *fucker* in the context of this book, for instance, doesn't carry the same weight as directing the word at an individual in a derogatory or harmful manner.

Nunberg concludes, "racists don't use slurs because they're derogative; slurs are derogative *because they're the words that racists use*." (Nunberg, 2018, p. 244)

Many disability rights theorists and organizations have rejected person-first language (e.g., *person with blindness*) and have advocated for identity-first language (e.g., *blind person*) or nominals (e.g., *the blind*).² For instance, the

²This is a case of an adjective functioning as a modifier in the nominal while, at the same

two main associations working for the social and economic inequality of blind people in Canada are called The Canadian Federation of the Blind and the Canadian National Institute for the Blind. One could argue that this name is simply an anachronism to be explained away by the fact that institutions can't change their name every time linguistic preferences change, but the CFB takes the position that "blindness is not a handicap, but a characteristic," and the blind and deaf communities

8.2.1 METAPHORICAL EXTENSIONS

A newer woke program aims to eliminate metaphorical extensions of the words when such extensions have negative connotations. "Blindness should never be equated with darkness. And, avoid using *blind* as an adjective (e.g. *She was blind to the idea*)" (Karapita, 2017, p. 8). The original meanings of *blind* in Old English include "confused", "dark", and "obscure". In this case, it is "sightless" which is the metaphorical extension, but I'll avoid the etymological fallacy. For most people today, the central meaning of *blind* is "lacking sight", and *being blind to something* is seen as a metaphorical use. Should we, then avoid it? One unanticipated outcome of doing so may be to deprecate use of the word in all its senses. If the only people using it are people who are bigots and the intolerant (or should that be people who are bigoted and intolerant), then the word will come to be associated only with such individuals and their attitudes. Paraphrasing Nunberg, bigots don't use these words because they're derogative; the words become derogative *when they're the words that bigots use*."

In 'Good Vibrations' Henry Schiller talks about Beaver & Stanley's ideas.

8.3 FASHION: INNOVATION AS GLAMOUR

Even it's never approved

In 2012, Cory Doctorow wrote on *Boing Boing*, "But if they test it on kids – even it's never approved for use on kids! – the folks at the FDA will extend their patent by another six months."

My English-language students say and write things like *even it's never approved* (meaning 'even if it's never approved') all the time, but it's ungrammatical for me – likely for you too – and I correct them. Doctorow's post was the first time I'd seen it from a proficient speaker of English, so I emailed him to ask if it was intentional (not to correct him). "Yup," he responded. "It's colloquial, but expressive, unambiguous, and colorful."³

time functioning as its head (a Modifier–Head) in what the *Cambridge grammar of the English language* called fusion of functions.

³Personal communication, July 6, 2012.

There's an *even* + pronoun construction that does work, one with stress on the pronoun, as in *even I know that's ungrammatical*. But that's not happening here. In this innovation, *even* takes on a conditional meaning and becomes a preposition that takes a clausal complement. *If* is missing, but the sense and syntax are otherwise unchanged.

You may accuse me of arbitrary discrimination against English-language learners, but if you did (even you did?) I would protest. If Doctorow had written *even if it's never approve*, I would have instantly recognized it as a typo. Same if he'd said, *even if he's never approved*. Same for a million other mistakes. It's simply not the case that a proficient speaker of a language can get away with anything.

A grammatical innovation will only work if it achieves some kind of semantic or pragmatic goal, something different from what the established form achieves, an effect intended by the innovator. That could be humour, group membership, irony, emphasis, (in)formality, (in)directness, intimacy, immediacy, skepticism, or any number of other effects. And to do that, there needs to be some established connection between the innovation and the effect, some linguistic or cultural point of reference.

A grammatical innovation that fails to do these things is ungrammatical.

Tracksuits

By the year 2000, the tracksuit was clearly unfashionable, a relic of a past era of bold and sporty looks that no longer resonated with the sleek, minimalist trends of the new millennium. Since the start of the pandemic, though, the matching sweatpants-and-sweatshirt look has gained currency (V. Friedman, 2024).

As with most things, the details matter. Wear a sweatsuit with a sweatband and sneakers and you look like a victim of the endless cycle of New Year's workout resolutions. Wear it with a nice handbag and a trench coat and you look like a modern Jean Harlow out to take the air.

(Vanessa Friedman)

Just as a fashion influencer can take this once-derided outfit and transform it into a statement of effortless chic, a linguistically savvy individual can take a phrase that might be considered a grammatical misstep and turn it into a mark of stylistic flair. The phrase *a pair glasses*, when uttered by someone grappling with the intricacies of English, will likely be dismissed as ungrammatical. Yet, when used by a proficient speaker with the linguistic sense to use it intentionally to achieve some goal, it becomes a deliberate stylistic choice, an intentional innovation.

I have a pair glasses, for instance, might signal irony in reply to “I have a dozen tracksuits,” *dozen* establishing the template on which the new *pair* construction is built.⁴

The tracksuit’s journey from the back of the wardrobe to the forefront of fashion showcases how context and the wearer’s choices can completely alter the perception of an item. When paired with the right accessories and worn with purpose, the once unfashionable item becomes glamorous, the once ungrammatical expression becomes the hot new slang.

The glory of ⟨enshittification⟩

Enshittification was the American Dialect Society 2023 selection for Word of the Year (Roberts, 2024), and Cory Doctorow is credited with popularizing it. In early 2023, he wrote,

Here is how platforms die: first, they are good to their users; then they abuse their users to make things better for their business customers; finally, they abuse those business customers to claw back all the value for themselves. Then, they die. I call this *enshittification*, (Doctorow, 2023)

It’s a perfectly cromulent word, the kind any linguist loves: new, creative, flexible, and evocative. When it came to the attention of linguist Nancy Friedman, she said it was “primed for neologistic success” (N. Friedman, 2024). Ben Zimmer wrote, “From the time that it first appeared in Doctorow’s posts and articles, the word had all the markings of a successful neologism, being instantly memorable and adaptable to a variety of contexts.”

Part of what makes *enshittification* so appealing is its anchoring in a linguistic and cultural history. Morphologically, it starts with an everyday term and then accessorizes it with a gaudy array of familiar affixes, each elevating its lexical glory.

The base word here is *shit*, a noun with Old English roots, specifically *scitte*, meaning ‘diarrhea’, which is itself derived from the Proto-Germanic *skit-*. Over centuries, *shit* has evolved in English as a vulgar term for feces, but also as a versatile word used in a variety of idiomatic expressions and colloquialisms.

⁴Most collectives, like *group*, *bunch*, *lot*, *pile*, *heap*, and *shitload* license an *of* preposition phrase complement (e.g., *a group of words*). *Dozen* is different: **a dozen of words* is ungrammatical, though *a dozen of them* is fine. In this way, *a dozen* is like the determinatives *a little* and *a few*, and *a dozen of them* is partitive, where the object of the *of* preposition phrase has to be definite.

To this indelicate base, he adds *-ify*, from the Latin *-ficare*, which in turn is derived from *facere*, meaning ‘make’ or ‘do’. This suffix turns nouns or adjectives into verbs, denoting the process of making something have a particular quality or be in a particular state. Doctorow could have simply said the platforms *shittify* themselves, but it’s unlikely that the word would have had the appeal of *enshittification*. It’s a little too literal, a little too sharp, a little too base. It needs more decoration.

-ic, an adjective-forming suffix denoting a characteristic or quality, attaches readily to *-ify*, as in *terrific*, but an adjective won’t do. So on he goes.

By the curious logic of English, *-ific* attracts the suffix *-ate*, resulting in *shittificate*, the opposite of *beatificate*, re-verbing the word. “Three suffixes is grand,” says Doctorow, “but I can do more. *Shittificate* has three, but I can do four” (with apologies to Dr. Seuss).

-tion is a noun-forming suffix that turns verbs into nouns. When attached to *-ificate*, as in *magnification*, it creates a noun that indicates the action or process of making something have a particular quality. He’s now gone full circle from the noun *shit* to the noun *shittification*, and *shittification* could denote what he’s after. Doctorow, though, is still not done.

In a bold step, audacious even, he adds the completely redundant *en-* prefix. This prefix, typically used to turn nouns into verbs (as in *enlighten* from *light*), here seems superfluous. Yet, its inclusion is a masterstroke in neologism, enhancing the word’s dramatic flair. *Enshittification* isn’t just about becoming shit or the act of making something shitty – it’s about the willful and relentless driving of a platform into the ground by stuffing it with shit.

Doctorow uses the familiar to produce the sublime. But it’s not just the use of familiar affixes but the particular affixes used that have resonance with other successful neologisms. *Embiggen* appeared on “The Simpsons” in 1996 and stuck around, finding a spot in the *Oxford English dictionary* in 2017 and in the *Merriam-Webster dictionary* in 2018. Doctorow was almost certainly aware of this when he chose the *en-/em-* prefix.

Language aside, *enshittification* perfectly captures the tech world’s slide from revolutionary to reviled. We’ve all experienced that moment when a platform, once a beacon of digital innovation, becomes just crummy, betraying its original ethos, turning into something users hardly recognize. We’ve all experienced it. Doctorow crystallized it.

He’s a blogger, a journalist, a science fiction author, an expert language user, and somebody really tapped into the social zeitgeist. He knows what he’s doing. The final form, *enshittification*, thus becomes a linguistic ornament, an embellished word capturing a complex social phenomenon in a single, extravagant

term. And all of that skill and intentionality lead to the acceptance and recognition that the word has received.

8.4 CODESWITCHING

- “Now the whole earth had one language and the same words.” So begins the story of the tower of Babel in the book of Genesis.
- Most primates live in a pre-Babel world. A ‘leopard’ call means the same thing to every vervet and green monkey from Senegal to South Africa (Price et al., 2014).
- This shows that animals don’t have a language. Instead they have a system of signs (semiotics) where there’s a one-to-one relationship between a given sign and a given meaning.
- In this view of language, humans never could have lived in a pre-Babel world because what we want to communicate is contingent. It depends on the communicative situation.
- That means that which constructions are grammatical depends on the communicative situation.
- The most general communicative situation, the kind of thing you assume when you write a book like this which could be read by anybody even well into the future is something like “English” or “French” or “Japanese”.
- What’s grammatical in one may not be in another. This leads to dialectal differences and register differences.
- If two people share the context of “English” and “Japanese”, they can mix constructions from both in the same sentence and it will be grammatical.
- Swearing feels right in some context and wrong in others. Perhaps this is because it belongs to some communicative situations – it’s grammatical there – and not to others, where it’s ungrammatical.

The ultimate point of the chapter, eventually, will be that a language is simply a communicative situation. When we’re guessing how best to convey our intentions, the language is part of that. But languages aren’t just English, Japanese, Urdu, Apache, etc. They’re much smaller – swearing has different effects in different communicative situations/languages – and they’re much more mixed – a bilingual conversation can mix languages within clauses and it will be fully

grammatical. It all depends on what the sender knows and what they believe the audience knows.

It was 2003. We – me, Yoko, and our toddler Tomo – had just moved to Toronto from Tokyo. Bad timing, I know. That was the year of the first SARS outbreak, Toronto was SARS central, and the city felt empty. Finding a teaching job was impossible when all the international students were understandably panicking.

Because of this, for several months, I was mostly at home with Yoko, our son Tomo, and my mom. We were lucky we could stay with her while we got our feet under us.

Yoko used Japanese exclusively with Tomo. I did too, except for when I'd read to him, which was always in English. My mom spoke only English, so he had a great bilingual environment and Tomo was picking up both Japanese and English as well as any toddler could.

I can't pinpoint when exactly, maybe six months after we arrived, but there was one day in the kitchen. Yoko and I were talking in Japanese, and Tomo and my mom were also there. During a pause, Tomo turned to her and started explaining what Yoko and I had been saying. I was amazed. It wasn't just that he knew there was a difference between Japanese and English; he also knew who understood what language.

Smith and Hoefler (2016) explain that when we communicate, whether using words or other methods like gestures, it's not just about the message itself. Instead, effective communication involves two key steps: showing and guessing. First, the person speaking or signalling tries to make their intentions clear (showing). Then, the person listening or observing uses these clues to understand what's being communicated (guessing). In this way, communication is like a game where one person has to think about the best way to get their message across, considering how the audience will interpret their words or actions, and the audience has to think about the best way to interpret them.

More languages than you think

Heike Wiese (2023) tells a similar story about her daughter and communicative situations, what she calls *com-sits*.

A somewhat extreme, but not uncommon, case was when at the age of three our younger daughter, Inya, seemed to consider English a linguistic quirk of her father, restricted to com-sits with him alone. Her experience had been that Daddy spoke that way, and most people also used this way of speaking when interacting with

him, but not in other situations. So, when our British in-laws came visiting, Inya refused to speak English to them because she found it awkward to use the “Daddy register” with anyone else – even when it became clear they did not understand German. While this was an idiosyncratic identification of relevant com-sit characteristics, it in fact underlines the conventional and hence community-oriented nature of com-sits: such deviations put a spotlight on learner hypotheses that will be revised in later development, with more exposure to language use reflecting the conventional patterns, similarly to grammatical hypotheses from earlier acquisitional stages. (Wiese, 2023, pp. 42–43)

Is it a step too far to say that Inya experiences speaking to her grandparents as ungrammatical?

Grieve (2023) claims

that languages can vary in their situational diversity. Languages that have been adapted for a wider range of communicative contexts are more situationally complex than languages that have been adapted for a narrower range of communicative contexts. To support this claim, I consider examples of situational diversity from across a range of different languages and varieties of languages, drawing on empirical research from linguistics and anthropology. Second, I claim that situational diversity can help explain variation in grammatical complexity. I propose that increasing situational diversity in a language over time should lead to decreasing grammatical complexity. Furthermore, I argue that this trade-off between situational and grammatical complexity could explain how overall linguistic complexity could be maintained across languages and over time.

8.5 SWEARING

Linguistic innovation is often used by subgroups in society.

“We were the people who were not in the papers. We lived in the blank white spaces at the edges of print. It gave us more freedom. We lived in the gaps between the stories.”

“It’s impossible to say a thing exactly the way it was, because of what you say can never be exact, you always have to leave something out, there are too many parts, sides, crosscurrents, nuances; too many gestures, which could mean this

or that, too many shapes which can never be fully described, too many flavors, in the air or on the tongue, half-colors, too many.”

Swearing in certain contexts is fine and in others, it's norm breaking, as are nudity, sex, and bodily fluids. Is swearing a norm violation because it's obscene, like a couple lustily copulating in a McDonalds booth or a pile of shit on the floor of an elevator, or because it's ungrammatical?

Imagine swearing isn't about the shock value of certain words, but instead exists as a secret language hidden within our own. It has its own twisted grammar, a code of expletives and crude turns of phrase that flout everything your school-teacher drilled into you. Just like slang, or the cryptic lingo of a trade, swearing becomes a badge of belonging.

Those within the circle have mastered this rebellious grammar. They hurl a curse with casual precision, knowing exactly when it adds emphasis or earns knowing laughter. But those on the outside fumble, either too shocked to use the words at all or dropping them clumsily into a sentence, revealing their outsider status. It's not about morality, but about fluency in this forbidden dialect.

This hidden language enforces its own sense of order. A well-timed profanity among familiar faces sparks camaraderie - its violation of “polite” rules becomes a wink of shared understanding. Yet, that same word unleashed against a stranger is an attack, precisely because they exist outside the imagined tribe where this twisted grammar is allowed. Swearing doesn't signal a failure of vocabulary, but rather a pointed choice to speak - or not speak - the subversive language that draws lines between those in the know, and those firmly excluded.

Across language boundaries

We speak in many ways. Some of us use academic language for scholarly articles and academic settings with specialized terminology and formal structures differing significantly from our everyday conversations. At the same time, we recognize the similarities and differences between that and the precise and structured language of legal English, a form reflecting the often adversarial nature of law and an accumulated body of precedence.

Literary English showcases a broad spectrum of styles, from classical literature's elevated diction and meandering clause structure to the colloquialisms in contemporary works, and we traverse among them, usually unhampered by any linguistic borders. In film and television, writers combine colloquial speech with scripted language, accommodating a wide range of social contexts and character archetypes, the language differences evoking for us myriad elements of character and setting, at least when done well.

Bill Withers moves us with the lamentation, *ain't no sunshine when she's gone*, even though its grammatical structures might not be ones we typically use – though we could if we would. We can even do dialectal time travel: somehow, even today, we know to avoid contractions when producing “old” English, reflecting a change that took place more than three centuries ago (Castillo González, 2007).

In poetry, too, we dance in ways we don't elsewhere. Advertising and marketing English is persuasive and catchy, often employing creative wordplay to appeal to diverse audiences. Sports commentary uses a specific set of terms unique to each sport, highlighting the dynamic nature of sports culture, and its idioms often enjoy something like home-field advantage in many non-sporting contexts.

The Scottish use of Scots, a Germanic language closely related to English, in

conjunction with Standard English, showcases the kind of dual linguistic identity seen in many places. Singlish, a colloquial form of English in Singapore, amalgamates English with elements from Malay, Mandarin, and other Chinese dialects. Beyond these come the pidgins and creoles, where English blends with other languages in multilingual communities, reflecting a mix of linguistic heritages. Many of us are multilingual, sometimes frighteningly so.

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From the viewpoint of Standard English, only (1a) is grammatical. The ' marks (1b) as belonging to a non-Standard English, and, as usual * means it's just ungrammatical: nobody speaks like that.

- (1) a. *Nobody has come to help us.*
- b. ' *Ain't nobody come to help us.*
- c. * *Isn't nobody come to help us.*
- d. * *Hasn't nobody come to help us.*

But what about from the perspective of Singlish or of African American Vernacular English?¹ While I don't know of any studies on the point, surely speakers of Singlish and African American English have extensive exposure to Standard English, and, perhaps more importantly, the Standard varieties are generally perceived as being correct, while the non-standard varieties, or at least salient elements thereof, are stigmatized. As a result, it seems likely that from the perspective of non-Standard English, both Standard and their own non-Standard varieties are grammatical, while, from the point of view of Standard English, the non-Standard forms are mostly seen as ungrammatical.

If a Singlish speaker said, *This person's Singlish is very good*, it would be considered grammatical in both Singlish and Standard English. If they said, *This guy Singlish damn good leh*, that would still be grammatical in Singlish, but not in Standard English. And the non-Singlish speaker might not even judge *Wah lau! This guy Singlish si beh hiong sia* to be English, let alone have an opinion on its grammaticality.²

Constructions such as (1c) are ungrammatical in any dialect. Others, like (1b), aren't only ungrammatical in Standard English but highly stigmatized, despite being grammatical in other varieties. But some non-Standard constructions, like (2a), are camouflaged (Spears, 2015). A Standard-English speaker might take it to be (2b) and ignore it.

¹I'll just call it AFRICAN AMERICAN ENGLISH without "vernacular".

²The examples are taken from the Singlish article on [Wikipedia](#).

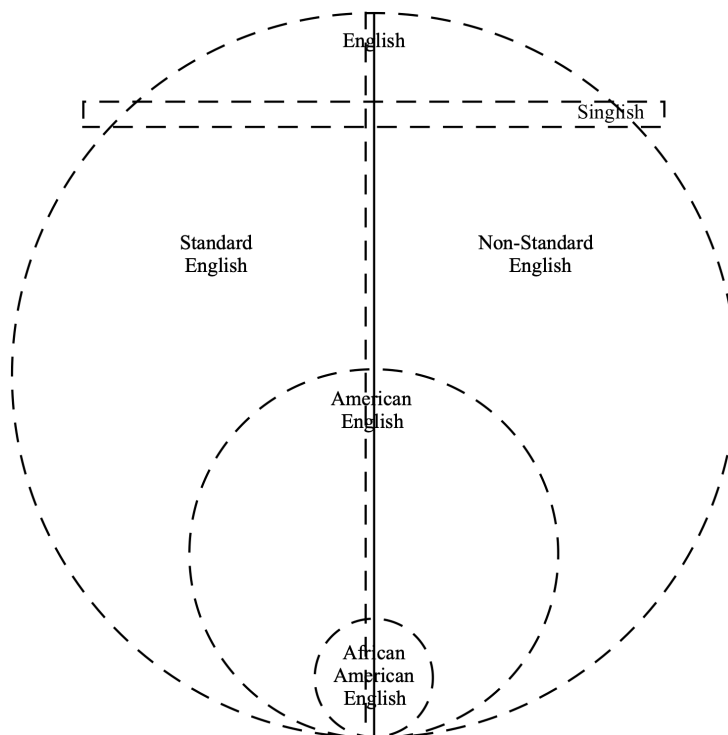


Figure 9.1. The relationships among various varieties of English. Dotted lines show a certain permeability, while the solid vertical line shows that non-Standard forms are less readily accepted from the perspective of the Standard forms.

- (2) a. ¹ *They've BIN living in Chicago.*
 b. *They've been living in Chicago.*

But (2a) isn't (2b). Where *they've been* could be any length of time, *they've BIN* can only mean 'for a long time' (Spears, 2015). This shows that, for something to be judged ungrammatical, it needs to be noticed.

In the all-Black schools I attended, for example, stigmatized vernacular language features elicited reprimand and sometimes punitive measures. Even students, during all-student gatherings, regimented the language of other students to conform to the student "standard", which tolerated a small set of vernacular forms (e.g., *ain't* and multiple negatives). Even students, who today might be labeled thugs or gang members, participated in language policing, upbraiding or ridiculing peers for using habitual *be*, *aks* (cp. *ask*), or pronunciations considered to diverge too far from the standard norm. Spears, 2015, p. 796

9.0.1 JUST HOW DIFFERENT IS IT?

The example in (1b) likely feels familiar to you, and no doubt *ain't* says clearly, "this isn't Standard English", which is true, as far as it goes. But it doesn't go nearly far enough. The next step is to notice that the Standard English construction employs *have*, not *be*, which is what we normally associate with *ain't*.

A third departure is the location of the subject *nobody*. In most English sentences, the subject precedes a verb, whether it be *ain't* (*I ain't eatin' that*) or something else. The major exception to this is in questions, such as *Is it OK?*, where the subject swaps places with the auxiliary verb. Nothing about (1b), though, would allow this kind of inversion in Standard English.

Finally, both *ain't* and *nobody* are negatives. Now, despite what you may have been told, having two negatives in a sentence is perfectly normal in the world's languages (see §6.1). In Standard Italian, we find

- (3) *Non è venuto nessuno ad aiutarci*
 not is come nobody to help-us
 'Nobody came to help us.'

But even though it's logically possible, it's still ungrammatical in Standard English. Those first two words alone contain four grammatical differences between *Ain't nobody come to help us* and *Nobody has come to help us*.

9.1 GRAMMAR IN BILINGUAL SPEECH

We've been considering the implications of using Standard vs non-Standard language, but that's far from exhausting the possibilities. When bilinguals speak amongst themselves, they often draw on both languages. By this point, you may not be surprised to find that there seem to be grammatical and ungrammatical ways to do this.

Consider the position of adjectives. German syntax typically places adjectives (underlined) before the noun, as in *eine glückliche Frau* ('a happy woman'). In contrast, Italian favors post-nominal adjectives, yielding *una donna felice*. This might suggest that either order would be possible, but that's not the case. *Una glückliche Frau* is grammatical for German-Italian bilinguals, but *i verdi Augen* ('the green eyes'), for instance, isn't (Cantone & MacSwan, 2009).

I won't try to generalize from these two examples. The data seems inconsistent, and various positions have been staked out and defended (it depends on the language of the adjective!). Suffice it to say that independent bilingual speakers and communities across a range of language pairs have a surprising level of agreement on what's grammatical and what's not. It's not perfectly consistent, but it is predictable: there's a grammar behind it.

9.2 ONE LANGUAGE, MULTIPLE GRAMMARS

Balinese

1. INTENTIONALITY
2. CLARITY
3. FAMILIARITY
4. GROUP ATTITUDES TOWARDS NOVELTY
5. IDEAS OF PROPRIETY
6. STATUS MAINTENANCE

9.3 GENDERED LANGUAGE

Even as far back as 1944, Furfey (1944) observed that men and women had characteristically different ways of speaking. He cited expressions like *Oh, dear!* and *How perfectly sweet!* as being feminine coded. Today, those might also feel a little dated, but we still have gender-coded language. I expect you have no problem determining which of the following are associated with which gender.

- (4) a. *I feel so blessed!*



Figure 9.2. A Venn diagram by Matt Shirley showing the contextual factors behind many simple expressions.

- b. *I'm stoked for this.*
- c. *I need to get laid.*
- d. *Isn't that adorable?*
- e. *Man up!*
- f. *Oh my gosh!*
- g. *Sick, dude!*
- h. *That's so sweet (of you).*

The question is, is your feeling upon hearing a masculine person say *Oh my gosh!* fundamentally the same as when you hear an Anglophone say *I no like that*? Each presents a mismatch between the persona and the signal encoded in the language. One is a gender mismatch, while the other is a competence mismatch. But is the competence mismatch somehow special?

9.4 MICRODIALECTS

In the early 1950s, two graduate students at Catholic University set out to study the dialect of African American residents living in the inhabited alleys of Washington D.C. Father George N. Putnam and Sister Rosina O'Hern were students of the sociologist Rev. Paul Hanly Furfey, who had introduced them to the field of the sociology of language.³

The alleys, with their substandard living conditions, had for decades formed insular communities that developed their own distinctive linguistic features. Putnam and O'Hern interviewed 74 of the 88 known adult alley residents, carefully transcribing their speech phonetically. They also made tape recordings of five residents reading sentences and identifying objects, and even recounting one of *Aesop's Fables*.

The researchers subjected these recordings to detailed phonetic and spectrographic analysis, examining vowel and consonant qualities, stress, intonation and timing. They also noted non-standard grammar and vocabulary used by the residents based on the interviews and recordings.

But Putnam and O'Hern took things a step further with an innovative experiment. They recorded twelve African American speakers, three from the alleys and nine of higher social status, all telling the same story. They then played these recordings to 70 listeners and asked them to rank the speakers by social status. Remarkably, the rankings correlated very closely with the actual status of the speakers, showing that dialect alone, even in the absence of any other cues, strongly signaled social standing.

³I got this story from Joseph (2002, pp. 121–122).

This study, published in 1955 in the journal *Language*, demonstrated that we all track innumerable linguistic features, assigning all kinds of non-linguistic meanings to them. It's not likely that the judges had met or heard any residents of these alleys, so they wouldn't have been able to form associations between their particular linguistic features and their socio-economic status.

To the extent that we can trust their data,⁴ this means that it must have been the residents themselves who extrapolated from the differences they observed in the language around them, subconsciously applying them to their own speech in ways that reflected their self-perceptions of their place in society. And they must have done this in ways that resonated for the judges. It's a truly remarkable feat of coordination, one that likely benefits from innate attention to various linguistic and social factors, but not one that relies on any "Universal Grammar".

⁴"Labov perhaps not inaccurately but surely incompletely characterized" it as unsystematic, saying "and it is not clear what the judges were reacting to, or how representative their judgments were" (Labov 1966:19), as cited in Joseph (2002, p. 125).

Impossible languages

Linguists have ventured into the realm of impossible languages, crafting systems that are ungrammatical – ingrammatical, antigrammatical – by their very nature. These languages defy the syntactic structures we're familiar with, presenting sentences where words are arranged in random sequences, obliterating any semblance of conventional order. They experiment with grammatical rules based on the numerical position of words or on fully or partially backward sequences. These linguistic constructs stretch the definition of language, inhabiting a space that's scarcely linguistic at all.

In one language, for instance, any string is grammatical as long as its length is prime: any five-word string is in the language, any six-word string is out. Or consider the language where the rule is that any word that occurs in a sentence must occur an even number of times. So, if we use different letters to represent different words, the sequence *abbcabcb* is grammatical, while *bbbcabcb* is ungrammatical.

(Sampson, 1975, p. 54) ...The trouble is that if we use transformations, we can define any kind of language that can be defined at all. I have been talking about ps grammars, transformations, and so on in quite a woolly, intuitive way, but one can make these notions precise and rigorous. If one defines the notion transformational grammar precisely, there is a theorem that any rel can be defined by some transformational grammar. In other words, even the ridiculous case of the prime number language I discussed earlier, which, we agreed, could never occur as a natural language, could be specified by some transformational grammar. ...So, although the theory of transformational grammar does not partition the set of rel's into a set of possible and a set of

impossible languages, it does impose an ordering on the languages, so that some of the rel's will be relatively easy to describe with transformational notation and other rel's will be much harder to define that way (even though they might be quite easy to define in some other notation system). There are all kinds of notation systems one could imagine, each of which would permit all the rel's to be defined; but, depending which notation system one chooses, one gets a different complexity ordering among the various rel's. If it is empirically true that, in the ordering defined by Chomsky's particular notation system, all the natural languages are relatively near the simple end of the range, then Chomsky's notation must correspond to Ejiirt contingent fact about the nature of natural language. Now it might be the case that there is some feature of the environment, outside the human organism, which forces this structure on the languages that humans use. But (Sampson, 1975, p. 56)

To make a negative clause, place *no* after the third word of the clause. The first article of the clause has to agree with the last noun. A closed interrogative clause is formed by inverting the order of the corresponding declarative clause. (Moro, 2016, pp. 55–56)

What could this be for?

Sometimes, the best way to clarify something is to imagine what it's not. The hugely influential linguist Noam Chomsky (1957) believed that some sentences belong to a particular language and other sequences of words don't. But beyond that, he believed the strings that don't can be further subdivided into those that *could* be part of the particular language, and those that simply *couldn't*. To make that more concrete, various ideas have been floated about what these impossible languages might be like.

For example, in (1b), all the even-numbered words are fronted, preserving their original order. In (1c), a special character is inserted, after which everything is reversed. And in (1d), the tense-marker on any verb attaches to the word four positions to the right of the verb (or less if fewer words are available). Figure 10.1 puts these and other perturbations on an impossible–possible continuum of these languages.

- (1) a. *He cleans all the books on his very messy bookshelf.* [English]
- b. *He all books his messy cleans the on very bookshelf.* [Even-Odd]
- c. *He cleans all the books on | bookshelf messy very his.* [Partial reverse]

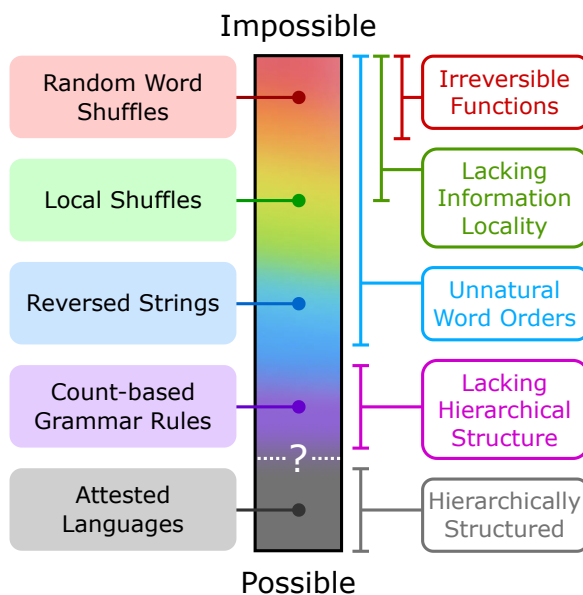


Figure 10.1. Partial impossibility continuum of languages based on complexity. Reproduced from Kallini et al. (2024) Figure 1.

d. *He clean all the books on his very messy bookshelf.* [Tense hop]

(2) a. *Colorless green ideas sleep furiously.*

b. **Furiously ideas sleep green colorless.*

Chomsky (1957, p. 16) claims that neither (2a) nor (2b), nor their subparts, would occur in a corpus of English, they would be “equally ‘remote’ from English [...] in any statistical model of English” and that they’re equally ungrammatical. But Fernando Pereira (2000) showed that (2a) is much more statistically likely than (2b), which matches my intuition (and yours too, perhaps).

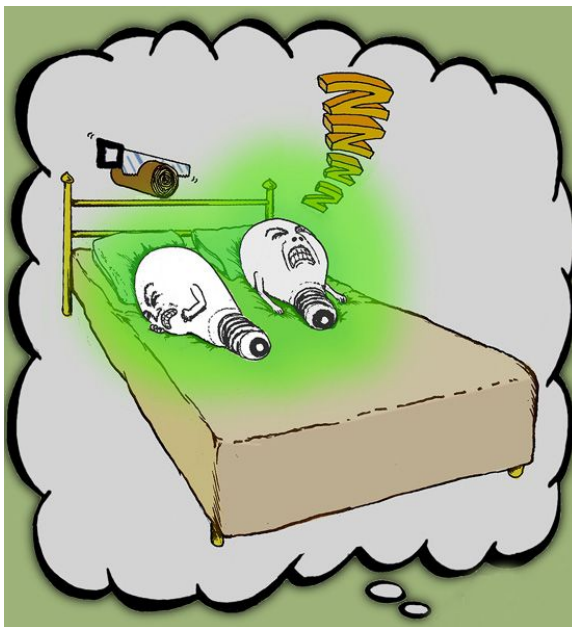


Figure 10.2. A cartoon showing angry lightbulbs with green glows sleeping, from Mikael Parkvall's *Limits of Language*.

Communicative efficiency

11.0.1 KEEPING RIGHT AND KEEPING LEFT

Languages aren't arbitrary in their structure but are shaped by a balance between the need to convey information effectively and the cognitive constraints of human brains.¹ The EFFICIENCY HYPOTHESIS posits that languages evolve to achieve this balance. A key principle underlying this hypothesis is DEPENDENCY LOCALITY, which suggests that languages tend to organize words in a way that reduces the distance between related elements, thereby making communication more efficient.

Consider the organization of a workspace: tools and materials frequently used together are kept close to each other to minimize the effort needed to reach them, drill bits with the drill, coffee mugs near the coffee maker, reading glasses by the computer. Similarly, in languages, related words are arranged to minimize their separation. This principle can be observed across different language families, indicating a universal tendency towards efficiency in human communication.

Where we drive on the right, highway exits are generally to the right, and where we drive on the left, the exits tend to follow suit. This keeps things simple, reduces the number of lanes requiring overpasses, and shortens the length of on and off ramps. We see a similar principle in grammar: in languages like English, where the object usually follows the head verb, the dependent in a preposition phrase also usually follows the head preposition. For example, in the verb phrase *take a rest*, the object *a rest* follows the head verb *take*, and in the preposition phrase *with a rest*, it follows the head preposition *with*. You don't find **a rest take* or **a rest with*. The same regularity occurs in languages like Japanese, but with all the orders inverted: the dependent precedes the head, whether it is in a

¹This chapter draws heavily on Futrell et al. (2020).

verb phrase or a preposition phrase.

In both cases, this principle minimizes the dependency distance between heads and their dependents. We can see this in (1 & 2), where the heads are underlined. The distance is minimized in (a–c), but in the ungrammatical (d), where the dependent orders are mis-matched, they are much farther apart.

- (1) a. *I take a rest [before].*
 b. *I take a rest [before meetings].*
 c. *I take a rest [before I take part in an important meeting].*
 d. **I take a rest [I take part in an important meeting before].*
- (2) a. [前に]休む
 b. [会議前に]休む
 c. [大切な会議に参加する前に]休む
 d. * [前に大切な会議に参加する]休む

To be clear, it's not likely that the (d) cases are ungrammatical *because* the heads are far apart – sometimes, heads can be much farther apart than that. It's more likely that the ordering has become conventionalized because it is the one that usually minimizes dependency distances. And it's the unconventional aspect of (d), not the dependency distance, that registers as ungrammatical ... unless it can be connected to some semantic or syntactic motivation. But what could motivate such an unconventional structure?

Occasionally, a highway interchange is set up with a contra-lateral exit. One such case is to be found on Ontario's Highway 427 southbound to Eglinton Avenue. As the highway crosses Eglinton, it jogs sideways, and Highway 401 happens to intersect at just this point, with the mess in Figure 11.1 (OpenStreetMap contributors, 2024) being the result. But this left-side exit wasn't capricious or accidental; it's constructed this way because the usual rightwards design would have been more complex to execute.

We find similar adjustments for complexity in language. One is the so-called HEAVY NP (NOUN PHRASE) SHIFT. A “heavy” NP is one with a lot of words in it: *a book that I had found in a used-book store in Paris* is a heavy NP, *a book* is light. And the shift is a shift away from the verb, to the right in English, reflecting a “short-before-long” preference.

In heavy NP shift, the general ordering tendency is suppressed. The result, again, is to minimize the dependency distance between heads. In (3), we can count the number of words intervening between *gave* and the head of each of the dependents (*book*, *to*, and *for*) to find the dependency distance.

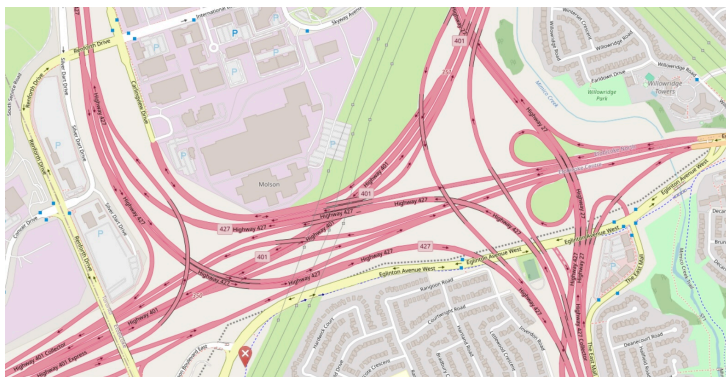
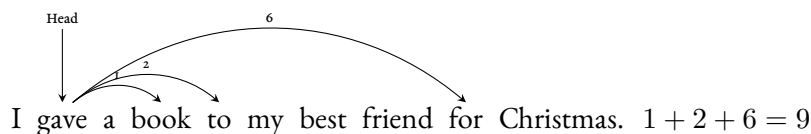


Figure 11.1. The interchange complexity that results in a left-side exit from Hwy 427 southbound to Eglinton Avenue.

(3) a.



- b. *I gave [a book that I had found in a used-book store in Paris] to my best friend for Christmas.* $1 + 12 + 16 = 29$
- c. *I gave to my best friend for Christmas [a book that I had found in a used-book store in Paris].* $0 + 4 + 6 = 10$

The basic version in (a) has a total dependency distance of nine words. The (b) version is grammatical, but with the heavy noun phrase plugged into the basic sentence structure, the dependency distance more than doubles, from nine words to twenty. On the other hand, under the heavy NP shift, the dependency distance drops back to 10 words. Just as in the highway 427 exit example, there's a good reason to adjust the structure to accommodate an unusual situation.

But if we try to do the same thing with a “light” NP (4), there's no justification, and so the unusual structure fails. It's ungrammatical. The shift is only allowed when there's a good reason for it.

(4) * *I gave to my best friend for Christmas a book.* $0 + 4 + 6 = 10$

The passive voice is another way to repackage things to reduce dependency distances. Consider the passive (5a) from Wikipedia with a dependency distance of zero between the underlined head of the subject and head verb. Now compare the active counterpart (b) with a dependency distance of 21.

- (5) a. *A convolutional neural network architecture is formed by a stack of distinct layers that transform the input volume into an output volume (e.g. holding the class scores) through a differentiable function.*
[PASSIVE, 0]
- b. *A stack of distinct layers that transform the input volume into an output volume (e.g. holding the class scores) through a differentiable function forms a convolutional neural network architecture.*
[ACTIVE, 21]

No matter how many times you've been told to avoid the passive voice, I think you'll agree that the passive is better here.

One more example illustrates the point. Consider the difference between *I threw out the trash* and *I threw the trash out*. Both are fine, but as the *trash* noun phrase becomes weighed down with more and more words in (6), the pressure for the *out*-first version builds. Notice the feeling of resolution going from (f) to (g).

- (6) a. *I threw it out.*
b. *I threw the trash out.*
c. *I threw the kitchen trash out.*
d. *I threw the smelly kitchen trash out.*
e. [?] *I threw the smelly kitchen trash from under the sink out.*
f. [?] *I threw the smelly kitchen trash from under the sink that had been sitting there for two weeks out.*
g. *I threw out the smelly kitchen trash from under the sink that had been sitting there for two weeks.*

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As compact as a consistent head - dependent order is, there's a more efficient one. Gildea and Temperley (2007) show that alternating dependents minimizes dependency length. Under this ideal structure, a head with an even number of dependents would have one on each side. The odd dependents would still tend to the post-head position, at least in languages like English. This alternating-dependent structure is most effective at minimizing dependency distance in phrases with multiple dependents, such as verb phrases and noun phrases.

English turns out to be quite close to this ideal. Verb phrases mix head-before-dependent as in *take a break* with dependent-before-head, as in *I rested*. In the

noun phrase, too, any determiner always precedes the noun – we have *the world*, *this year*, *some people*, *many times*, *my life*, and millions more.² But modifiers like relative clauses or preposition phrases typically follow the noun, as in *the world that we live in* or *a person with knowledge of the changes*.

Another factor is phrase length. If one dependent is pre-head and one post, which should go first? In a predominantly head-first language, like English, we've already seen that heavier, longer phrases gravitate right. Shorter phrases, conversely, tend to occupy the pre-head positions.

You may have noticed that the determiner is a single word in all these examples, and typically a fairly short one at that. A determiner can be longer *too many people* or *almost all cases*, but almost all determiners are short, with the articles *the* and *a* dominating. The opposite holds for relative clauses (*the things that he said*) and preposition phrases (*the girl with the green eyes*). Shorter modifiers, like adjectives, tend to be pre-head (*other people*).

²By *always*, I don't literally mean 'always'. There are cases like *I grant you wishes three* or Shakespeare's *o mistress mine*.

What grammaticality is

- 12.1 THE CLUSTER, NAMED
- 12.2 THE DETECTOR
- 12.3 WHAT HPC IS NOT
- 12.4 THE MECHANISMS
- 12.5 THE FOURTEEN-POINT MODEL, DEMOTED
- 12.6 THE STAKES

Getting grammaticality wrong

13.1 THE DIACHRONIC CONTEXT

Old English didn't have relative pronouns the way we have them. It just used *þe* to mark subordinate clauses of various kinds, including relative clauses.

- (1) þonne ealra oþra kyninga þe in middangearde æfre wæron ...
 Then all other kings that in Middle-Earth ever were ...
 'Then all the kings who were ever on earth ...'

There was *hwa* 'who', but it was mainly interrogative. It had a special construction meaning roughly 'whoever', as in *whoever enters remains*, but the *þe* in (1) couldn't have been replaced with *hwa*. It was only around 1325, in the midst of Middle English, that we start to see a real relative *who* in the way we know it today.

- (2) He nadde bote an doȝter wo miȝte is eir be.
 He hadn't but one daughter who might his heir be.
 'He had no one but a daughter who could be his heir.'

Similarly, there wasn't a relative *whose* in Old English. Instead you get constructions like *þe his* 'that his' in (3).

- (3) Eaðig bið se wer, þe his tohopa bið to Drihtne
 Blessed be the man that his hope is in the Lord.
 'Blessed be the man whose hope is in the Lord'

13.2 THE HANKAMER-POSTAL MOMENT

- 13.3 WHOSE, FOUND IN THE WILD
- 13.4 THE GPT-4 MOMENT
- 13.5 PAYNE'S REVISION
- 13.6 WHAT THE CLUSTER SHOWS

Coda

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